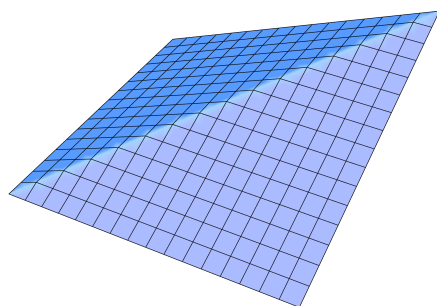




# PMATH 450: Lebesgue Integration and Fourier Analysis

Prof. Mahya Ghandehari • Winter 2015 • University of Waterloo  
TR 12:30–2:20PM in PHY 150

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Transcribed,  $\text{\LaTeX}$ ed, and edited by Rob Zimmerman

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## Lecture 1 — Tuesday, January 06, 2015

## 1. Measure Theory Foundations

## Riemann Integration and the Need for Measure

To begin with, we briefly review Riemann integration. Let  $f : [a, b] \rightarrow \mathbb{R}$  be given. Partition  $[a, b]$  by

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

Define

$$U(f, P) = \sum_{i=1}^n \sup_{[x_{i-1}, x_i]} f \cdot (x_i - x_{i-1})$$

and

$$L(f, P) = \sum_{i=1}^n \inf_{[x_{i-1}, x_i]} f \cdot (x_i - x_{i-1}),$$

the **upper and lower Riemann sums**.

**Fact 1.1** For any two partitions  $P$  and  $Q$ ,  $U(f, P) \geq L(f, Q)$ .

**Proof.** Let  $P : a = x_0 < x_1 < \cdots < x_n = b$  and  $Q : a = y_0 < y_1 < \cdots < y_m = b$  be two partitions of  $[a, b]$ . Let  $R$  be the common refinement of  $P$  and  $Q$ , which is the partition obtained by taking the union of the points in  $P$  and  $Q$ . Then  $U(f, P) \geq U(f, R)$  and  $L(f, Q) \leq L(f, R)$ , since the upper sum decreases and the lower sum increases when we refine the partition. Therefore,  $U(f, P) \geq U(f, R) \geq L(f, R) \geq L(f, Q)$ .  $\square$

Thus  $\inf_P U(f, P) \geq \sup_Q L(f, Q)$ . We say that  $f$  is **Riemann integrable** if equality holds.

**Theorem 1.2** If  $f$  is continuous on  $[a, b]$ , then  $f$  is Riemann integrable.

Figure 1 shows the geometric comparison behind the upper and lower sums: refinement makes the two stepwise estimates close in on the same area.

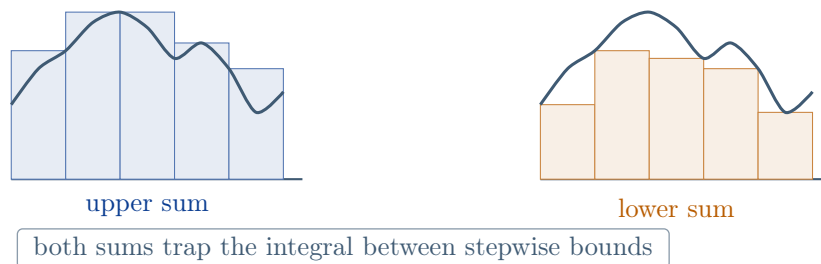


FIGURE 1

**Notation 1.3** Let  $E \subseteq \mathbb{R}$ . Then  $\mathbf{1}_E$  denotes the **characteristic function** of  $E$ . That is,

$$\mathbf{1}_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \notin E \end{cases}$$

**Example 1.4**  $\mathbf{1}_{\mathbb{Q} \cap [0,1]}$  is not Riemann integrable.

**Proof.** Let  $P : 0 < x_1 < x_2 < \cdots < x_n = 1$  be a partition of  $[0, 1]$ . Since  $\mathbb{Q}$  is dense in  $\mathbb{R}$ , for every  $i$ , there exists  $q_i \in \mathbb{Q} \cap [x_{i-1}, x_i]$ . Then  $\sup_{[x_{i-1}, x_i]} \mathbf{1}_{\mathbb{Q}} = 1$  and  $\inf_{[x_{i-1}, x_i]} \mathbf{1}_{\mathbb{Q}} = 0$ . Hence,

$$U(\mathbf{1}_{\mathbb{Q}}, P) = 1 \neq 0 = L(\mathbf{1}_{\mathbb{Q}}, P).$$

□

Let  $\{x_1, x_2, \dots\} = \mathbb{Q} \cap [0, 1]$ . For any  $n \in \mathbb{N}$ , define

$$f_n(x) = \begin{cases} 1 & \text{if } x = x_j \text{ and } 1 \leq j \leq n \\ 0 & \text{otherwise} \end{cases}.$$

Then  $f_n(x) \rightarrow \mathbf{1}_{\mathbb{Q} \cap [0,1]}(x)$  for every  $x \in [0, 1]$ . However,

$$\lim_{n \rightarrow \infty} \int f_n(x) \, dx = 0.$$

Then

$$\int \lim_{n \rightarrow \infty} f_n(x) \, dx \neq \lim_{n \rightarrow \infty} \int f_n(x) \, dx.$$

If  $E_1, \dots, E_n$  are disjoint intervals of  $\mathbb{R}$  and

$$f = \sum_{i=1}^n a_i \mathbb{1}_{E_i}$$

for  $a_i \in \mathbb{R}$ , then

$$\int f(x) \, dx = \sum_{i=1}^n a_i \cdot (\text{length of } E_i).$$

For a function  $f : [a, b] \rightarrow \mathbb{R}$ , partition an interval containing  $f([a, b])$  as

$$A = y_0 < y_1 < \dots < y_n = B.$$

For  $i = 0, \dots, n-1$ , let  $E_i = f^{-1}([y_i, y_{i+1}))$ ; then

$$f \approx \sum_{i=0}^{n-1} y_i \mathbb{1}_{E_i}.$$

In other words,  $f(x)$  is close to

$$\sum_{i=0}^{n-1} y_i \mathbb{1}_{E_i}(x).$$

When the Lebesgue integral is defined, we want

$$\int_a^b \left( \sum_{i=0}^{n-1} y_i \mathbb{1}_{E_i}(x) \right) \, dx = \sum_{i=0}^{n-1} y_i (\text{length of } E_i).$$

This suggests the approximation

$$\int_a^b f(x) \, dx \approx \int \sum_{i=0}^{n-1} y_i \mathbb{1}_{E_i}(x) \, dx.$$

However,  $f^{-1}([y_i, y_{i+1}))$  is not necessarily an interval; we need a better concept of “length” for a general arbitrary set. This motivates the concept of a measure.

We ideally want to have a function  $m : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$  which satisfies the following properties:

1.  $m(I) = \text{length}(I)$  for any interval  $I$ .

2. If  $E_1, E_2, \dots$  are disjoint subsets of  $\mathbb{R}$ , then

$$m\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} m(E_i)$$

(**countable additivity**).

3. Let  $E \subseteq \mathbb{R}$  and  $y \in \mathbb{R}$ . Then the translation of  $E$  with  $y$  is  $E + y = \{e + y \mid e \in E\}$ . So we want  $m(E + y) = m(E)$  (**translation invariance**).

4. If  $A \subseteq B$ , then  $m(A) \leq m(B)$  (**monotonicity**).

However, there are storm clouds on the horizon.

**Proposition 1.5** There is no function  $m : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$  satisfying Properties (1)–(4) above.

**Proof.** We will construct a countable collection of sets which cannot satisfy (1)–(4) for any function  $m$ . On  $[0, 1)$ , define  $x \sim y$  if  $x - y \in \mathbb{Q}$ . This is an equivalence relation.

Using the Axiom of Choice, choose a set  $N \subseteq [0, 1)$  containing exactly one representative from each equivalence class. For  $r \in \mathbb{Q} \cap [0, 1)$ , let

$$N_r = \{r + n \mid n \in N \cap [0, 1 - r)\} \cup \{-1 + r + n \mid n \in N \cap [1 - r, 1)\}.$$

If  $m : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$  satisfies (1)–(4), then translation invariance and finite additivity give  $m(N_r) = m(N)$ . Moreover, the sets  $N_r$  are pairwise disjoint and

$$\bigcup_{r \in \mathbb{Q} \cap [0, 1)} N_r = [0, 1).$$

Therefore,

$$m([0, 1)) = \sum_{r \in \mathbb{Q} \cap [0, 1)} m(N_r).$$

The left-hand side is 1, while the right-hand side is 0 if  $m(N) = 0$  and  $\infty$  if  $m(N) > 0$ . Either way, we have a contradiction.  $\square$

## Lecture 2 — Thursday, January 08, 2015

## Lebesgue Outer Measure

We want to define the Lebesgue measure. Before that, though, we define the Lebesgue outer measure.

**Notation 2.1**

1. If  $a < b$ ,  $a, b \in \mathbb{R}$ , then

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$$

is an **open interval**. With  $a = b$ , the empty set  $\emptyset$  is also considered an open interval.

2.  $\mathcal{P}(\mathbb{R})$  is the collection of all subsets of  $\mathbb{R}$  and is called the **power set** of  $\mathbb{R}$ .
3. Let  $E$  be a subset of  $\mathbb{R}$ . The sequence of open intervals  $\{I_n\}_{n=1}^{\infty}$  is called an **open cover for  $E$**  if  $E \subseteq \bigcup_{i=1}^{\infty} I_i$ .

**Definition 2.2** The **Lebesgue outer measure** is the function  $\lambda^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$  defined as follows: for every  $E \subseteq \mathbb{R}$ ,

$$\lambda^*(E) = \inf \left\{ \sum_{i=1}^{\infty} \ell(I_i) \mid \{I_i\}_{i=1}^{\infty} \text{ is an open cover of } E \right\}$$

As Figure 2 suggests, the infimum searches through every countable open cover, not merely the first convenient one we happen to draw.

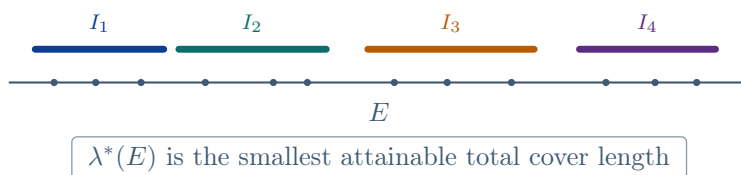


FIGURE 2

**Proposition 2.3** The following hold:

1.  $\lambda^*(E) \geq 0$  for every  $E \subseteq \mathbb{R}$  (**nonnegativity**).
2.  $\lambda^*(\emptyset) = 0$  and  $\lambda^*(\{x\}) = 0$ .
3. If  $A, B \subseteq \mathbb{R}$  and  $A \subseteq B$ , then  $\lambda^*(A) \leq \lambda^*(B)$  (**monotonicity**).
4.  $\lambda^*(E) = \lambda^*(E + x)$  for all  $E \subseteq \mathbb{R}$  and  $x \in \mathbb{R}$  (**translation invariance**).
5. If  $A_1, A_2, \dots \subseteq \mathbb{R}$ , then

$$\lambda^* \left( \bigcup_{i=1}^{\infty} A_i \right) \leq \sum_{i=1}^{\infty} \lambda^*(A_i)$$

( **$\sigma$ -subadditivity**).

6.  $\lambda^*(\text{interval}) = \ell(\text{interval})$  for any interval in  $\mathbb{R}$ .

**Proof.** (1) is obvious.

For (2), observe that  $\{x\} \subseteq (x - \varepsilon, x + \varepsilon)$ . Then  $\lambda^*(\{x\}) \leq 2\varepsilon$  by definition of  $\lambda^*$ . This holds for every  $\varepsilon > 0$ ; therefore,  $\lambda^*(\{x\}) = 0$ . The same argument applies for  $\emptyset$ .

For (3), first let  $\mathcal{C}(A)$  denote the collection of all open covers of  $A$ ; that is,

$$\mathcal{C}(A) = \{ \{I_i\}_{i=1}^{\infty} \mid \{I_i\}_{i=1}^{\infty} \text{ is an open cover for } A \}$$

If  $A \subseteq B$ , then  $\mathcal{C}(B) \subseteq \mathcal{C}(A)$ . Therefore,

$$\inf \left\{ \sum_{i=1}^{\infty} \ell(I_i) \mid \{I_i\} \in \mathcal{C}(A) \right\} \leq \inf \left\{ \sum_{i=1}^{\infty} \ell(I_i) \mid \{I_i\} \in \mathcal{C}(B) \right\},$$

which is the same as  $\lambda^*(A) \leq \lambda^*(B)$ .

For (4), let  $x \in \mathbb{R}$ . If  $\{I_i\}_{i=1}^{\infty}$  is an open cover of  $E$ , then  $\{I_i + x\}_{i=1}^{\infty}$  is an open cover of  $E + x$ , and  $\ell(I_i + x) = \ell(I_i)$ . Hence

$$\lambda^*(E + x) \leq \sum_{i=1}^{\infty} \ell(I_i + x) = \sum_{i=1}^{\infty} \ell(I_i).$$

Taking the infimum over all open covers of  $E$  gives  $\lambda^*(E + x) \leq \lambda^*(E)$ . Replacing  $x$  by  $-x$  yields

$\lambda^*(E) \leq \lambda^*(E + x)$ , so  $\lambda^*(E + x) = \lambda^*(E)$ .

For (5), if

$$\sum_{i=1}^{\infty} \lambda^*(A_i) = \infty,$$

then we are done, so assume otherwise. Pick  $\varepsilon > 0$ . For  $n \in \mathbb{N}$ , let  $\{I_{i,n}\}_{i=1}^{\infty}$  be an open cover for  $A_n$  such that

$$\lambda^*(A_n) + \frac{\varepsilon}{2^n} > \sum_{i=1}^{\infty} \ell(I_{i,n}).$$

Observe that the collection  $\{I_{i,n}\}_{i,n=1}^{\infty}$  is an open cover for  $\bigcup_{i=1}^{\infty} A_i$ . Then

$$\lambda^*\left(\bigcup A_i\right) \leq \sum_{n=1}^{\infty} \sum_{i=1}^{\infty} \ell(I_{i,n}) = \sum_{n=1}^{\infty} \sum_{i=1}^{\infty} \ell(I_{i,n}) \leq \sum_{n=1}^{\infty} [\lambda^*(A_n) + \varepsilon/2^n] = \sum_{n=1}^{\infty} \lambda^*(A_n) + \varepsilon,$$

which is true for all  $\varepsilon > 0$ . So

$$\lambda^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \lambda^*(A_i).$$

For (6), we want to prove  $\lambda^*(\text{interval}) = \ell(\text{interval})$ . We will prove it for closed intervals  $[a, b]$  where  $a, b \in \mathbb{R}$ , which amounts to showing  $\lambda^*([a, b]) = b - a$ . For the upper bound,  $[a, b] \subseteq (a - \varepsilon/2, b + \varepsilon/2)$  gives  $\lambda^*([a, b]) \leq b - a + \varepsilon$  for every  $\varepsilon > 0$ , and hence  $\lambda^*([a, b]) \leq b - a$ . For the reverse inequality, we want to show that if  $\{I_n\}_{n=1}^{\infty}$  is an open cover for  $[a, b]$ , then

$$\sum_{n=1}^{\infty} \ell(I_n) \geq b - a.$$

Now,  $[a, b]$  is compact, so that it admits a finite subcover of  $\{I_n\}_{n=1}^{\infty}$ , say  $I_1, \dots, I_N$ . We know

$$\sum_{n=1}^{\infty} \ell(I_n) \geq \ell(I_1) + \dots + \ell(I_N).$$

So assume our open cover is finite, say  $\{I_n\}_{n=1}^N$ . If  $a \in \bigcup_{n=1}^N I_n$ , then there exists  $n_1$  such that  $a \in I_{n_1} = (a_{n_1}, b_{n_1})$ . If  $b \in I_{n_1}$ , then

$$\sum_{n=1}^N \ell(I_n) \geq \ell(I_{n_1}) \geq b - a,$$

so we are done. Otherwise,  $b \notin I_{n_1}$ , so  $b_{n_1} \in [a, b]$ . Choose  $I_{n_2}$  with  $b_{n_1} \in I_{n_2} = (a_{n_2}, b_{n_2})$ . Since  $b_{n_1} \notin I_{n_1}$ , the interval  $I_{n_2}$  is distinct from  $I_{n_1}$ . Continue in this way: at step  $j$ , choose  $I_{n_j}$

containing  $b_{n_{j-1}}$ . Since the cover is finite and the right endpoints strictly move to the right until  $b$  is covered, this process terminates in a finite number of steps. Thus,

$$\sum_{n=1}^N \ell(I_n) \geq \sum_{j=1}^k \ell(I_{n_j}) = (b_{n_1} - a_{n_1}) + \cdots + (b_{n_k} - a_{n_k}) \geq b_{n_k} - a_{n_1} \geq b - a.$$

Hence,  $\lambda^*([a, b]) = b - a$ . □

**Example 2.4**  $\lambda^*(\mathbb{Q}) = 0$ . In general, if  $A$  is countable, then  $\lambda^*(A) = 0$  since

$$A = \{x_1, x_2, \dots\} = \bigcup_{n=1}^{\infty} \{x_n\}$$

and

$$\lambda^*(A) \leq \sum_{n=1}^{\infty} \lambda^*({x_n}) = \sum_{n=1}^{\infty} 0 = 0$$

by  $\sigma$ -subadditivity.

## Lebesgue Measurable Sets

**Definition 2.5** We say that  $A \subseteq \mathbb{R}$  is **Lebesgue measurable** if, for every  $E \subseteq \mathbb{R}$ ,

$$\lambda^*(E) = \lambda^*(E \cap A) + \lambda^*(E \setminus A).$$

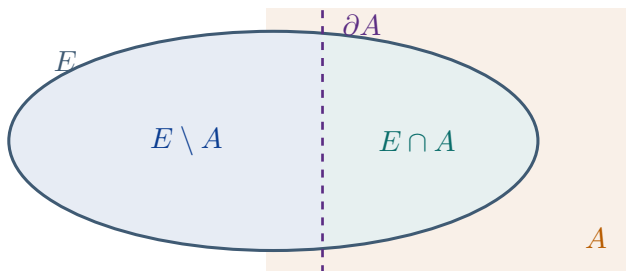
**Remark 2.6** Because  $E = (E \cap A) \cup (E \setminus A)$ , we always have

$$\lambda^*(E) \leq \lambda^*(E \cap A) + \lambda^*(E \setminus A)$$

by  $\sigma$ -subadditivity. It is the reverse inequality that is the nontrivial part of the definition of measurability.

Figure 3 depicts Carathéodory's criterion: a measurable set  $A$  cuts every test set  $E$  into two pieces without creating any extra outer measure.

**Notation 2.7** Let  $\mathcal{L}(\mathbb{R})$  denote the collection of all Lebesgue measurable subsets of  $\mathbb{R}$ .



the two measured pieces add back to the whole

FIGURE 3

**Example 2.8** Here are some examples of Lebesgue measurable sets:

1.  $\emptyset \in \mathcal{L}(\mathbb{R})$ .
2.  $\{x\} \in \mathcal{L}(\mathbb{R})$  for  $x \in \mathbb{R}$ . Why? Let  $E \subseteq \mathbb{R}$ . Obviously,  $\lambda^*(E \cap \{x\}) \leq \lambda^*(\{x\}) = 0$  because  $E \cap \{x\} \subseteq \{x\}$ . Also,

$$\lambda^*(E) \leq \lambda^*(E \setminus \{x\}) + \lambda^*(\{x\}) = \lambda^*(E \setminus \{x\}).$$

Hence,  $\lambda^*(E) = \lambda^*(E \cap \{x\}) + \lambda^*(E \setminus \{x\})$ . In general, if  $A$  is such that  $\lambda^*(A) = 0$ , then  $A$  is measurable.

### Lecture 3 — Tuesday, January 13, 2015

#### Sigma-Algebras and Borel Sets

Recall that the outer measure  $\lambda^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$  is translation invariant, satisfies  $\lambda^*(\text{interval}) = \ell(\text{interval})$ , is  $\sigma$ -subadditive, and is monotone. We defined a set  $A \subseteq \mathbb{R}$  to be measurable when  $\lambda^*(E) = \lambda^*(E \cap A) + \lambda^*(E \setminus A)$  for every  $E \subseteq \mathbb{R}$ .

#### Theorem 3.1

1.  $\emptyset, \mathbb{R} \in \mathcal{L}(\mathbb{R})$ .

2. If  $A \in \mathcal{L}(\mathbb{R})$ , then  $A^c \in \mathcal{L}(\mathbb{R})$ .
3. If  $A_1, A_2, \dots \in \mathcal{L}(\mathbb{R})$ , then  $\bigcup_{n=1}^{\infty} A_n \in \mathcal{L}(\mathbb{R})$ .

**Proof.** For (1), let  $E \subseteq \mathbb{R}$ . Then

$$\lambda^*(E) = \lambda^*(E) + \lambda^*(\emptyset) = \lambda^*(E \cap \emptyset) + \lambda^*(E \setminus \emptyset),$$

so  $\emptyset \in \mathcal{L}(\mathbb{R})$ . Similarly,

$$\lambda^*(E) = \lambda^*(E \cap \mathbb{R}) + \lambda^*(E \setminus \mathbb{R}),$$

so  $\mathbb{R} \in \mathcal{L}(\mathbb{R})$ .

For (2), let  $E \subseteq \mathbb{R}$ . Because  $A \in \mathcal{L}(\mathbb{R})$ , we have

$$\lambda^*(E) = \lambda^*(E \cap A) + \lambda^*(E \setminus A) = \lambda^*(E \cap A) + \lambda^*(E \cap A^c),$$

so  $A^c \in \mathcal{L}(\mathbb{R})$ .

For (3), we first check that  $\mathcal{L}(\mathbb{R})$  is closed under finite unions. If  $A, B \in \mathcal{L}(\mathbb{R})$  and  $E \subseteq \mathbb{R}$ , then

$$\begin{aligned} \lambda^*(E) &= \lambda^*(E \cap A) + \lambda^*(E \setminus A) \\ &= \lambda^*(E \cap A) + \lambda^*((E \setminus A) \cap B) + \lambda^*(E \setminus (A \cup B)) \\ &\geq \lambda^*(E \cap (A \cup B)) + \lambda^*(E \setminus (A \cup B)). \end{aligned}$$

The reverse inequality follows from subadditivity, so  $A \cup B$  is measurable. Induction now gives closure under finite unions.

Let  $A = \bigcup_{n=1}^{\infty} A_n$  and disjointify the sequence by setting

$$B_1 = A_1 \quad \text{and} \quad B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k, \quad n \geq 2.$$

Each  $B_n$  is measurable, the sets  $B_n$  are pairwise disjoint, and  $\bigcup_{n=1}^{\infty} B_n = A$ . Repeatedly applying measurability gives, for every  $N$  and every  $E \subseteq \mathbb{R}$ ,

$$\lambda^*(E) = \sum_{n=1}^N \lambda^*(E \cap B_n) + \lambda^*\left(E \setminus \bigcup_{n=1}^N B_n\right).$$

Since  $E \setminus A \subseteq E \setminus \bigcup_{n=1}^N B_n$ , we may let  $N \rightarrow \infty$  to obtain

$$\lambda^*(E) \geq \sum_{n=1}^{\infty} \lambda^*(E \cap B_n) + \lambda^*(E \setminus A) \geq \lambda^*(E \cap A) + \lambda^*(E \setminus A).$$

Subadditivity gives the reverse inequality. Hence  $A$  is measurable.  $\square$

**Definition 3.2** Let  $X$  be any set. Let  $\mu$  be a collection of some subsets of  $X$ . We say  $\mu$  is an **algebra** on  $X$  if the following hold:

1.  $\emptyset, X \in \mu$ .
2. If  $A \in \mu$ , then  $A^c \in \mu$ .
3. If  $A_1, \dots, A_n \in \mu$ , then  $\bigcup_{i=1}^n A_i \in \mu$ .

The collection  $\mu$  is called a  **$\sigma$ -algebra on  $X$**  if it is an algebra closed under countable unions; that is, if  $A_1, A_2, \dots \in \mu$ , then  $\bigcup_{n=1}^{\infty} A_n \in \mu$ .

**Example 3.3** Examples of  $\sigma$ -algebras on  $\mathbb{R}$  include the following:

1.  $\{\emptyset, \mathbb{R}\}$ .
2.  $\mathcal{L}(\mathbb{R})$ .
3. The **Borel  $\sigma$ -algebra**, which is the smallest  $\sigma$ -algebra on  $\mathbb{R}$  containing all open sets, and denoted by

$$\mathcal{B}(\mathbb{R}) := \bigcap \{ \mathcal{M} \mid \mathcal{M} \text{ is a } \sigma\text{-algebra and } \mathcal{M} \text{ contains all open sets} \}.$$

4.  $\bigcap_{\alpha \in A} \mathcal{M}_{\alpha}$ , where  $\{\mathcal{M}_{\alpha}\}_{\alpha \in A}$  is a family of  $\sigma$ -algebras on  $\mathbb{R}$ .

**Definition 3.4** Every subset of  $\mathbb{R}$  belonging to  $\mathcal{B}(\mathbb{R})$  is called a **Borel set**.

**Example 3.5** Examples of Borel sets include the following:

1. Open sets, closed sets (hence compact sets), countable sets, and  $(a, b] = (a, b) \cup \{b\}$  (hence all intervals).
2. Let  $\mathcal{G}$  denote the family of open sets, and let  $\mathcal{F}$  denote the family of closed sets. Let

$$\mathcal{G}_\delta := \{\text{all countable intersections of open sets}\}$$

and

$$\mathcal{F}_\sigma := \{\text{all countable unions of closed sets}\}.$$

Then  $\mathcal{G}_\delta, \mathcal{F}_\sigma \subseteq \mathcal{B}(\mathbb{R})$ .

We now look into the relationship between  $\mathcal{L}(\mathbb{R})$  and  $\mathcal{B}(\mathbb{R})$ .

**Proposition 3.6** The following hold:

1. Every open interval  $(a, b)$  with  $a < b$  is Lebesgue measurable.
2. Every interval  $(a, +\infty)$  or  $(-\infty, b)$  is Lebesgue measurable.
3. Every open set is Lebesgue measurable.

**Proof.** For (1), let  $E \subseteq \mathbb{R}$  and  $\varepsilon > 0$ . By definition of outer measure, there exist open intervals  $\{I_n\}_{n=1}^\infty$  such that

$$E \subseteq \bigcup_{n=1}^\infty I_n \quad \text{and} \quad \sum_{n=1}^\infty \ell(I_n) < \lambda^*(E) + \varepsilon.$$

Fix  $n$ , and define

$$I_n^- := I_n \cap (-\infty, a], \quad I_n^0 := I_n \cap (a, b), \quad \text{and} \quad I_n^+ := I_n \cap [b, \infty).$$

Each of these sets is an interval (possibly empty), and

$$\ell(I_n^-) + \ell(I_n^0) + \ell(I_n^+) = \ell(I_n).$$

The family  $\{I_n^0\}_{n=1}^\infty$  covers  $E \cap (a, b)$ , and the family  $\{I_n^-, I_n^+\}_{n=1}^\infty$  covers  $E \setminus (a, b)$ . Therefore,

$$\lambda^*(E \cap (a, b)) \leq \sum_{n=1}^\infty \ell(I_n^0)$$

and

$$\lambda^*(E \setminus (a, b)) \leq \sum_{n=1}^{\infty} (\ell(I_n^-) + \ell(I_n^+)).$$

Adding these inequalities gives

$$\lambda^*(E \cap (a, b)) + \lambda^*(E \setminus (a, b)) \leq \sum_{n=1}^{\infty} \ell(I_n) < \lambda^*(E) + \varepsilon.$$

Since  $\varepsilon > 0$  is arbitrary,

$$\lambda^*(E \cap (a, b)) + \lambda^*(E \setminus (a, b)) \leq \lambda^*(E).$$

The reverse inequality always holds by subadditivity because  $E = (E \cap (a, b)) \cup (E \setminus (a, b))$ . Hence

$$\lambda^*(E) = \lambda^*(E \cap (a, b)) + \lambda^*(E \setminus (a, b)),$$

so  $(a, b)$  is Lebesgue measurable.

For (2),

$$(a, +\infty) = \bigcup_{n=1}^{\infty} (a, a + n) \in \mathcal{L}(\mathbb{R}).$$

For (3), let  $O$  be an open set in  $\mathbb{R}$ . Then  $O = \bigcup_{n=1}^{\infty} (a_n, b_n)$ , where  $a_n, b_n \in \mathbb{R} \cup \{-\infty, \infty\}$ . Every  $(a_n, b_n)$  is Lebesgue measurable by (1) and (2), and hence their countable union is as well.  $\square$

By definition of Borel  $\sigma$ -algebra,  $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{L}(\mathbb{R})$ . We can summarize everything so far:

**Theorem 3.7** The set of Lebesgue measurable sets  $\mathcal{L}(\mathbb{R})$  is a  $\sigma$ -algebra which contains the Borel sets  $\mathcal{B}(\mathbb{R})$  and all  $E \subseteq \mathbb{R}$  such that  $\lambda^*(E) = 0$ . The restriction  $\lambda^*|_{\mathcal{L}(\mathbb{R})} =: \lambda$  is called the **Lebesgue measure** and satisfies the following properties:

1.  $\lambda(E) = \lambda(E + x)$  for all  $E \in \mathcal{L}(\mathbb{R})$  and  $x \in \mathbb{R}$  (**translation invariance**).
2. If  $A, B \in \mathcal{L}(\mathbb{R})$  and  $A \subseteq B$ , then  $\lambda(A) \leq \lambda(B)$  (**monotonicity**).
3.  $\lambda(I) = \ell(I)$  for any interval  $I$
4. If  $A_1, A_2, \dots \in \mathcal{L}(\mathbb{R})$  and  $A_i \cap A_j = \emptyset$  for  $i \neq j$ , then

$$\lambda\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \lambda(A_n)$$

(countable additivity).

**Proof.** All statements except (4) have already been established:  $\mathcal{L}(\mathbb{R})$  is a  $\sigma$ -algebra,  $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{L}(\mathbb{R})$ , null sets are measurable, and  $\lambda = \lambda^*|_{\mathcal{L}(\mathbb{R})}$  inherits (1), (2), and (3) from  $\lambda^*$ . So it remains to prove (4). Let

$$A = \bigcup_{n=1}^{\infty} A_n.$$

By countable subadditivity of outer measure,

$$\lambda(A) = \lambda^*(A) \leq \sum_{n=1}^{\infty} \lambda^*(A_n) = \sum_{n=1}^{\infty} \lambda(A_n).$$

For the reverse inequality, first note that if  $B, C \in \mathcal{L}(\mathbb{R})$  are disjoint, then measurability of  $B$  gives

$$\lambda(B \cup C) = \lambda^*((B \cup C) \cap B) + \lambda^*((B \cup C) \setminus B) = \lambda(B) + \lambda(C).$$

By induction, finite additivity holds for every finite family of pairwise disjoint Lebesgue measurable sets. Therefore, for every  $N \geq 1$ ,

$$\lambda(A) \geq \lambda\left(\bigcup_{n=1}^N A_n\right) = \sum_{n=1}^N \lambda(A_n).$$

Letting  $N \rightarrow \infty$  gives

$$\lambda(A) \geq \sum_{n=1}^{\infty} \lambda(A_n)$$

, and hence equality holds. □

This means that every Lebesgue measurable set is “almost” Borel, but  $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{L}(\mathbb{R})$ .

## Properties of Lebesgue Measure

We now collect several useful properties of the Lebesgue measure  $\lambda$ .

**Proposition 3.8** The following hold:

1. If  $A \subseteq B$ ,  $A, B \in \mathcal{L}(\mathbb{R})$ , and  $\lambda(A) < \infty$ , then  $\lambda(B \setminus A) = \lambda(B) - \lambda(A)$ .
2. If  $A \subseteq \mathbb{R}$  is compact, then  $\lambda(A) < \infty$ .
3.  $\lambda$  is  **$\sigma$ -subadditive**. That is, if  $A_1, A_2, \dots \in \mathcal{L}(\mathbb{R})$ , then

$$\lambda\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \lambda(A_i).$$

Note that  $A_1, A_2, \dots$  do not need to be pairwise disjoint!

4. **Continuity of measure**: if  $A_1, A_2, \dots \in \mathcal{L}(\mathbb{R})$  and  $A_1 \subseteq A_2 \subseteq \dots$ , then

$$\lambda\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \lambda(A_n).$$

5. **Downwards continuity**: let  $A_1, A_2, \dots \in \mathcal{L}(\mathbb{R})$  and  $A_1 \supseteq A_2 \supseteq \dots$ , then

$$\lambda\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \lambda(A_n)$$

if  $\lambda(A_1) < \infty$ .

**Proof.** For (1), since  $A \subseteq B$ , we have  $B = A \cup (B \setminus A)$ , where  $A$  and  $B \setminus A$  are disjoint. By countable additivity,

$$\lambda(B) = \lambda(A) + \lambda(B \setminus A).$$

Rearranging gives  $\lambda(B \setminus A) = \lambda(B) - \lambda(A)$ .

For (2), if  $A$  is compact, then  $A$  is bounded by the Heine–Borel theorem, so there exist  $m < M$  with  $A \subseteq [m, M]$ . By monotonicity and  $\lambda([m, M]) = M - m$ ,

$$\lambda(A) \leq \lambda([m, M]) = M - m < \infty.$$

For (3),

$$\lambda\left(\bigcup_{i=1}^{\infty} A_i\right) = \lambda^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \lambda^*(A_i) = \sum_{i=1}^{\infty} \lambda(A_i),$$

where we used countable subadditivity of  $\lambda^*$ .

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**Proof continued.** For (4), let  $A = \bigcup_{n=1}^{\infty} A_n$  and define  $B_1 = A_1$ ,  $B_2 = A_2 \setminus A_1, \dots, B_n = A_n \setminus A_{n-1}$ . Then  $\{B_i\}_{i=1}^{\infty}$  is a collection of pairwise disjoint sets in  $\mathcal{L}(\mathbb{R})$ , so that

$$\lambda\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} \lambda(B_i) = \lambda(A).$$

Thus

$$\lim_{n \rightarrow \infty} \lambda(A_n) = \sum_{i=1}^{\infty} \lambda(B_i) = \lambda\left(\bigcup_{i=1}^{\infty} B_i\right) = \lambda(A).$$

For (5), let  $B_n := A_1 \setminus A_n$ . Then  $B_1 \subseteq B_2 \subseteq \dots$ , and

$$\bigcup_{n=1}^{\infty} B_n = A_1 \setminus \bigcap_{n=1}^{\infty} A_n.$$

By (4),

$$\lambda\left(A_1 \setminus \bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \lambda(B_n).$$

Also, since  $A_n \subseteq A_1$  and  $\lambda(A_1) < \infty$ ,

$$\lambda(B_n) = \lambda(A_1) - \lambda(A_n).$$

Therefore

$$\lambda(A_1) - \lambda\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} (\lambda(A_1) - \lambda(A_n)),$$

so  $\lambda(\bigcap_{n=1}^{\infty} A_n) = \lim_{n \rightarrow \infty} \lambda(A_n)$ . □

**Remark 4.1** The condition  $\lambda(A_1) < \infty$  in (5) is necessary. For example, if  $A_n = [n, \infty)$ , then  $\{A_n\}_{n=1}^{\infty}$  forms a decreasing chain, but  $\lambda(A_n) = \infty$ . Then  $\bigcap_{n=1}^{\infty} A_n = \emptyset$ , and therefore

$$\lambda\left(\bigcap_{n=1}^{\infty} A_n\right) = 0,$$

so

$$\lim_{n \rightarrow \infty} \lambda(A_n) \neq \lambda\left(\bigcap_{n=1}^{\infty} A_n\right).$$

**Proposition 4.2 (Outer regularity)** If  $E \in \mathcal{L}(\mathbb{R})$ , then

$$\lambda(E) = \inf\{\lambda(U) : E \subseteq U, U \text{ open}\}.$$

**Proof.** For every open  $U \supseteq E$ , monotonicity gives  $\lambda(E) \leq \lambda(U)$ , so

$$\lambda(E) \leq \inf\{\lambda(U) : E \subseteq U, U \text{ open}\}.$$

Conversely, let  $\varepsilon > 0$ . By the definition of outer measure, there exist open intervals  $\{I_n\}_{n=1}^{\infty}$  such that

$$E \subseteq \bigcup_{n=1}^{\infty} I_n \quad \text{and} \quad \sum_{n=1}^{\infty} \ell(I_n) < \lambda^*(E) + \varepsilon = \lambda(E) + \varepsilon.$$

Set  $U := \bigcup_{n=1}^{\infty} I_n$ . Then  $U$  is open,  $E \subseteq U$ , and

$$\lambda(U) = \lambda^*(U) \leq \sum_{n=1}^{\infty} \ell(I_n) < \lambda(E) + \varepsilon.$$

Hence

$$\inf\{\lambda(U) : E \subseteq U, U \text{ open}\} \leq \lambda(E) + \varepsilon.$$

Letting  $\varepsilon \downarrow 0$  yields the reverse inequality. □

**Proposition 4.3** If  $E \in \mathcal{L}(\mathbb{R})$ , then

$$\lambda(E) = \sup\{\lambda(K) \mid K \subseteq E, K \text{ compact}\}.$$

**Proof.** Let  $E_n := E \cap [-n, n]$ . Then  $E_n \nearrow E$ , so by continuity from below,

$$\lambda(E) = \lim_{n \rightarrow \infty} \lambda(E_n) = \sup_n \lambda(E_n).$$

Thus it suffices to treat bounded measurable sets. Fix  $n$  and  $\varepsilon > 0$ , and set

$$F_n := [-n, n] \setminus E_n.$$

By [Proposition 4.2](#), choose an open set  $O \supseteq F_n$  such that

$$\lambda(O) < \lambda(F_n) + \varepsilon.$$

Define

$$K_n := [-n, n] \setminus O.$$

Then  $K_n$  is compact and  $K_n \subseteq E_n$ . Also,

$$E_n \setminus K_n = E_n \cap O \subseteq O \setminus F_n,$$

so

$$\lambda(E_n \setminus K_n) \leq \lambda(O \setminus F_n) = \lambda(O) - \lambda(F_n) < \varepsilon.$$

Hence

$$\lambda(K_n) = \lambda(E_n) - \lambda(E_n \setminus K_n) > \lambda(E_n) - \varepsilon.$$

Since  $\varepsilon$  is arbitrary,

$$\lambda(E_n) = \sup\{\lambda(K) : K \subseteq E_n, K \text{ compact}\}.$$

Taking the supremum over  $n$  gives the claim for  $E$ . □

**Corollary 4.4** If  $E \in \mathcal{L}(\mathbb{R})$  with  $\lambda(E) < \infty$  and  $\varepsilon > 0$ , then there exist an open set  $U$  and a compact set  $K$  such that

$$K \subseteq E \subseteq U \quad \text{and} \quad \lambda(U \setminus K) < \varepsilon.$$

**Proof.** By [Proposition 4.2](#), choose an open  $U \supseteq E$  with

$$\lambda(U) < \lambda(E) + \frac{\varepsilon}{2}.$$

Then

$$\lambda(U \setminus E) = \lambda(U) - \lambda(E) < \frac{\varepsilon}{2}.$$

By [Proposition 4.3](#), choose a compact  $K \subseteq E$  with

$$\lambda(E \setminus K) < \frac{\varepsilon}{2}.$$

Therefore,

$$\lambda(U \setminus K) \leq \lambda(U \setminus E) + \lambda(E \setminus K) < \varepsilon.$$

□

## Riemann Integrability and Discontinuities

**Theorem 4.5 (Characterization of Riemann integrability)** A bounded function  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable if and only if

$$\lambda(\{x \in [a, b] \mid f \text{ is not continuous at } x\}) = 0.$$

**Proof.** Let  $E = \{x \in [a, b] \mid f \text{ is not continuous at } x\}$ .

To begin with, we need to show that  $E$  is Lebesgue measurable in the first place. We do this by verifying several claims. For each  $k \in \mathbb{N}$ , let

$$E_k = \left\{ x \in [a, b] \mid \begin{array}{l} \text{for every } \delta > 0, \text{ there exist } y, z \in [a, b] \cap [x - \delta, x + \delta] \\ \text{such that } |f(y) - f(z)| \geq 1/k \end{array} \right\}.$$

1. *Claim 1:*  $\bigcup_{k=1}^{\infty} E_k \subseteq E$ . *Verification.* If  $z \in E_k$ , then  $f$  is not continuous at  $z$ . Indeed, if  $f$  were continuous at  $z$ , then for  $\varepsilon = 1/(2k)$  there would be a  $\delta > 0$  such that  $|z - x| < \delta$  implies  $|f(z) - f(x)| < 1/(2k)$ . Hence  $|f(x) - f(y)| < 1/k$  for all  $x, y \in [z - \delta, z + \delta]$ , contradicting  $z \in E_k$ . Therefore  $\bigcup_{k=1}^{\infty} E_k \subseteq E$ .
2. *Claim 2:*  $E = \bigcup_{k=1}^{\infty} E_k$ . *Verification.* Conversely, if  $z \in E$ , then there exists  $\varepsilon_0 > 0$  such that for every  $\delta > 0$  we can find  $y$  with  $|y - z| < \delta$  and  $|f(y) - f(z)| \geq \varepsilon_0$ . Choose  $k$  with  $1/k \leq \varepsilon_0$ . Then  $z \in E_k$ . Hence  $E \subseteq \bigcup_{k=1}^{\infty} E_k$ , and together with Claim (1) we get

$$E = \bigcup_{k=1}^{\infty} E_k.$$

3. *Claim 3:*  $E \in \mathcal{L}(\mathbb{R})$ . *Verification.* We first show that each  $E_k$  is closed in  $[a, b]$ . Let  $x \in [a, b] \setminus E_k$ . Then there exists  $\delta_0 > 0$  such that

$$|f(y) - f(z)| < 1/k \quad \text{for all } y, z \in [a, b] \cap [x - \delta_0, x + \delta_0].$$

Now let  $u \in [a, b]$  satisfy  $|u - x| < \delta_0/2$ . If  $y, z \in [a, b] \cap [u - \delta_0/2, u + \delta_0/2]$ , then  $y, z \in [a, b] \cap [x - \delta_0, x + \delta_0]$ , so  $|f(y) - f(z)| < 1/k$ . Hence  $u \notin E_k$ . Therefore  $[a, b] \setminus E_k$  is open in  $[a, b]$ , and  $E_k$  is closed in  $[a, b]$ .

Since  $[a, b]$  is compact, each closed subset  $E_k$  is compact, hence Borel, and therefore Lebesgue

measurable. By Claim (2),

$$E = \bigcup_{k=1}^{\infty} E_k,$$

so  $E$  is a countable union of measurable sets and thus  $E \in \mathcal{L}(\mathbb{R})$ .

We can now prove the main statement.

*Reverse implication.* Assume first that  $\lambda(E) = 0$ . We prove that  $f$  is Riemann integrable by making the difference between its upper and lower Riemann sums arbitrarily small. Let  $\varepsilon > 0$ , and choose  $C \geq 0$  such that  $|f| \leq C$ . Since  $\lambda^*(E) = 0$ , there is a countable open cover of  $E$  by intervals whose total length is less than  $\varepsilon/(4C + 1)$ . Choose  $k \in \mathbb{N}$  so large that  $(b - a)/k < \varepsilon/2$ . Because  $E_k$  is compact, finitely many intervals  $I_1, \dots, I_N$  from this cover suffice to cover  $E_k$ . Let

$$X = [a, b] \setminus \bigcup_{n=1}^N I_n.$$

If  $z \in X$ , then  $z \notin E_k$ , so we can choose a relative open interval  $V_z$  containing  $z$  such that the oscillation of  $f$  on  $\overline{V_z}$  is less than  $1/k$ . Compactness gives a finite subcover  $V_1, \dots, V_m$  of  $X$ . The intervals

$$I_1, \dots, I_N, V_1, \dots, V_m$$

form an open cover of  $[a, b]$ , so this cover has a Lebesgue number. Choose a partition  $P = \{a = x_0 < \dots < x_n = b\}$  whose mesh is smaller than that number and which includes every endpoint of the  $I_j$  lying in  $[a, b]$ . Each partition interval is contained in one of the displayed intervals. If it is contained in some  $V_j$ , its oscillation is less than  $1/k$ ; the remaining partition intervals lie in the union of the  $I_j$  and have oscillation at most  $2C$ . These are the only possible cases (exercise!). Therefore

$$U(f, P) - L(f, P) \leq 2C \sum_{j=1}^N \ell(I_j) + \frac{b - a}{k} < \frac{2C\varepsilon}{4C + 1} + \frac{\varepsilon}{2} < \varepsilon.$$

Since  $\varepsilon$  was arbitrary,  $f$  is Riemann integrable.

*Forward implication.* Assume that  $f$  is Riemann integrable. It is enough to show that  $\lambda(E_k) = 0$  for every  $k \in \mathbb{N}$ , because

$$\lambda(E) \leq \sum_{k=1}^{\infty} \lambda(E_k) = 0$$

once this is known. Fix  $k \in \mathbb{N}$  and  $\varepsilon > 0$ . Choose a partition  $P$  of  $[a, b]$  such that  $U(f, P) - L(f, P) < \varepsilon/k$ . Write

$$P = \{x_0 = a < x_1 < \dots < x_n = b\}.$$

Then

$$\frac{\varepsilon}{k} > U(f, P) - L(f, P) = \sum_{i=0}^{n-1} (M_i - m_i)(x_{i+1} - x_i).$$

Let  $J = \{i \mid M_i - m_i \geq 1/k\}$ . Every point of  $E_k$  that is not a partition point lies in the interior of a unique partition interval, and that interval must have oscillation at least  $1/k$ . Thus

$$E_k \subseteq \bigcup_{i \in J} (x_i, x_{i+1}) \cup \{x_0, \dots, x_n\}.$$

Moreover,

$$\frac{\varepsilon}{k} \geq \sum_{i \in J} (M_i - m_i)(x_{i+1} - x_i) \geq \frac{1}{k} \sum_{i \in J} (x_{i+1} - x_i).$$

Thus

$$\sum_{i \in J} \ell([x_i, x_{i+1}]) < \varepsilon,$$

By monotonicity,

$$\lambda(E_k) \leq \sum_{i \in J} \lambda((x_i, x_{i+1})) + \lambda(\{x_0, \dots, x_n\}) < \varepsilon.$$

Since  $\varepsilon > 0$  was arbitrary,  $\lambda(E_k) = 0$ . □

Many functions are not Riemann integrable. For example, the characteristic function  $\mathbb{1}_{\mathbb{Q} \cap [0,1]}$  is discontinuous at every point of  $[0, 1]$ , and hence is not Riemann integrable.

**Example 4.6** Every bounded function on  $\mathbb{R}$  with finitely many discontinuities is Riemann integrable on every compact interval.

**Proof.** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be bounded and suppose that  $f$  has only finitely many discontinuities, say at points  $x_1, \dots, x_m \in \mathbb{R}$ . To prove Riemann integrability, fix any compact interval  $[a, b]$  and consider the restriction  $f|_{[a,b]}$ .

Its set of discontinuities in  $[a, b]$  is contained in

$$\{x_1, \dots, x_m\} \cap [a, b],$$

which is a finite set. Every finite set has Lebesgue measure 0, so

$$\lambda(\{x \in [a, b] \mid f|_{[a,b]} \text{ is not continuous at } x\}) = 0.$$

By Theorem 4.5,  $f|_{[a,b]}$  is Riemann integrable on  $[a, b]$ . Since  $[a, b]$  was arbitrary,  $f$  is Riemann integrable on every compact interval.

□

## 2. Measurable Functions and Integration

**Definition 4.7** A real-valued function  $f$  is said to be **Lebesgue measurable** (or simply **measurable**) if

$$f^{-1}((\alpha, \infty)) = \{x \mid f(x) > \alpha\}$$

is measurable for every  $\alpha \in \mathbb{R}$ . A complex-valued function  $f$  is called measurable if both its real and imaginary parts are measurable.

**Example 4.8** Let  $E$  be a measurable set in  $\mathbb{R}$ , and let  $\mathbf{1}_E$  be its characteristic function. Then  $\mathbf{1}_E$  is measurable. Indeed, if  $\alpha \geq 1$ , then  $\mathbf{1}_E^{-1}((\alpha, \infty)) = \emptyset$ . If  $0 < \alpha \leq 1$ , then  $\mathbf{1}_E^{-1}((\alpha, \infty)) = E$ . If  $\alpha < 0$ , then  $\mathbf{1}_E^{-1}((\alpha, \infty)) = \mathbb{R}$ . All three sets are measurable.

**Example 4.9** Let  $E_1, \dots, E_n \in \mathcal{L}(\mathbb{R})$  be pairwise disjoint, and let  $a_1, \dots, a_n \in \mathbb{R}$ . The function

$$\sum_{i=1}^n a_i \mathbf{1}_{E_i}$$

is called a **simple function**.

**Proposition 4.10** Simple functions are measurable.

**Proof.** Let  $f = \sum_{i=1}^n a_i \mathbf{1}_{E_i}$  be a simple function. For every  $\alpha \in \mathbb{R}$ , we have

$$f^{-1}((\alpha, \infty)) = \bigcup_{i: a_i > \alpha} E_i,$$

which is a finite union of measurable sets, and hence measurable.

□

## Lecture 5 — Tuesday, January 20, 2015

## Measurable Functions and Closure Properties

Recall that a function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is Lebesgue measurable if  $f^{-1}((\alpha, \infty)) \in \mathcal{L}(\mathbb{R})$  for every  $\alpha \in \mathbb{R}$ .

**Proposition 5.1** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$ . The following are equivalent:

1.  $f$  is Lebesgue measurable.
2.  $f^{-1}((\alpha, \infty)) \in \mathcal{L}(\mathbb{R})$  for all  $\alpha \in \mathbb{R}$ .
3.  $f^{-1}((-\infty, \alpha)) \in \mathcal{L}(\mathbb{R})$  for all  $\alpha \in \mathbb{R}$ .
4.  $f^{-1}((-\infty, \alpha]) \in \mathcal{L}(\mathbb{R})$  for all  $\alpha \in \mathbb{R}$ .

**Proof.** (1) *implies* (2). This is the definition.

(2) *implies* (3). Let  $\alpha \in \mathbb{R}$ . Then

$$f^{-1}((-\infty, \alpha)) = \bigcup_{n=1}^{\infty} f^{-1}((-\infty, \alpha - 1/n]) = \bigcup_{n=1}^{\infty} f^{-1}((\alpha - 1/n, \infty))^c,$$

which is measurable.

(3) *implies* (4). Let  $\alpha \in \mathbb{R}$ . Then

$$f^{-1}((-\infty, \alpha]) = \{x \mid f(x) \leq \alpha\} = \bigcap_{n=1}^{\infty} \{x \mid f(x) < \alpha + 1/n\} = \bigcap_{n=1}^{\infty} f^{-1}((-\infty, \alpha + 1/n)),$$

which is a countable intersection of measurable sets, and hence measurable.

(4) *implies* (1). Let  $\alpha \in \mathbb{R}$ . Then

$$f^{-1}((\alpha, \infty)) = \{x \mid f(x) > \alpha\} = \{x \mid f(x) \leq \alpha\}^c = f^{-1}((-\infty, \alpha])^c,$$

which is measurable. □

**Corollary 5.2** A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is measurable if and only if  $f^{-1}(B) \in \mathcal{L}(\mathbb{R})$  for every  $B \in \mathcal{B}(\mathbb{R})$ .

**Proof.** Suppose first that  $f$  is measurable, and let

$$\mathcal{M} = \{B \subseteq \mathbb{R} \mid f^{-1}(B) \in \mathcal{L}(\mathbb{R})\}.$$

Preimages preserve complements and countable unions, so  $\mathcal{M}$  is a  $\sigma$ -algebra. It contains every interval  $(\alpha, \infty)$  by measurability, and these intervals generate  $\mathcal{B}(\mathbb{R})$ . Hence  $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{M}$ .

Conversely, if the inverse image of every Borel set is measurable, then this holds in particular for every open interval  $(\alpha, \infty)$ . Thus  $f$  is measurable by definition.  $\square$

**Proposition 5.3** Let  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  be measurable, let  $\phi : \mathbb{R} \rightarrow \mathbb{R}$  continuous, and let  $c \in \mathbb{R}$ . Then the following functions are measurable:

1.  $cf$ .
2.  $f + g$ .
3.  $\phi \circ f$ .
4.  $|f|$ ,  $f^+$ , and  $f^-$ , where  $f^+(x) = \max\{f(x), 0\}$  and  $f^-(x) = \max\{-f(x), 0\}$ .

**Proof.** For (1), let  $\alpha \in \mathbb{R}$ .

We have  $(cf)^{-1}((\alpha, \infty)) = \{x \mid cf(x) > \alpha\}$ . If  $c > 0$ , this is  $\{x \mid f(x) > \alpha/c\} = f^{-1}((\alpha/c, \infty)) \in \mathcal{L}(\mathbb{R})$ , since  $f$  is measurable. If  $c < 0$ , this is

$$\{x \mid f(x) < \alpha/c\} = f^{-1}((-\infty, \alpha/c)) \in \mathcal{L}(\mathbb{R}),$$

since  $f$  is measurable. If  $c = 0$ , this is  $\{x \mid 0 > \alpha\}$ , which is either  $\emptyset$  or  $\mathbb{R}$ , both of which are measurable.

For (2), let  $\alpha \in \mathbb{R}$ . Then

$$\begin{aligned} (f + g)^{-1}((\alpha, \infty)) &= \{x \mid f(x) + g(x) > \alpha\} \\ &= \{x \mid f(x) > \alpha - g(x)\} \end{aligned}$$

$$\begin{aligned}
&= \{x \mid \text{there exists } r \in \mathbb{Q} \text{ such that } f(x) > r \text{ and } r > \alpha - g(x)\} \\
&= \bigcup_{r \in \mathbb{Q}} \{x \mid f(x) > r\} \cap \{x \mid g(x) > \alpha - r\} \\
&= \bigcup_{r \in \mathbb{Q}} (f^{-1}((r, \infty)) \cap g^{-1}((\alpha - r, \infty))),
\end{aligned}$$

which is a countable union of measurable sets, and hence measurable.

For (3), let  $\alpha \in \mathbb{R}$ . Then  $(\phi \circ f)^{-1}((\alpha, \infty)) = f^{-1}(\phi^{-1}((\alpha, \infty)))$ . Since  $\phi$  is continuous,  $\phi^{-1}((\alpha, \infty)) \in \mathcal{B}(\mathbb{R})$  (exercise!), so  $f^{-1}(\phi^{-1}((\alpha, \infty))) \in \mathcal{L}(\mathbb{R})$  by [Corollary 5.2](#).

For (4), let  $\phi(x) = |x|$ , which is continuous. Then by (3),  $|f|$  is measurable. Thus  $f^+ = (|f| + f)/2$  and  $f^- = (|f| - f)/2$  are both measurable.  $\square$

**Definition 5.4** A function  $f : \mathbb{R} \rightarrow [-\infty, \infty]$  is called an **extended real function**. We say that  $f : \mathbb{R} \rightarrow [-\infty, \infty]$  is **Lebesgue measurable** if  $f^{-1}((\alpha, \infty)) \in \mathcal{L}(\mathbb{R})$  for all  $\alpha \in \mathbb{R}$  and  $f^{-1}(\{-\infty\}), f^{-1}(\{\infty\}) \in \mathcal{L}(\mathbb{R})$ .

**Proposition 5.5** Let  $f_1, f_2, \dots$  be a sequence of extended real-valued measurable functions. Then the following are also measurable:

1.  $\sup_{n \in \mathbb{N}} f_n$ .
2.  $\inf_{n \in \mathbb{N}} f_n$ .
3.  $\limsup_{n \rightarrow \infty} f_n$ .
4.  $\liminf_{n \rightarrow \infty} f_n$ .
5.  $f$ , where  $f_n \rightarrow f$  pointwise.

**Proof.** For (1), let  $g = \sup_{n \in \mathbb{N}} f_n$ . Let  $\alpha \in \mathbb{R}$ . Then

$$\begin{aligned}
g^{-1}((\alpha, \infty)) &= \{x \mid g(x) > \alpha\} \\
&= \{x \mid \sup_{n \in \mathbb{N}} f_n(x) > \alpha\} \\
&= \{x \mid \text{there exists } n \in \mathbb{N} \text{ such that } f_n(x) > \alpha\} \\
&= \bigcup_{n \in \mathbb{N}} \{x \mid f_n(x) > \alpha\}
\end{aligned}$$

$$= \bigcup_{n \in \mathbb{N}} f_n^{-1}((\alpha, \infty)) \in \mathcal{L}(\mathbb{R})$$

since  $f_n$  is measurable for all  $n$ .

For (2), let  $h = \inf_{n \in \mathbb{N}} f_n$ . Then

$$h^{-1}((\alpha, \infty)) = \bigcup_{m=1}^{\infty} \bigcap_{n=1}^{\infty} f_n^{-1}((\alpha + 1/m, \infty)),$$

which is measurable. In both the supremum and infimum cases, the fibres over  $\infty$  and  $-\infty$  are obtained from the sets above by countable unions, intersections, and complements. Thus both extended real functions are measurable.

For (3), define  $g_n = \sup_{k \geq n} f_k$  for every  $n \in \mathbb{N}$ . Each  $g_n$  is measurable by (1), and  $\limsup_{n \rightarrow \infty} f_n = \lim_{n \rightarrow \infty} g_n = \inf_{n \in \mathbb{N}} g_n$  is measurable by (2).

For (4), use the same argument with infima and suprema interchanged.

For (5), let  $f = \lim_{n \rightarrow \infty} f_n$ . Then  $f = \liminf_{n \rightarrow \infty} f_n$  is measurable by (4). □

## Simple Functions and Approximation

**Theorem 5.6** Let  $f : \mathbb{R} \rightarrow \mathbb{R}^{\geq 0}$  be a function. Then  $f$  is measurable if and only if there exists a sequence of simple functions  $\{\phi_n\}_{n \in \mathbb{N}}$  such that  $\phi_n \leq \phi_{n+1}$  and  $\phi_n \rightarrow f$  pointwise.

**Proof.** *Reverse implication.* The pointwise limit of measurable functions is measurable by [Proposition 5.5](#), and simple functions are measurable by [Proposition 4.10](#).

*Forward implication.* Fix  $n \in \mathbb{N}$ . Define  $E_{k,n} = f^{-1}([k/2^n, (k+1)/2^n))$  for  $k = 0, 1, \dots, n2^n - 1$ , which is measurable because  $f$  is measurable. Let  $E_n = \bigcup_{k=0}^{n2^n-1} E_{k,n}$  and define

$$\phi_n(x) = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mathbf{1}_{E_{k,n}} + n \mathbf{1}_{E_n^c},$$

which is a simple function. Then  $\phi_n \leq \phi_{n+1} \leq f$ , and  $\lim_{n \rightarrow \infty} \phi_n(x) = f(x)$  for every  $x \in \mathbb{R}$ . Indeed, fix  $x \in \mathbb{R}$ . There exists  $N \in \mathbb{N}$  such that  $f(x) \leq N$ , so  $x \in E_n$  for all  $n \geq N$  (since

$E_n = \{x \mid f(x) < n\}$ . From the way we defined  $\phi_n$ , we have

$$|\phi_n(x) - f(x)| \leq \frac{1}{2^n} \xrightarrow{n \rightarrow \infty} 0.$$

□

Figure 4 shows the construction at one fixed resolution: the simple function rounds  $f$  downward, with a vertical error no larger than the mesh size.

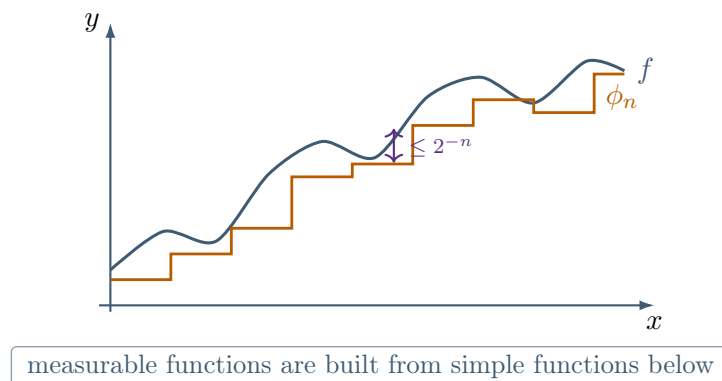


FIGURE 4

## Lebesgue Integrals of Nonnegative Functions

**Definition 5.7** If

$$\phi = \sum_{i=1}^n a_i \mathbf{1}_{E_i},$$

is a simple function and  $A \in \mathcal{L}(\mathbb{R})$ , then the **Lebesgue integral** over  $A$  is

$$\int_A \phi \, d\lambda := \sum_{i=1}^n a_i \lambda(E_i \cap A),$$

with the convention that  $0 \cdot \infty = 0$ .

**Example 5.8**  $\mathbf{1}_{\mathbb{Q} \cap [0,1]}$  is not Riemann integrable, but it is very Lebesgue integrable:

$$\int_{\mathbb{R}} \mathbf{1}_{\mathbb{Q} \cap [0,1]} \, d\lambda = \lambda(\mathbb{Q} \cap [0,1]) = 0.$$

**Definition 5.9** If  $f : \mathbb{R} \rightarrow [0, \infty]$ , then for  $E \in \mathcal{L}(\mathbb{R})$ , the **Lebesgue integral** over  $E$  is

$$\int_E f \, d\lambda = \sup_{\substack{0 \leq \phi \leq f \\ \phi \text{ simple}}} \int_E \phi \, d\lambda.$$

## Lecture 6 — Thursday, January 22, 2015

**Proposition 6.1** Let  $A \in \mathcal{L}(\mathbb{R})$ , and let  $f, g : \mathbb{R} \rightarrow [0, \infty]$  be measurable.

1. If  $f \leq g$  on  $A$ , then

$$\int_A f \, d\lambda \leq \int_A g \, d\lambda.$$

2. If  $\emptyset \subseteq B \subseteq A$  and  $B \in \mathcal{L}(\mathbb{R})$ , then

$$\int_B f \, d\lambda = \int_A f \cdot \mathbf{1}_B \, d\lambda.$$

3. The two integrations defined above coincide for simple functions.

**Proof.** For (1), if  $\phi \leq f$  and  $\phi$  is simple, then  $\phi \leq g$ .

Therefore the admissible class

$$\{\phi : 0 \leq \phi \leq f, \phi \text{ simple}\}$$

is contained in

$$\{\psi : 0 \leq \psi \leq g, \psi \text{ simple}\}.$$

Taking suprema of the corresponding simple integrals gives

$$\int_A f \, d\lambda \leq \int_A g \, d\lambda.$$

For (2), let

$$\mathcal{S}_B := \{\phi : 0 \leq \phi \leq f, \phi \text{ simple on } B\}$$

and

$$\mathcal{S}_A := \{\psi : 0 \leq \psi \leq f\mathbf{1}_B, \psi \text{ simple on } A\}.$$

If  $\phi \in \mathcal{S}_B$ , define  $\tilde{\phi} := \phi\mathbf{1}_B$ . Then  $\tilde{\phi} \in \mathcal{S}_A$  and

$$\int_A \tilde{\phi} \, d\lambda = \int_B \phi \, d\lambda.$$

Conversely, if  $\psi \in \mathcal{S}_A$ , then  $\psi = \psi\mathbf{1}_B$ , and the restriction  $\psi|_B$  belongs to  $\mathcal{S}_B$  with

$$\int_A \psi \, d\lambda = \int_B \psi \, d\lambda.$$

Hence the two sets of simple integral values are identical, so their suprema are equal:

$$\int_B f \, d\lambda = \int_A f\mathbf{1}_B \, d\lambda.$$

For (3), let  $s$  be a nonnegative simple function. In the definition

$$\int_A s \, d\lambda = \sup \left\{ \int_A \phi \, d\lambda : 0 \leq \phi \leq s, \phi \text{ simple} \right\},$$

the admissible family contains  $\phi = s$ , so the supremum is at least the integral of  $s$  previously defined for simple functions. On the other hand, if  $0 \leq \phi \leq s$  are simple, monotonicity of the simple integral gives

$$\int_A \phi \, d\lambda \leq \int_A s \, d\lambda$$

for the old simple function definition. Thus the supremum cannot exceed that value. Therefore both definitions coincide for simple functions.

□

## Lecture 7 — Tuesday, January 27, 2015

**Definition 7.1** Let  $f : \mathbb{R} \rightarrow [-\infty, \infty]$  be a function. We say that  $f$  is **Lebesgue integrable** if the following two conditions hold:

1.  $f$  is measurable; that is,  $f^+$  and  $f^-$  are measurable.

2.

$$\int f^+ d\lambda < \infty \quad \text{and} \quad \int f^- d\lambda < \infty.$$

In that case, we define

$$\int f d\lambda := \int f^+ d\lambda - \int f^- d\lambda.$$

If  $\emptyset \neq A \in \mathcal{L}(\mathbb{R})$ , then  $f$  is integrable over  $A$  if  $f \cdot \mathbb{1}_A$  is integrable.

**Proposition 7.2** If  $B \subseteq A$  are Lebesgue measurable and  $f$  is integrable on  $B$ , then

$$\int_B f d\lambda = \int_A f \cdot \mathbb{1}_B d\lambda.$$

**Proof.** Write  $f = f^+ - f^-$ . Then

$$(f\mathbb{1}_B)^+ = f^+\mathbb{1}_B \quad \text{and} \quad (f\mathbb{1}_B)^- = f^-\mathbb{1}_B.$$

By [Proposition 6.1\(2\)](#) applied to the nonnegative measurable functions  $f^+$  and  $f^-$ ,

$$\int_B f^+ d\lambda = \int_A f^+\mathbb{1}_B d\lambda \quad \text{and} \quad \int_B f^- d\lambda = \int_A f^-\mathbb{1}_B d\lambda.$$

Therefore,

$$\begin{aligned} \int_B f d\lambda &= \int_B f^+ d\lambda - \int_B f^- d\lambda \\ &= \int_A f^+\mathbb{1}_B d\lambda - \int_A f^-\mathbb{1}_B d\lambda \\ &= \int_A (f^+ - f^-)\mathbb{1}_B d\lambda \end{aligned}$$

$$= \int_A f \mathbf{1}_B \, d\lambda.$$

□

## Monotone Convergence Theorem and Fatou's Lemma

**Theorem 7.3 (Monotone convergence theorem)** Let  $f_1, f_2, \dots : \mathbb{R} \rightarrow [0, \infty]$  be measurable functions and  $f_1 \leq f_2 \leq \dots$  pointwise. Then

$$\int_A \lim_{n \rightarrow \infty} f_n \, d\lambda = \lim_{n \rightarrow \infty} \int_A f_n \, d\lambda$$

for  $A \in \mathcal{L}(\mathbb{R})$ . In particular,

$$\text{if } \sup_{n \in \mathbb{N}} \int_A f_n \, d\lambda < \infty, \text{ then } \lim_{n \rightarrow \infty} f_n < \infty \text{ almost everywhere on } A.$$

The picture in Figure 5 captures both monotonicities in the theorem: the graphs rise pointwise, and the shaded areas rise with them.

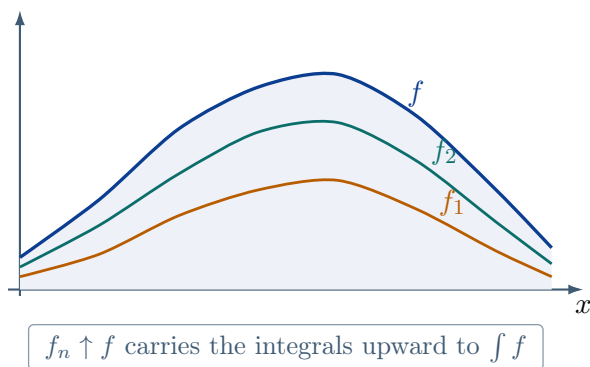


FIGURE 5

**Proof.** Fix  $A \in \mathcal{L}(\mathbb{R})$  and write  $f = \lim_{n \rightarrow \infty} f_n$ . Since

$$0 \leq f_1 \leq f_2 \leq \dots,$$

we have

$$0 \leq \int_A f_1 \, d\lambda \leq \int_A f_2 \, d\lambda \leq \dots$$

by Proposition 6.1(1). Thus

$$\lim_{n \rightarrow \infty} \int_A f_n \, d\lambda = \sup_n \int_A f_n \, d\lambda.$$

The function  $f$  is measurable, and  $f_n \leq f$  for every  $n$ . Monotonicity therefore gives

$$\sup_{n \in \mathbb{N}} \int_A f_n \, d\lambda \leq \int_A f \, d\lambda.$$

It remains to prove the reverse inequality

$$\int_A f \, d\lambda \leq \lim_{n \rightarrow \infty} \int_A f_n \, d\lambda.$$

Let  $0 \leq \phi \leq f$  be simple, and let  $0 < \delta < 1$ . Define

$$A_n = \{x \in A \mid f_n(x) > \delta\phi(x)\}.$$

Then  $A_1 \subseteq A_2 \subseteq \dots$ , and their union contains  $\{x \in A \mid \phi(x) > 0\}$ . Moreover,

$$\int_A f_n \, d\lambda \geq \int_{A_n} f_n \, d\lambda \geq \int_{A_n} \delta\phi \, d\lambda,$$

because  $\delta\phi(x) \leq f_n(x)$  on  $A_n$ . Write

$$\phi = \sum_{i=1}^m a_i \mathbf{1}_{E_i}.$$

Then

$$\int_{A_n} \delta\phi \, d\lambda = \delta \sum_{i=1}^m a_i \lambda(E_i \cap A_n)$$

for every  $n \in \mathbb{N}$ . Now,

$$\lim_{n \rightarrow \infty} \int_A f_n \, d\lambda \geq \delta \sum_{i=1}^m a_i \lambda(E_i \cap A)$$

because the union of the  $A_n$  contains every  $E_i \cap A$  for which  $a_i > 0$ . Hence

$$\lim_{n \rightarrow \infty} \int_A f_n \, d\lambda \geq \delta \int_A \phi \, d\lambda.$$

Letting  $\delta \rightarrow 1$ , we have

$$\lim_{n \rightarrow \infty} \int_A f_n \, d\lambda \geq \int_A \phi \, d\lambda$$

for all simple functions  $\phi$  such that  $0 \leq \phi \leq f$ . Hence

$$\lim_{n \rightarrow \infty} \int_A f_n \, d\lambda \geq \int_A f \, d\lambda.$$

Finally, if the common value of these integrals is finite and

$$N = \{x \in A \mid f(x) = \infty\},$$

then  $m\mathbf{1}_N \leq f$  for every  $m$ . Consequently,

$$m\lambda(N) \leq \int_A f \, d\lambda < \infty$$

for every  $m$ , so  $\lambda(N) = 0$ . □

**Proposition 7.4** Let  $f, g : \mathbb{R} \rightarrow [0, \infty]$  be measurable,  $A \in \mathcal{L}(\mathbb{R})$ , and  $c \geq 0$ . Then:

1.

$$\int_A (f + g) = \int_A f + \int_A g \quad \text{and} \quad \int_A cf = c \int_A f.$$

2. If  $f_1, f_2, \dots : \mathbb{R} \rightarrow [0, \infty]$  are measurable, then

$$\int_A \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} \int_A f_n.$$

**Proof.** For (1), if  $f, g$  are measurable and nonnegative, then there exists  $\phi_n \rightarrow f$ ,  $0 \leq \phi_n \nearrow f$  simple and then there exists  $\psi_n \rightarrow g$ ,  $0 \leq \psi_n \nearrow g$  simple and therefore  $\phi_n + \psi_n \rightarrow f + g$  with  $\{\phi_n + \psi_n\}$  an increasing sequence of nonnegative simple functions. Then, by the [monotone convergence theorem](#),

$$\int f + \int g = \lim_{n \rightarrow \infty} \int \phi_n + \lim_{n \rightarrow \infty} \int \psi_n = \lim_{n \rightarrow \infty} \left( \int \phi_n + \int \psi_n \right).$$

By definition of the Lebesgue integral for simple functions, we know  $\int \phi_n + \int \psi_n = \int (\phi_n + \psi_n)$  for every  $n$ . Passing to the limit gives

$$\int f + \int g = \int (f + g).$$

Similarly, if  $c \geq 0$  and  $0 \leq \phi_n \nearrow f$  are simple, then  $0 \leq c\phi_n \nearrow cf$  and

$$\int c\phi_n = c \int \phi_n.$$

By the [monotone convergence theorem](#),

$$\int cf = c \int f.$$

For (2), let

$$g_n = \sum_{i=1}^n f_i,$$

for all  $n \in \mathbb{N}$ . Then

$$\lim_{n \rightarrow \infty} g_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n f_i,$$

and

$$0 \leq g_1 \leq g_2 \leq \dots$$

By the [monotone convergence theorem](#),

$$\lim_{n \rightarrow \infty} \int_A g_n = \int_A \sum_{i=1}^{\infty} f_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n \int_A f_i = \sum_{i=1}^{\infty} \int_A f_i,$$

because  $g_n = f_1 + \dots + f_n$ . □

**Theorem 7.5 (Fatou's lemma)** Let  $f_n : \mathbb{R} \rightarrow [0, \infty]$  be measurable. Then

$$\int_A \liminf_{n \rightarrow \infty} f_n \leq \liminf_{n \rightarrow \infty} \int_A f_n$$

for any  $A \in \mathcal{L}(\mathbb{R})$ .

**Proof.** Let  $n \in \mathbb{N}$  and let  $g_n = \inf_{k \geq n} f_k$ . Then  $\liminf_{n \rightarrow \infty} f_n = \lim_{n \rightarrow \infty} g_n$ . Now,

$$0 \leq g_1 \leq g_2 \leq \dots,$$

so by the [monotone convergence theorem](#),

$$\lim_{n \rightarrow \infty} \int_A g_n = \int_A \lim_{n \rightarrow \infty} g_n = \int_A \liminf_{n \rightarrow \infty} f_n.$$

Now,  $g_n \leq f_k$  for all  $k \geq n$ , so  $\int_A g_n \leq \int_A f_k$  for all  $k \geq n$ . Hence

$$\int_A g_n \leq \inf_{k \geq n} \int_A f_k$$

for every  $n$ . Hence,

$$\lim_{n \rightarrow \infty} \int_A g_n = \liminf_{n \rightarrow \infty} \int_A g_n \leq \liminf_{n \rightarrow \infty} \int_A f_n.$$

□

**Remark 7.6** Let  $f_n = n\mathbf{1}_{[0,1/n]}$ . Then  $f_n \rightarrow 0$  pointwise, but

$$\lim_{n \rightarrow \infty} \int f_n = 1 \neq \int \lim_{n \rightarrow \infty} f_n = 0.$$

Lecture 8 — Thursday, January 29, 2015

## Integrable Functions and Almost Everywhere Equivalence

**Definition 8.1** We say  $f$  is integrable on  $A$  if  $f \cdot \mathbf{1}_A$  is integrable. If  $f : \mathbb{R} \rightarrow \mathbb{C}$ , we say  $f$  is integrable if  $\operatorname{Im}(f)$  and  $\operatorname{Re}(f)$  are both integrable; in that case, we write

$$\int f = \int \operatorname{Re}(f) + i \int \operatorname{Im}(f).$$

**Fact 8.2** Let  $f : \mathbb{R} \rightarrow [-\infty, \infty]$  be integrable. Then  $\lambda(f^{-1}(\{-\infty, \infty\})) = 0$ .

**Proof.** Let  $A = f^{-1}(\{-\infty, \infty\})$ , which is a measurable subset of  $\mathbb{R}$ . Let

$$g_n = n\mathbf{1}_A \leq |f|.$$

Then

$$n\lambda(A) = \int g_n \leq \int |f| < \infty$$

because  $f$  is integrable. Since this holds for all  $n \in \mathbb{N}$ , we must have  $\lambda(A) = 0$ . □

**Definition 8.3** Let  $f, g : \mathbb{R} \rightarrow [-\infty, \infty]$ . We say  $f = g$  **almost everywhere** if

$$\lambda(\{x \mid f(x) \neq g(x)\}) = 0.$$

**Fact 8.4** Let  $f : \mathbb{R} \rightarrow [-\infty, \infty]$  be integrable, and let  $f = g$  almost everywhere for some other function  $g : \mathbb{R} \rightarrow [-\infty, \infty]$ . Then  $g$  is integrable as well, with

$$\int f = \int g.$$

**Proof.** Let  $N = \{x \mid f(x) \neq g(x)\}$ . This is a measurable null set. Lebesgue measure is complete, so every subset of  $N$  is measurable. For each  $\alpha \in \mathbb{R}$ ,

$$\{g > \alpha\} = (\{f > \alpha\} \setminus N) \cup (\{g > \alpha\} \cap N),$$

and the same decomposition works for the fibres over  $\infty$  and  $-\infty$ . Thus  $g$  is measurable.

Every nonnegative measurable function supported on  $N$  has integral zero: any simple function below it is also supported on  $N$ , so its simple integral is zero. Since  $f^\pm = g^\pm$  off  $N$ , it follows that

$$\int f^+ d\lambda = \int g^+ d\lambda \quad \text{and} \quad \int f^- d\lambda = \int g^- d\lambda.$$

The two integrals on the left are finite because  $f$  is integrable. Hence  $g$  is integrable, and subtracting the displayed equalities gives

$$\int f d\lambda = \int g d\lambda.$$

□

**Definition 8.5** Let  $f_n : \mathbb{R} \rightarrow \mathbb{R}$  be measurable and  $f : \mathbb{R} \rightarrow \mathbb{R}$ . We say  $f_n \rightarrow f$  **almost everywhere** if there exists  $N \in \mathcal{L}(\mathbb{R})$  such that  $\lambda(N) = 0$  and  $\lim_{n \rightarrow \infty} f_n(x) = f(x)$  for all  $x \in \mathbb{R} \setminus N$  (that is,  $f_n$  converges pointwise everywhere outside of  $N$ ).

**Example 8.6** If  $f_n \rightarrow f$  almost everywhere and  $f_n$  is measurable, then  $f$  is measurable.

**Proof.** Let  $N$  be such that  $\lambda(N) = 0$  and  $f_n \rightarrow f$  pointwise on  $\mathbb{R} \setminus N$ . Then  $f$  is measurable on  $\mathbb{R} \setminus N$  by [Proposition 5.5\(5\)](#). Since  $N$  is measurable,  $f$  is measurable on  $\mathbb{R}$ . □

**Proposition 8.7** Let  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  be integrable and let  $c \in \mathbb{R}$ . Then the following hold:

1.  $cf$  is integrable and

$$\int cf = c \int f.$$

2.  $f + g$  is integrable with

$$\int (f + g) = \int f + \int g.$$

3.  $|f|$  is integrable, and

$$\left| \int f \right| \leq \int |f|.$$

**Proof.** For (1), if  $c \geq 0$ , then  $(cf)^+ = cf^+$  and  $(cf)^- = cf^-$ , so

$$\int (cf)^+ = c \int f^+ < \infty \quad \text{and} \quad \int (cf)^- = c \int f^- < \infty.$$

Hence  $cf$  is integrable, and

$$\int cf = \int (cf)^+ - \int (cf)^- = c \left( \int f^+ - \int f^- \right) = c \int f.$$

If  $c < 0$ , write  $cf = -|c|f$  and use linearity with  $-f$ .

For (2), we know that if  $f$  and  $g$  are measurable, then  $f + g$  is measurable too. Since  $(f + g)^+ \leq f^+ + g^+$ , we have

$$\int (f + g)^+ \leq \int f^+ + \int g^+ < \infty$$

and similarly, since  $(f + g)^- \leq f^- + g^-$ ,

$$\int (f + g)^- \leq \int f^- + \int g^- < \infty.$$

Thus  $f + g$  is integrable. We want to show

$$\int (f + g) = \int f + \int g.$$

Now

$$(f + g)^+ - (f + g)^- = f + g = (f^+ - f^-) + (g^+ - g^-),$$

so

$$(f + g)^+ + f^- + g^- = (f + g)^- + f^+ + g^+,$$

and

$$\int (f + g)^+ + \int f^- + \int g^- = \int (f + g)^- + \int f^+ + \int g^+$$

and hence

$$\int (f + g) = \int (f + g)^+ - \int (f + g)^- = \int f^+ - \int f^- + \int g^+ - \int g^- = \int f + \int g.$$

For (3),  $f$  being integrable implies that  $f^+, f^-$  are measurable and hence  $|f| = f^+ + f^-$  is measurable as well. Also,

$$\int |f| = \int f^+ + \int f^- < \infty$$

implies that  $|f|$  is integrable. Furthermore,

$$\left| \int f \right| = \left| \int f^+ - \int f^- \right| \leq \left| \int f^+ \right| + \left| \int f^- \right| = \int f^+ + \int f^- = \int (f^+ + f^-) = \int |f|.$$

□

**Corollary 8.8** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$ . Then  $f$  is integrable if and only if  $|f|$  is integrable.

**Warning** It may be tempting to think that if  $f + g$  is integrable, then so are  $f$  and  $g$  individually. However, this is not necessarily true. For example, consider the function  $f = \mathbb{1}_E + \mathbb{1}_{[0,1] \setminus E}$  where  $[0, 1] \supset E \notin \mathcal{L}(\mathbb{R})$ . Then  $f = 1$  is integrable on  $[0, 1]$ , but  $\mathbb{1}_E$  is not even measurable, let alone integrable.

## Dominated Convergence Theorem

We will use [Fatou's lemma](#) here. Recall its statement: if  $\{f_n\}_{n=1}^\infty$  is a sequence of measurable functions with  $f_n : \mathbb{R} \rightarrow [0, \infty]$  and  $A \in \mathcal{L}(\mathbb{R})$ , then

$$\int_A \liminf_{n \rightarrow \infty} f_n \leq \liminf_{n \rightarrow \infty} \int_A f_n.$$

**Theorem 8.9 (Dominated convergence theorem)** Let  $f_n : E \rightarrow \mathbb{R}$  be measurable, where  $E \in \mathcal{L}(\mathbb{R})$ . Let  $g : \mathbb{R} \rightarrow [0, \infty)$  be integrable on  $E$ , and suppose that  $g$  dominates  $\{f_n\}$  on  $E$  almost everywhere; that is,  $|f_n| \leq g$  almost everywhere on  $E$  for every  $n \in \mathbb{N}$ . If  $f_n \rightarrow f$  pointwise almost everywhere on  $E$ , then  $f$  is integrable on  $E$  and

$$\lim_{n \rightarrow \infty} \int_E f_n = \int_E f.$$

**Proof.** We prove this for  $E = \mathbb{R}$ ; the general case follows by extending the functions by zero outside  $E$ . Define

$$N = \{x \in \mathbb{R} \mid \lim_{n \rightarrow \infty} f_n(x) \neq f(x) \text{ or the limit does not exist}\} \cup \bigcup_{n=1}^{\infty} \{x \in \mathbb{R} \mid |f_n(x)| > g(x)\},$$

which is the set of points that are up to no good. By assumption,  $\lambda(N) = 0$ . Let  $A = \mathbb{R} \setminus N$ . We have  $|f_n| \leq g$  and  $|f| \leq g$  on  $A$ . Therefore,

$$\int_A |f_n| d\lambda \leq \int_{\mathbb{R}} g d\lambda < \infty \quad \text{and} \quad \int_A |f| d\lambda \leq \int_{\mathbb{R}} g d\lambda < \infty.$$

The restrictions of the  $f_n$  to  $A$  are measurable, and their pointwise limit  $f|_A$  is measurable by [Proposition 5.5\(5\)](#). Hence  $f_n \mathbb{1}_A$  and  $f \mathbb{1}_A$  are integrable. They agree almost everywhere with  $f_n$  and  $f$ , respectively, so [Fact 8.4](#) shows that every  $f_n$  and  $f$  is integrable and that

$$\int_{\mathbb{R}} f_n d\lambda = \int_A f_n d\lambda \quad \text{and} \quad \int_{\mathbb{R}} f d\lambda = \int_A f d\lambda.$$

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**Proof continued.** Since  $|f_n| \leq g$  on  $A$ , the functions  $g + f_n$  and  $g - f_n$  are nonnegative on  $A$  for every  $n$ .

Apply [Fatou's lemma](#) to  $h_n := g + f_n$ :

$$\int_A \liminf_{n \rightarrow \infty} h_n \leq \liminf_{n \rightarrow \infty} \int_A h_n.$$

Because  $f_n \rightarrow f$  pointwise on  $A$ , we have  $h_n \rightarrow g + f$ , so

$$\int_A (g + f) \leq \liminf_{n \rightarrow \infty} \left( \int_A g + \int_A f_n \right) = \int_A g + \liminf_{n \rightarrow \infty} \int_A f_n.$$

Subtracting  $\int_A g$  from both sides gives

$$\int_A f \leq \liminf_{n \rightarrow \infty} \int_A f_n.$$

Now apply [Fatou's lemma](#) to  $k_n := g - f_n$ :

$$\int_A \liminf_{n \rightarrow \infty} k_n \leq \liminf_{n \rightarrow \infty} \int_A k_n.$$

Again  $k_n \rightarrow g - f$ , so

$$\int_A (g - f) \leq \liminf_{n \rightarrow \infty} \left( \int_A g - \int_A f_n \right) = \int_A g - \limsup_{n \rightarrow \infty} \int_A f_n.$$

Subtracting  $\int_A g$  and multiplying by  $-1$  give

$$\limsup_{n \rightarrow \infty} \int_A f_n \leq \int_A f.$$

Therefore,

$$\limsup_{n \rightarrow \infty} \int_A f_n \leq \int_A f \leq \liminf_{n \rightarrow \infty} \int_A f_n.$$

Hence

$$\lim_{n \rightarrow \infty} \int_A f_n = \int_A f.$$

Since  $\int_{\mathbb{R}} f_n = \int_A f_n$  and  $\int_{\mathbb{R}} f = \int_A f$ , this also gives

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n = \int_{\mathbb{R}} f.$$

□

### 3. $L^p$ Spaces and Hilbert Spaces

#### $L^p$ and $L^\infty$ Spaces

**Definition 9.1** Let  $A \in \mathcal{L}(\mathbb{R})$  with  $\lambda(A) > 0$ . Let  $f : A \rightarrow \mathbb{R}$  be a measurable function. For  $1 \leq p < \infty$ , define

$$\|f\|_{L^p(A)} = \left( \int_A |f|^p \right)^{1/p}.$$

When  $A$  is clear from context, we write  $\|f\|_p$  instead of  $\|f\|_{L^p(A)}$ .

Note that if  $f = g$  almost everywhere on  $A$ , then  $\|f - g\|_p = 0$ . We thus define an equivalence relation  $\sim$  by  $f \sim g$  when  $f = g$  almost everywhere. Observe that if  $f \sim g$ , then  $\|f\|_p = \|g\|_p$ .

**Definition 9.2** We define  $L^p(A)$  as the set of equivalence classes of measurable functions  $f : A \rightarrow \mathbb{R}$  such that  $\|f\|_p < \infty$ , where two functions are considered equivalent if they are equal almost everywhere.

**Example 9.3**  $C([a, b])$ , the set of continuous functions on  $[a, b]$ , is a subset of  $L^p([a, b])$  for  $1 \leq p < \infty$ . For example,  $C([1, 2]) \subset L^1([1, 2])$ , but  $C(\mathbb{R}) \not\subset L^1(\mathbb{R})$ .

Even in two dimensions, changing  $p$  changes the geometry of the norm. Figure 6 shows the three standard unit balls that foreshadow this distinction in function spaces.

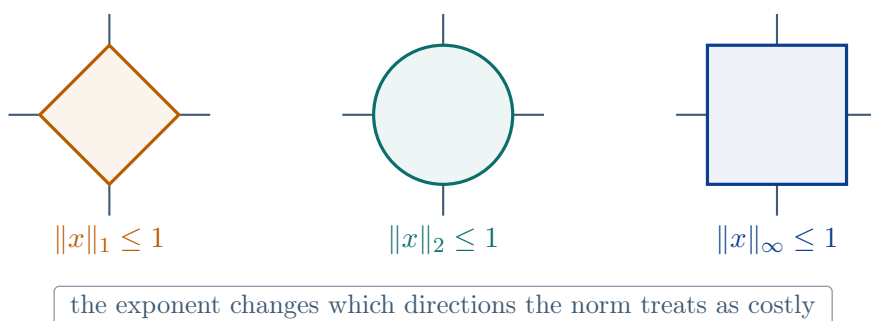


FIGURE 6

**Proposition 9.4** Let  $1 \leq p < \infty$  and let  $E \in \mathcal{L}(\mathbb{R})$  with  $\lambda(E) > 0$ . Then the following hold:

1.  $L^p(E)$  is a vector space.
2.  $\|\cdot\|_p$  satisfies the following:
  - (a)  $\|f\|_p = 0$  if and only if  $f = 0$  almost everywhere
  - (b)  $\|cf\|_p = |c| \cdot \|f\|_p$  for every  $c \in \mathbb{C}$ .
  - (c)  $\|f + g\|_p \leq \|f\|_p + \|g\|_p$ .

**Proof.** For (1), if  $f \in L^p(E)$  and  $c \in \mathbb{C}$ , then  $cf \in L^p(E)$  because  $cf$  is measurable and

$$\|cf\|_p^p = \int_E |cf|^p = \int_E |c|^p |f|^p = |c|^p \int_E |f|^p < \infty.$$

If  $f, g \in L^p(E)$ , then  $f + g$  is measurable and

$$\|f + g\|_p^p = \int_E |f + g|^p \leq \int_E 2^p(|f|^p + |g|^p) = 2^p \left[ \int_E |f|^p + \int_E |g|^p \right] < \infty.$$

The inequality follows from the convexity of  $x \mapsto x^p$ , namely  $((a + b)/2)^p \leq (a^p + b^p)/2$ . Thus,  $f + g \in L^p(E)$ , proving (1).

For (2), (2b) is proven above. For (2a),

$$\|f\|_p = 0 \iff \int_E |f|^p = 0$$

If the integral is zero, let

$$A_n = \{x \in E \mid |f(x)|^p \geq 1/n\}.$$

Then  $0 \geq \lambda(A_n)/n$ , so every  $A_n$  is null. Since  $\{f \neq 0\} = \bigcup_{n=1}^{\infty} A_n$ , we have  $f = 0$  almost everywhere. The converse is immediate.

We will handle (2c) in the next lecture after we have established Hölder's inequality. □

**Definition 9.5** Let  $E \in \mathcal{L}(\mathbb{R})$  with  $\lambda(E) > 0$  and let  $f : E \rightarrow \mathbb{C}$ . Define the **essential supremum**

$$\|f\|_\infty = \inf \left\{ \sup_{x \in A} |f(x)| : A \subseteq E, \lambda(E \setminus A) = 0 \right\}.$$

We define  $L^\infty(E)$  as the set of equivalence classes of measurable functions  $f : E \rightarrow \mathbb{C}$  such that  $\|f\|_\infty < \infty$ , where two functions are considered equivalent if they are equal almost everywhere.

**Example 9.6** If  $f = 0$  almost everywhere, then  $\|f\|_\infty = 0$ , and conversely. If  $E$  is a compact interval and  $f$  is continuous on  $E$ , then  $\|f\|_\infty = \sup_E |f(x)|$ .

**Proposition 9.7**  $L^\infty(E)$  is a vector space, and  $\|\cdot\|_\infty$  is a norm on  $L^\infty(E)$ .

**Proof.** If  $f, g \in L^\infty(E)$ , choose sets  $N_f, N_g$  such that  $\lambda(N_f) = \lambda(N_g) = 0$  and  $|f(x)| \leq \|f\|_\infty + \varepsilon$  on  $E \setminus N_f$  and  $|g(x)| \leq \|g\|_\infty + \varepsilon$  on  $E \setminus N_g$ . Then on  $E \setminus (N_f \cup N_g)$ ,

$$|f + g| \leq |f| + |g| \leq \|f\|_\infty + \|g\|_\infty + 2\varepsilon \quad \text{and} \quad |cf| \leq |c|(\|f\|_\infty + \varepsilon),$$

so  $f + g, cf \in L^\infty(E)$ . For the norm axioms, nonnegativity is immediate,  $\|f\|_\infty = 0$  if and only if  $f = 0$  almost everywhere (from the preceding example),  $\|cf\|_\infty = |c|\|f\|_\infty$ , and

$$\|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty$$

from the pointwise bound above after removing a set of Lebesgue measure zero. Thus  $\|\cdot\|_\infty$  is a norm.  $\square$

## Hölder and Minkowski Inequalities

**Theorem 10.1 (Hölder's inequality)** Let  $p, q$  be conjugate indices (that is,  $1/p + 1/q = 1$ ),  $f \in L^p(E)$  and  $g \in L^q(E)$ . Then

$$\int_E |fg| \leq \|f\|_p \|g\|_q.$$

In particular,  $fg \in L^1(E)$ .

**Proof.** *Case:*  $p = 1, q = \infty$  (or vice versa). Define  $A = \{x \in E : |g(x)| > \|g\|_\infty\}$ . We first show that  $\lambda(A) = 0$ . Let  $A_n = \{x \in E : |g(x)| > \|g\|_\infty + 1/n\}$ , so  $A = \bigcup_{n=1}^\infty A_n$ . It suffices to show that  $\lambda(A_n) = 0$  for each  $n$ . Let  $B \subseteq E$  satisfy  $\lambda(E \setminus B) = 0$  and  $\sup_{x \in B} |g(x)| \leq \|g\|_\infty + 1/n$ . Then  $A_n \subseteq E \setminus B$ , so  $\lambda(A_n) = 0$ . Thus  $\lambda(A) = 0$ . Therefore, for  $f \in L^1(E)$  and  $g \in L^\infty(E)$ ,

$$\int_E |fg| = \int_{E \setminus A} |fg| \leq \|g\|_\infty \int_E |f| = \|g\|_\infty \|f\|_1.$$

*Case:*  $1 < p, q < \infty$ . If  $\|f\|_p = 0$ , then  $f = 0$  almost everywhere, so  $fg = 0$  almost everywhere and hence  $\int |fg| = 0$ , so we are done. Similarly, if  $\|g\|_q = 0$ , then we are done. Therefore, we may assume that  $\|f\|_p \neq 0, \|g\|_q \neq 0$ . It is enough to show

$$\int_E \frac{|f|}{\|f\|_p} \frac{|g|}{\|g\|_q} \leq 1.$$

Let  $a = |f|/\|f\|_p$  and  $b = |g|/\|g\|_q$ . Young's inequality gives

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Therefore

$$\int_E \frac{|f|}{\|f\|_p} \frac{|g|}{\|g\|_q} \leq \int_E \left( \frac{1}{p} \frac{|f|^p}{\|f\|_p^p} + \frac{1}{q} \frac{|g|^q}{\|g\|_q^q} \right) = \frac{1}{p} \frac{\|f\|_p^p}{\|f\|_p^p} + \frac{1}{q} \frac{\|g\|_q^q}{\|g\|_q^q} = \frac{1}{p} + \frac{1}{q} = 1.$$

□

**Theorem 10.2 (Minkowski's inequality)** Let  $1 \leq p \leq \infty$ ,  $f, g \in L^p(E)$ . Then

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

**Proof.** *Case:*  $p = \infty$ . We have  $|f + g| \leq |f| + |g|$  by the usual triangle inequality, and moreover  $|f| \leq \|f\|_\infty$  and  $|g| \leq \|g\|_\infty$  almost everywhere. Therefore, we have

$$|f + g| \leq \|f\|_\infty + \|g\|_\infty$$

almost everywhere, and taking the essential supremum on both sides gives

$$\|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty.$$

*Case:*  $1 \leq p < \infty$ . It is enough to show that

$$\int_E |f + g|^p \leq \|f + g\|_p^{p-1} (\|f\|_p + \|g\|_p).$$

We have

$$\int_E |f + g|^p = \int_E |f + g|^{p-1} |f + g| \leq \int_E |f + g|^{p-1} |f| + \int_E |f + g|^{p-1} |g|.$$

Let  $q$  be such that  $1/p + 1/q = 1$ . Then

$$\|(f + g)^{p-1}\|_q = \left( \int_E |f + g|^{(p-1)q} \right)^{1/q} = \left( \int_E |f + g|^p \right)^{1/q} = \|f + g\|_p^{p/q} < \infty$$

because  $L^p(E)$  is a vector space. Then by [Hölder's inequality](#),

$$\int_E |f + g|^{p-1} |f| \leq \|f + g\|_p^{p/q} \|f\|_p = \|f + g\|_p^{p-1} \|f\|_p.$$

Similarly,

$$\int_E |f + g|^{p-1} |g| \leq \|f + g\|_p^{p-1} \|g\|_p.$$

Therefore,

$$\|f + g\|_p^p \leq \|f + g\|_p^{p-1} (\|f\|_p + \|g\|_p),$$

which implies that  $\|f + g\|_p \leq \|f\|_p + \|g\|_p$ . (If  $\|f + g\|_p = 0$ , the inequality is immediate.) □

**Corollary 10.3** For  $1 \leq p \leq \infty$ ,  $L^p(E)$  is a normed vector space.

**Definition 10.4** A complete normed vector space is called a **Banach space**.

**Theorem 10.5** Let  $1 \leq p \leq \infty$ . Then  $L^p(E)$  is a Banach space.

**Proof.** We want to show that  $L^p(E)$  is complete. Our strategy is to take a Cauchy sequence  $\{f_n\}$  and find a subsequence converging to some  $f \in L^p(E)$ . The Cauchy property will then force the entire sequence to converge to  $f$ . Let's get into it!

*Case:*  $1 \leq p < \infty$ . Let  $\{f_n\}$  be a Cauchy sequence in  $L^p(E)$ . This means that  $\|f_n - f_m\|_p \rightarrow 0$  as  $m, n \rightarrow \infty$ . Let  $\{f_{n_k}\}$  be a subsequence such that  $\|f_{n_{k+1}} - f_{n_k}\|_p < 1/2^k$ . Let

$$g_k = \sum_{i=1}^k |f_{n_{i+1}} - f_{n_i}|$$

and

$$g = \sum_{i=1}^{\infty} |f_{n_{i+1}} - f_{n_i}|.$$

For every finite  $k$ , [Minkowski's inequality](#) gives

$$\|g_k\|_p \leq \sum_{i=1}^k \|f_{n_{i+1}} - f_{n_i}\|_p \leq \sum_{i=1}^{\infty} \frac{1}{2^i} = 1.$$

Now  $g_k \nearrow g$  pointwise. By the [monotone convergence theorem](#),

$$\int_E |g|^p = \lim_{k \rightarrow \infty} \int_E |g_k|^p \leq 1,$$

so  $g \in L^p(E)$ . In particular,  $g$  is finite outside a set  $A$  with  $\lambda(A) = 0$ . Hence

$$f = f_{n_1} + \sum_{i=1}^{\infty} (f_{n_{i+1}} - f_{n_i})$$

is well-defined on  $A^c$  as the pointwise limit of

$$h_k = f_{n_1} + \sum_{i=1}^k (f_{n_{i+1}} - f_{n_i}).$$

Set  $f(x) = 0$  on  $A$ . Then  $f$  is measurable and  $f_{n_k} \rightarrow f$  almost everywhere. The inequality  $|f| \leq |f_{n_1}| + g$  and [Minkowski's inequality](#) show that  $f \in L^p(E)$ . Moreover,  $|f_{n_k} - f|^p \rightarrow 0$  almost everywhere and  $|f_{n_k} - f|^p \leq |g|^p$ . The [dominated convergence theorem](#) therefore gives

$$\lim_{k \rightarrow \infty} \int_E |f_{n_k} - f|^p = 0.$$

Thus  $\|f_{n_k} - f\|_p \rightarrow 0$ . Since  $\{f_n\}$  is Cauchy, the triangle inequality now gives  $\|f_n - f\|_p \rightarrow 0$ .

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**Proof continued.** *Case:  $p = \infty$ .* Let  $\{f_n\}_{n=1}^\infty$  be a  $\|\cdot\|_\infty$ -Cauchy sequence in  $L^\infty(E)$ . For  $k \in \mathbb{N}$ , set

$$A_k = \{x \in E : |f_k(x)| > \|f_k\|_\infty\}.$$

We have already proved  $\lambda(A_k) = 0$ . For  $m, n \in \mathbb{N}$ , set

$$B_{n,m} = \{x \in E : |f_n(x) - f_m(x)| > \|f_n - f_m\|_\infty\},$$

and  $\lambda(B_{n,m}) = 0$ . Let

$$A = \left( \bigcup_k A_k \right) \cup \left( \bigcup_{n,m} B_{n,m} \right).$$

Then  $\lambda(A) = 0$ . For all  $x \in A^c$ , we have  $|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_\infty$ , so  $\{f_n\}$  is uniformly Cauchy on  $A^c$ . Hence  $f_n \rightarrow f$  uniformly on  $A^c$ . Define  $f(x) = 0$  for  $x \in A$ . Then

$$\|f_n - f\|_\infty \leq \sup_{x \in A^c} |f_n(x) - f(x)| \rightarrow 0.$$

Also,  $f \in L^\infty(E)$  because  $f$  is measurable and there exists  $N$  such that  $\|f_N - f\|_\infty < 1$ . Therefore,

$$\|f\|_\infty \leq \|f - f_N\|_\infty + \|f_N\|_\infty < 1 + \|f_N\|_\infty < \infty.$$

□

**Remark 11.1** We have proved the following:

1. If  $f \in L^\infty(E)$ , then  $\lambda(\{x \in E : |f(x)| > \|f\|_\infty\}) = 0$ .
2. If  $f_n \xrightarrow{\|\cdot\|_p} f$ , then there exists a subsequence  $\{f_{n_k}\}$  such that  $f_{n_k} \rightarrow f$  pointwise almost everywhere.
3.  $f_n \xrightarrow{\|\cdot\|_\infty} f$  if and only if there exists  $A$  such that  $\lambda(A) = 0$  and  $f_n \rightarrow f$  uniformly on  $A^c$  (we really only proved one direction, but the other is trivial).

**Warning** Norm convergence does not force pointwise convergence of the chosen representatives. For example, let  $f_n(x) = x^n$  on  $[0, 1]$ . For every  $1 \leq p < \infty$ ,

$$\|f_n\|_p = (np + 1)^{-1/p} \rightarrow 0,$$

but  $f_n(1) = 1$  for every  $n$ , so this representative does not converge pointwise to the zero function.

Let us record some facts about  $L^p$  spaces.

**Proposition 11.2** Let  $[a, b]$  be a compact interval and  $1 \leq p_1 < p_2 < \infty$ . Then

$$L^\infty([a, b]) \subsetneq L^{p_2}([a, b]) \subsetneq L^{p_1}([a, b]).$$

**Proof.** Let  $1 \leq p_1 < p_2 < \infty$  and  $f \in L^{p_2}([a, b])$ . By Hölder's inequality with exponents

$$r = \frac{p_2}{p_1} \quad \text{and} \quad r' = \frac{p_2}{p_2 - p_1},$$

we have

$$\int_a^b |f|^{p_1} = \int_a^b |f|^{p_1} \cdot 1 \leq \left( \int_a^b |f|^{p_2} \right)^{p_1/p_2} (b-a)^{1-p_1/p_2} < \infty.$$

Hence  $f \in L^{p_1}([a, b])$ , so  $L^{p_2}([a, b]) \subseteq L^{p_1}([a, b])$ . Also, if  $f \in L^\infty([a, b])$ , then

$$\|f\|_{p_1}^{p_1} = \int_a^b |f|^{p_1} \leq (b-a) \|f\|_\infty^{p_1},$$

so  $L^\infty([a, b]) \subseteq L^{p_1}([a, b])$  for every finite  $p_1$ .

To prove strictness, fix  $1 \leq p_1 < p_2 < \infty$  and define

$$h(x) := (x - a)^{-1/p_2} \mathbb{1}_{(a, (a+b)/2]}(x).$$

Then

$$\int_a^b |h|^{p_2} dx = \int_a^{(a+b)/2} (x - a)^{-1} dx = \infty,$$

so  $h \notin L^{p_2}([a, b])$ , while

$$\int_a^b |h|^{p_1} dx = \int_a^{(a+b)/2} (x - a)^{-p_1/p_2} dx < \infty$$

because  $p_1/p_2 < 1$ . Thus  $h \in L^{p_1}([a, b]) \setminus L^{p_2}([a, b])$ .

Finally, for each finite  $p \geq 1$ , the function

$$u_p(x) := (x - a)^{-1/(2p)} \mathbb{1}_{(a, (a+b)/2]}(x)$$

belongs to  $L^p([a, b])$  since

$$\int_a^{(a+b)/2} (x - a)^{-1/2} dx < \infty,$$

but  $u_p \notin L^\infty([a, b])$ . Therefore all inclusions in the chain are strict.

□

**Proposition 11.3** Let  $[a, b]$  be a compact interval. For  $1 \leq p < \infty$ , let  $C([a, b])$  be the set of continuous functions on  $[a, b]$ . Then the following hold:

1.  $C([a, b])$  is  $\|\cdot\|_p$ -dense in  $L^p([a, b])$ . In other words, for every  $f \in L^p([a, b])$  and every  $\varepsilon > 0$ , there exists  $h \in C([a, b])$  such that  $\|f - h\|_p < \varepsilon$ .
2.  $C([a, b]) \subsetneq L^\infty([a, b])$ , and  $C([a, b])$  is  $\|\cdot\|_\infty$ -closed in  $L^\infty([a, b])$  (in particular,  $C([a, b])$  is not  $\|\cdot\|_\infty$ -dense in  $L^\infty([a, b])$ ).

**Proof.** For (1), let  $f \in L^p([a, b])$  and  $\varepsilon > 0$ . We first treat real-valued  $f$ ; the complex case follows by applying the same argument to the real and imaginary parts.

*Step 1: reduction to a bounded function.* For  $M > 0$ , define

$$f_M(x) := \max\{-M, \min\{f(x), M\}\}.$$

Then  $|f_M| \leq |f|$  and  $|f - f_M|^p \rightarrow 0$  pointwise as  $M \rightarrow \infty$ . Since  $|f - f_M|^p \leq |f|^p \in L^1([a, b])$ , the [dominated convergence theorem](#) gives

$$\|f - f_M\|_p \rightarrow 0.$$

Choose  $M$  so that  $\|f - f_M\|_p < \varepsilon/3$ .

*Step 2: approximation of  $f_M$  by simple functions.* Choose  $n \in \mathbb{N}$  and define

$$s_n(x) := 2^{-n} \lfloor 2^n f_M(x) \rfloor.$$

Since  $f_M$  is bounded,  $s_n$  takes only finitely many values, so  $s_n$  is simple. Moreover,

$$|f_M(x) - s_n(x)| \leq 2^{-n} \quad \text{for all } x \in [a, b].$$

Hence

$$\|f_M - s_n\|_p \leq (b - a)^{1/p} 2^{-n}.$$

Choose  $n$  large enough so that  $(b - a)^{1/p} 2^{-n} < \varepsilon/3$ , and set  $s := s_n$ . Write

$$s = \sum_{j=1}^m a_j \mathbb{1}_{E_j}$$

for measurable sets  $E_j$ .

*Step 3: approximation of each indicator by a continuous cutoff.* Fix

$$\delta := \left( \frac{\varepsilon}{3 \sum_{j=1}^m (|a_j| + 1)} \right)^p.$$

By [Corollary 4.4](#) (applied to  $E_j \subseteq [a, b]$ ), for each  $j$  there exist a compact set  $K_j$  and an open set  $U_j$  with

$$K_j \subseteq E_j \subseteq U_j \quad \text{and} \quad \lambda(U_j \setminus K_j) < \delta.$$

Since  $[a, b]$  is normal, there exists  $\eta_j \in C([a, b])$  such that  $0 \leq \eta_j \leq 1$ ,  $\eta_j = 1$  on  $K_j$ , and  $\eta_j = 0$  on  $[a, b] \setminus U_j$ . Then

$$|\eta_j - \mathbb{1}_{E_j}| \leq \mathbb{1}_{U_j \setminus K_j},$$

so

$$\|\eta_j - \mathbf{1}_{E_j}\|_p^p \leq \lambda(U_j \setminus K_j) < \delta.$$

Define

$$h := \sum_{j=1}^m a_j \eta_j \in C([a, b]).$$

Using the triangle inequality,

$$\|h - s\|_p \leq \sum_{j=1}^m |a_j| \|\eta_j - \mathbf{1}_{E_j}\|_p \leq \delta^{1/p} \sum_{j=1}^m |a_j| < \varepsilon/3.$$

Therefore,

$$\|f - h\|_p \leq \|f - f_M\|_p + \|f_M - s\|_p + \|s - h\|_p < \varepsilon.$$

Hence  $C([a, b])$  is dense in  $L^p([a, b])$ .

For (2), first note that every continuous function on a compact interval is bounded, so  $C([a, b]) \subseteq L^\infty([a, b])$ .

Now let  $\{f_n\} \subseteq C([a, b])$  and assume  $f_n \xrightarrow{\|\cdot\|_\infty} f$  in  $L^\infty([a, b])$ . By Remark 11.1(3), there is a set  $A \subseteq [a, b]$  with  $\lambda([a, b] \setminus A) = 0$  such that  $f_n \rightarrow f$  uniformly on  $A$ . Define  $g = f$  on  $A$ . For  $x \in [a, b] \setminus A$ , choose a sequence  $\{x_m\} \subseteq A$  with  $x_m \rightarrow x$ , and define

$$g(x) := \lim_{m \rightarrow \infty} g(x_m).$$

This is well-defined: if  $x_m, y_m \in A$  both converge to  $x$ , then uniform convergence on  $A$  and uniform continuity of each continuous approximant imply

$$\lim_{m \rightarrow \infty} g(x_m) = \lim_{m \rightarrow \infty} g(y_m).$$

Hence  $g$  is continuous on  $[a, b]$ , and  $g = f$  almost everywhere on  $[a, b]$ . Therefore the  $L^\infty$  class of  $f$  belongs to  $C([a, b])$ , so  $C([a, b])$  is  $\|\cdot\|_\infty$ -closed in  $L^\infty([a, b])$ .

Finally, the characteristic function  $\mathbf{1}_{[a, (a+b)/2]}$  belongs to  $L^\infty([a, b])$  but does not agree almost everywhere with any continuous function. So  $C([a, b]) \subsetneq L^\infty([a, b])$ , and in particular  $C([a, b])$  is not  $\|\cdot\|_\infty$ -dense in  $L^\infty([a, b])$ .

□

**Proposition 11.4** Let  $[a, b]$  be a compact interval. For  $1 \leq p < \infty$ ,  $L^p([a, b])$  is separable.

**Proof.** By Proposition 11.3(1),  $C([a, b])$  is dense in  $L^p([a, b])$ . Next, by the Stone–Weierstrass theorem, the space  $\mathbb{C}[x]$  of polynomials in  $x$  with complex coefficients is  $\|\cdot\|_\infty$ -dense in  $C([a, b])$ . Indeed, the real version gives the complex case by writing  $f \in C([a, b])$  as  $f = f_1 + if_2$  with  $f_1, f_2$  real-valued and continuous:

$$\|f - (p_1 + ip_2)\|_\infty = \|(f_1 - p_1) + i(f_2 - p_2)\|_\infty \leq \|f_1 - p_1\|_\infty + \|f_2 - p_2\|_\infty < \varepsilon$$

for some  $p_1, p_2 \in \mathbb{R}[x]$  such that  $\|f_1 - p_1\|_\infty < \varepsilon/2$  and  $\|f_2 - p_2\|_\infty < \varepsilon/2$ .

Next, let  $\mathbb{Q}[x]$  be the space of polynomials on  $[a, b]$  with rational coefficients. Then  $\mathbb{Q}[x]$  is  $\|\cdot\|_\infty$ -dense in  $\mathbb{R}[x]$ . To see this, let  $p(x) = a_n x^n + \cdots + a_0$ , where  $a_n, \dots, a_0 \in \mathbb{R}$ . Choose  $r_n, \dots, r_0 \in \mathbb{Q}$  such that  $|a_i - r_i| < \varepsilon$ , and set  $Q(x) = r_n x^n + \cdots + r_0$ . If  $c = \max(|a|, |b|)$ , then

$$\|p - Q\|_\infty \leq |a_n - r_n|c^n + \cdots + |a_0 - r_0|.$$

Letting the rational approximations improve proves the claim.

Finally, the set of polynomials with rational real and imaginary coefficients,  $\mathbb{Q}[x] + i\mathbb{Q}[x]$ , are  $\|\cdot\|_\infty$ -dense in  $\mathbb{C}[x]$ . To see this, let  $p(x) = a_n x^n + \cdots + a_0$ , where  $a_n, \dots, a_0 \in \mathbb{C}$ . Write  $a_i = r_i + is_i$  with  $r_i, s_i \in \mathbb{R}$ . Choose  $r'_n, \dots, r'_0, s'_n, \dots, s'_0 \in \mathbb{Q}$  such that  $|r_i - r'_i| < \varepsilon$  and  $|s_i - s'_i| < \varepsilon$ , and set

$$q(x) = (r'_n + is'_n)x^n + \cdots + (r'_0 + is'_0).$$

Then

$$\begin{aligned} \|p - q\|_\infty &\leq |a_n - (r'_n + is'_n)|c^n + \cdots + |a_0 - (r'_0 + is'_0)| \\ &= |(r_n - r'_n) + i(s_n - s'_n)|c^n + \cdots + |(r_0 - r'_0) + i(s_0 - s'_0)| \\ &\leq \sqrt{(r_n - r'_n)^2 + (s_n - s'_n)^2}c^n + \cdots + \sqrt{(r_0 - r'_0)^2 + (s_0 - s'_0)^2} \\ &< \varepsilon\sqrt{2}(c^n + \cdots + 1). \end{aligned}$$

Therefore  $\mathbb{Q}[x] + i\mathbb{Q}[x]$  is  $\|\cdot\|_p$ -dense in  $C([a, b])$ , and hence  $\|\cdot\|_p$ -dense in  $L^p([a, b])$ . The set  $\mathbb{Q}[x] + i\mathbb{Q}[x]$  is countable, so  $L^p([a, b])$  is separable.

□

## Lecture 12 — Thursday, February 12, 2015

**Theorem 12.1**  $L^\infty([a, b])$  is not separable.

**Proof.** Assume  $a = 0, b = 1$ . Let  $\alpha = (\alpha_i)_{i=1}^\infty \in \{0, 1\}^\mathbb{N}$  be a binary sequence and let

$$f_\alpha = \sum_{n=1}^{\infty} \alpha_n \mathbf{1}_{[\frac{1}{n+1}, \frac{1}{n}]}.$$

If  $\alpha \neq \beta$  are distinct binary sequences, then there exists  $n_0 \in \mathbb{N}$  such that  $\alpha_{n_0} \neq \beta_{n_0}$ , which implies  $\|f_\alpha - f_\beta\|_\infty = 1$ . Suppose for a contradiction that  $L^\infty([0, 1])$  is separable. Let  $\{u_n\}_{n=1}^\infty$  be a countable dense subset of  $L^\infty([0, 1])$ . Since  $\{u_n\}$  is dense in  $L^\infty([0, 1])$ , for all  $\alpha \in \{0, 1\}^\mathbb{N}$ , there exists  $n_\alpha \in \mathbb{N}$  such that  $\|f_\alpha - u_{n_\alpha}\|_\infty < 1/2$ . Now, if  $\alpha \neq \beta$ , then  $n_\alpha \neq n_\beta$ . Then the map  $\{0, 1\}^\mathbb{N} \rightarrow \mathbb{N}$  defined by  $\alpha \mapsto n_\alpha$  is injective. Then  $|\{0, 1\}^\mathbb{N}| \leq |\mathbb{N}|$ , a contradiction.  $\square$

## Inner Products and Hilbert Spaces

The space  $L^2$  is special among the spaces  $L^p$ : its norm comes from an inner product, and this additional structure gives projection, orthogonality, and Fourier analyses.

**Definition 12.2** Let  $X$  be a  $\mathbb{C}$ -vector space (or  $\mathbb{R}$ -vector space). A map  $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{C}$  is called an **inner product** if it satisfies the following:

1.  $\langle v, v \rangle \geq 0$  (**nonnegativity**).
2.  $\langle v, v \rangle = 0$  if and only if  $v = 0$ .
3.  $\langle v, w \rangle = \overline{\langle w, v \rangle}$  (**conjugate symmetry**).
4.  $\langle \alpha v_1 + v_2, w \rangle = \alpha \langle v_1, w \rangle + \langle v_2, w \rangle$ .

We say that  $\langle \cdot, \cdot \rangle$  is **linear in the first coordinate** and **antilinear in the second coordinate**, because  $\langle v, \alpha w_1 + w_2 \rangle = \overline{\alpha} \langle v, w_1 \rangle + \langle v, w_2 \rangle$ .

**Example 12.3** In  $\mathbb{C}^2$ , define  $\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = x_1 \overline{y_1} + x_2 \overline{y_2}$ . This is an inner product on  $\mathbb{C}^2$ .

**Proposition 12.4** Let  $X$  be a vector space with an inner product. The formula

$$\|v\| = \sqrt{\langle v, v \rangle}$$

defines a nonnegative, definite, and homogeneous function on  $X$ .

**Proof.** By nonnegativity,  $\|v\|$  is a nonnegative real number, and definiteness gives  $\|v\| = 0$  if and only if  $v = 0$ . If  $\alpha$  is a scalar, then

$$\|\alpha v\|^2 = \langle \alpha v, \alpha v \rangle = |\alpha|^2 \langle v, v \rangle = |\alpha|^2 \|v\|^2,$$

so  $\|\alpha v\| = |\alpha| \|v\|$ . □

**Proposition 12.5 (Cauchy–Schwarz inequality)** Let  $v, w \in X$ , where  $X$  is a vector space with an inner product  $\langle \cdot, \cdot \rangle$ . Then

$$|\langle v, w \rangle| \leq \|v\| \|w\|.$$

Equality holds if and only if  $v$  and  $w$  are linearly dependent.

**Proof.** If  $w = 0$ , the inequality is immediate. Otherwise, set

$$\alpha = \frac{\langle v, w \rangle}{\|w\|^2}.$$

Then

$$0 \leq \|v - \alpha w\|^2 = \|v\|^2 - \frac{|\langle v, w \rangle|^2}{\|w\|^2},$$

which proves the inequality. Equality holds precisely when  $\|v - \alpha w\| = 0$ , or equivalently  $v = \alpha w$ . Together with the case where one vector is zero, this is exactly linear dependence. Conversely, direct substitution shows that linearly dependent vectors give equality. □

**Proposition 12.6** Let  $X$  be a vector space with inner product  $\langle \cdot, \cdot \rangle$ . Then  $\| \cdot \| = \sqrt{\langle \cdot, \cdot \rangle}$  is indeed a norm on  $X$ .

**Proof.** Nonnegativity and definiteness follow from the inner product axioms:

$$\|x\| = \sqrt{\langle x, x \rangle} \geq 0 \quad \text{and} \quad \|x\| = 0 \iff \langle x, x \rangle = 0 \iff x = 0.$$

For homogeneity,

$$\|\alpha x\|^2 = \langle \alpha x, \alpha x \rangle = |\alpha|^2 \langle x, x \rangle = |\alpha|^2 \|x\|^2,$$

so  $\|\alpha x\| = |\alpha| \|x\|$ . Triangle inequality follows from **Cauchy–Schwarz**:

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2. \end{aligned}$$

Hence  $\|x + y\| \leq \|x\| + \|y\|$ . □

**Definition 12.7** Let  $X$  be a vector space with inner product  $\langle \cdot, \cdot \rangle$  and  $\| \cdot \| = \sqrt{\langle \cdot, \cdot \rangle}$  as before. If  $X$  is complete with respect to  $\| \cdot \|$ , we call  $X$  a **Hilbert space**.

**Example 12.8** Let  $E$  be a Lebesgue measurable subset of  $\mathbb{R}$  with  $\lambda(E) > 0$ . Then  $L^2(E)$  is a Hilbert space with inner product

$$\langle f, g \rangle = \int_E f \bar{g}$$

for  $f, g \in L^2(E)$ . Note that the norm induced by this inner product is the  $L^2$  norm:

$$\sqrt{\langle f, f \rangle} = \left( \int_E |f|^2 \right)^{1/2} = \|f\|_2.$$

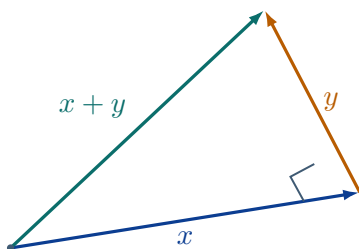
**Fact 12.9**  $L^p$  is a Hilbert space only when  $p = 2$ .

**Definition 12.10** Two vectors  $v, w$  in a Hilbert space  $H$  are called **orthogonal** if  $\langle v, w \rangle = 0$ ; we write  $v \perp w$ . A set  $S \subseteq H$  is called an **orthogonal set** if for every distinct pair  $v, w \in S$ ,  $v \perp w$ . A set  $S \subseteq H$  is called **orthonormal** if  $S$  is orthogonal and  $\|v\| = 1$  for every  $v \in S$ .

**Theorem 12.11 (Pythagorean theorem)** Let  $x, y \in \mathcal{H}$  with  $x \perp y$ . Then  $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ . More generally, if  $\{x_n\}_{n=1}^N$  is an orthogonal set, then

$$\left\| \sum_{n=1}^N x_n \right\|^2 = \sum_{n=1}^N \|x_n\|^2.$$

The familiar right triangle survives in any Hilbert space, as summarized in Figure 7; only the algebraic notions of length and perpendicularity are needed.



$x \perp y$  makes the squared lengths add

FIGURE 7

**Proof.** Observe that if  $x \perp y$ , then  $\langle x, y \rangle = 0$  and  $\langle y, x \rangle = 0$ . Therefore,

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle = \|x\|^2 + \|y\|^2.$$

The general case follows by induction on  $N$ . □

**Proposition 12.12** If  $\{x_n\}_{n=1}^\infty$  is an orthogonal set in a Hilbert space  $\mathcal{H}$ , then

$$\sum_{n=1}^{\infty} x_n$$

is convergent if and only if

$$\sum_{n=1}^{\infty} \|x_n\|^2 < \infty.$$

**Proof.** *Forward implication.* Assume that

$$\sum_{n=1}^{\infty} x_n$$

converges in  $\mathcal{H}$ . Let

$$S_n := \sum_{i=1}^n x_i.$$

Then  $\{S_n\}$  is Cauchy in  $\mathcal{H}$ . For  $n > m$ ,

$$\|S_n - S_m\|^2 = \left\| \sum_{i=m+1}^n x_i \right\|^2 = \sum_{i=m+1}^n \|x_i\|^2,$$

where the last equality is the finite [Pythagorean theorem](#) for orthogonal families. Since  $\|S_n - S_m\| \rightarrow 0$  as  $m, n \rightarrow \infty$ , the tails  $\sum_{i=m+1}^n \|x_i\|^2$  tend to 0. Hence the scalar series

$$\sum_{i=1}^{\infty} \|x_i\|^2$$

is Cauchy in  $\mathbb{R}$ , and therefore convergent.

*Reverse implication.* Assume

$$\sum_{i=1}^{\infty} \|x_i\|^2 < \infty.$$

With the same partial sums  $S_n$ , for  $n > m$  we again have

$$\|S_n - S_m\|^2 = \sum_{i=m+1}^n \|x_i\|^2.$$

Because the numerical series converges, its tails tend to 0, so  $\|S_n - S_m\| \rightarrow 0$ . Thus  $\{S_n\}$  is Cauchy

in  $\mathcal{H}$ . Since  $\mathcal{H}$  is complete,  $S_n \rightarrow x$  for some  $x \in \mathcal{H}$ , so

$$\sum_{n=1}^{\infty} x_n$$

converges.

Finally, if  $S_n \rightarrow x$ , then by continuity of the norm,

$$\|x\|^2 = \lim_{n \rightarrow \infty} \|S_n\|^2 = \lim_{n \rightarrow \infty} \sum_{i=1}^n \|x_i\|^2 = \sum_{i=1}^{\infty} \|x_i\|^2.$$

□

**Definition 12.13** We define the space  $\ell^2$  to be the set of all sequences  $\beta = (\beta_i)_{i=1}^{\infty}$  of complex numbers such that  $\sum_{i=1}^{\infty} |\beta_i|^2 < \infty$ . In other words,

$$\ell^2 = \{(\beta_i)_{i=1}^{\infty} : \beta_i \in \mathbb{C} \text{ and } \|\beta\|_2^2 = \sum_{i=1}^{\infty} |\beta_i|^2 < \infty\}.$$

**Proposition 12.14** Let  $\{e_n\}_{n=1}^{\infty}$  be an orthonormal set. Let  $\beta \in \ell^2$ . Then there exists  $x \in \mathcal{H}$  such that  $\|x\| = \|\beta\|_2$  and  $\langle x, e_i \rangle = \beta_i$ .

**Proof.** Choose

$$x = \sum_{i=1}^{\infty} \beta_i e_i,$$

which is convergent because

$$\sum_{i=1}^{\infty} |\beta_i|^2 < \infty$$

by [Proposition 12.12](#) and

$$\|x\|^2 = \sum_{i=1}^{\infty} |\beta_i|^2 = \|\beta\|_2^2.$$

For each fixed  $j$ , continuity of the inner product gives

$$\langle x, e_j \rangle = \lim_{N \rightarrow \infty} \left\langle \sum_{i=1}^N \beta_i e_i, e_j \right\rangle = \beta_j.$$

□

## Lecture 13 — Thursday, February 19, 2015

## Orthogonal Series and Bessel's Inequality

**Remark 13.1** There are two useful relationships between Riemann and Lebesgue integration. First, suppose that  $[a, b]$  is a compact interval and that  $f$  is a bounded function on  $[a, b]$ . If  $f$  is Riemann integrable, then  $f \in L^1([a, b])$  and

$$\int_a^b f(x) \, dx = \int_{[a, b]} f.$$

Thus the Riemann and Lebesgue integrals agree in this setting. Second, if  $f$  is nonnegative and Riemann integrable on  $\mathbb{R}$ , possibly as an improper Riemann integral, then  $f \in L^1(\mathbb{R})$  and the Lebesgue integral of  $f$  agrees with its Riemann integral.

Recall that

$$\ell^2 = \{(a_n)_{n=1}^\infty : a_n \in \mathbb{C}, \|a\|_2 < \infty\}$$

and let  $\{x_n\}_{n=1}^\infty \subseteq \mathcal{H}$  be an orthogonal sequence in a Hilbert space  $\mathcal{H}$ . The orthogonal series convergence criterion ([Proposition 12.12](#)) says that

$$\sum_{n=1}^\infty x_n$$

converges if and only if

$$\sum_{n=1}^\infty \|x_n\|^2 < \infty.$$

If  $x = \sum_{n=1}^\infty x_n$ , then

$$\|x\| = \left( \sum_{n=1}^\infty \|x_n\|^2 \right)^{1/2}.$$

**Lemma 13.2** Let  $\eta \in \mathcal{H}$  and define  $T_\eta : \mathcal{H} \rightarrow \mathbb{C}$  by  $\xi \mapsto \langle \xi, \eta \rangle$ . Then  $T_\eta$  is linear and continuous.

**Proof.** Linearity follows from linearity of the inner product in the first variable. To prove continuity, suppose that  $\xi_n \rightarrow \xi$ . Then

$$|T_\eta(\xi_n) - T_\eta(\xi)| = |\langle \xi_n, \eta \rangle - \langle \xi, \eta \rangle| = |\langle \xi_n - \xi, \eta \rangle| \leq \|\xi_n - \xi\| \|\eta\| \rightarrow 0.$$

□

**Proposition 13.3** Let  $\mathcal{H}$  be a Hilbert space, and let  $\{e_i : i \in \mathbb{N}\}$  be an orthonormal subset of  $\mathcal{H}$ . If  $(\beta_i) \in \ell^2$ , then there exists  $x \in \mathcal{H}$  such that  $\langle x, e_i \rangle = \beta_i$  for all  $i$  and

$$\|x\| = \left( \sum_{i=1}^{\infty} |\beta_i|^2 \right)^{1/2} = \|(\beta_i)\|_2.$$

**Proof.** Consider the series

$$\sum_{i=1}^{\infty} \beta_i e_i.$$

Since  $\{e_i\}_{i=1}^{\infty}$  is orthogonal, [Proposition 12.12](#) shows that this series converges, because

$$\sum_{i=1}^{\infty} \|\beta_i e_i\|^2 = \sum_{i=1}^{\infty} |\beta_i|^2 \|e_i\|^2 = \sum_{i=1}^{\infty} |\beta_i|^2 = \|(\beta_i)\|_2^2 < \infty.$$

Set

$$x = \sum_{i=1}^{\infty} \beta_i e_i \in \mathcal{H}$$

so that  $\|x\| = \|(\beta_i)\|_2$ . Let  $j \in \mathbb{N}$  be arbitrary. Since  $T_{e_j}$  is continuous by [Lemma 13.2](#), we have

$$\langle x, e_j \rangle = \left\langle \lim_{N \rightarrow \infty} \sum_{i=1}^N \beta_i e_i, e_j \right\rangle = \lim_{N \rightarrow \infty} \left\langle \sum_{i=1}^N \beta_i e_i, e_j \right\rangle = \lim_{N \rightarrow \infty} \sum_{i=1}^N \beta_i \langle e_i, e_j \rangle = \beta_j.$$

□

**Proposition 13.4 (Bessel's inequality)** Let  $\mathcal{H}$  be a Hilbert space, and let  $\{e_i : i \in \mathbb{N}\}$  be an orthonormal subset of  $\mathcal{H}$ . If  $x \in \mathcal{H}$ , then  $(\langle x, e_i \rangle)_{i \in \mathbb{N}} \in \ell^2$ , and

$$\|(\langle x, e_i \rangle)_{i \in \mathbb{N}}\|_2 \leq \|x\|.$$

**Proof.** Let  $x \in \mathcal{H}$  be arbitrary. It is enough to show that

$$\sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2 < \infty.$$

Fix  $N \in \mathbb{N}$  and set

$$x_N = \sum_{i=1}^N \langle x, e_i \rangle e_i \in \mathcal{H}.$$

Then

$$\langle x - x_N, x_N \rangle = \langle x, x_N \rangle - \langle x_N, x_N \rangle = \langle x, \sum_{i=1}^N \langle x, e_i \rangle e_i \rangle - \sum_{i=1}^N |\langle x, e_i \rangle|^2 = 0.$$

Furthermore,

$$\|x\|^2 = \|(x - x_N) + x_N\|^2 = \|x - x_N\|^2 + \|x_N\|^2,$$

so  $\|x_N\|^2 \leq \|x\|^2$ . That is,

$$\sum_{i=1}^N |\langle x, e_i \rangle|^2 \leq \|x\|^2$$

for every  $N$ . Therefore,

$$\sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2 \leq \|x\|^2,$$

as desired. □

## Orthonormal Bases and Parseval Expansions

**Definition 13.5** Let  $\mathcal{H}$  be a Hilbert space. An orthonormal set  $S \subseteq \mathcal{H}$  is called a **basis** for  $\mathcal{H}$  if  $\overline{\text{span}(S)} = \mathcal{H}$  (i.e.,  $\text{span}(S)$  is dense in  $\mathcal{H}$ ).

**Example 13.6** The space  $\ell^2$  is a vector space: if  $(a_n), (b_n) \in \ell^2$ , then  $(a_n) + (b_n) = (a_n + b_n) \in \ell^2$ , and if  $c \in \mathbb{C}$ , then  $c(a_n) = (ca_n) \in \ell^2$ . It has the inner product

$$\langle (a_n), (b_n) \rangle = \sum_{n=1}^{\infty} a_n \overline{b_n}.$$

The norm  $\|\cdot\|_2$  comes from this inner product:

$$\|(a_n)\|_2 = \sqrt{\langle (a_n), (a_n) \rangle} = \sqrt{\sum_{n=1}^{\infty} |a_n|^2},$$

so  $\ell^2$  is a Hilbert space. For  $i \in \mathbb{N}$ , define  $e_i = (0, 0, \dots, 1, 0, \dots) \in \ell^2$ , with the 1 in the  $i$ th coordinate. Then  $\{e_i : i = 1, 2, \dots\}$  is an orthonormal basis for  $\ell^2$ .

**Fact 13.7** Let  $\mathcal{H}$  be an infinite-dimensional Hilbert space. There is no orthonormal subset  $S$  of  $\mathcal{H}$  such that  $\text{span}(S) = \mathcal{H}$ .

**Proof.** Suppose such an orthonormal set exists. Let  $\{x_n\}_{n=1}^{\infty} \subseteq S$ , and let

$$x = \sum_{n=1}^{\infty} \frac{1}{2^n} x_n,$$

which exists because

$$\sum_{n=1}^{\infty} \left(\frac{1}{2^n}\right)^2 \|x_n\|^2 = \sum_{n=1}^{\infty} \left(\frac{1}{2^n}\right)^2 < \infty.$$

Then  $x \in \mathcal{H}$ , but  $x \notin \text{span}(S)$ . Indeed, if

$$x = \sum_{j=1}^N a_j y_j$$

for some  $y_j \in S$  and  $a_j \in \mathbb{C}$ , then there exists  $i$  such that  $x_i \notin \{y_1, \dots, y_N\}$ . For this  $i$ ,

$$\langle x, x_i \rangle = \left\langle \sum_{n=1}^{\infty} \frac{1}{2^n} x_n, x_i \right\rangle = \frac{1}{2^i}$$

by continuity of  $\langle \cdot, \cdot \rangle$ . On the other hand,

$$\langle x, x_i \rangle = \left\langle \sum_{j=1}^N a_j y_j, x_i \right\rangle = 0,$$

which is a contradiction. □

**Definition 13.8** An orthonormal set  $S$  in a Hilbert space  $\mathcal{H}$  is called **complete** if it has the property that if  $\langle x, e \rangle = 0$  for every  $e \in S$ , then  $x = 0$ .

**Theorem 13.9** Let  $S = \{e_i : i \in \mathbb{N}\}$  be an orthonormal set in a Hilbert space  $\mathcal{H}$ . The following are equivalent:

1.  $S$  is a basis for  $\mathcal{H}$ .
2.  $S$  is complete.
3.  $x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i$  for every  $x \in \mathcal{H}$ .

**Proof.** (1) *implies* (2). Assume that  $S$  is a basis for  $\mathcal{H}$ . Let  $x \in \mathcal{H}$  and suppose  $\langle x, e_i \rangle = 0$  for every  $i \in \mathbb{N}$ . We prove that  $x = 0$ . Let  $\varepsilon > 0$ . Since  $\text{span}(S)$  is dense in  $\mathcal{H}$ , there exists a finite linear combination

$$\sum_{i=1}^N a_i e_i$$

such that

$$\left\| x - \sum_{i=1}^N a_i e_i \right\| < \varepsilon.$$

Then

$$\left\langle x, \sum_{i=1}^N a_i e_i \right\rangle = \sum_{i=1}^N \overline{a_i} \langle x, e_i \rangle = 0.$$

Thus

$$\langle x, x \rangle = \|x\|^2 = \left\langle x, x - \sum_{i=1}^N a_i e_i + \sum_{i=1}^N a_i e_i \right\rangle = \left| \left\langle x, x - \sum_{i=1}^N a_i e_i \right\rangle \right|$$

By the [Cauchy–Schwarz inequality](#),

$$\left| \left\langle x, x - \sum_{i=1}^N a_i e_i \right\rangle \right| \leq \|x\| \left\| x - \sum_{i=1}^N a_i e_i \right\| \leq \varepsilon \|x\|.$$

Then either  $\|x\| = 0$ , in which case  $x = 0$ , or

$$\|x\| \leq \varepsilon.$$

Since  $\varepsilon > 0$  was arbitrary,  $\|x\| = 0$ , and therefore  $x = 0$ . Thus (2) holds.

(2) *implies* (3). Assume that  $S$  is complete. Let  $x \in \mathcal{H}$ . We show that

$$x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i.$$

Since  $x \in \mathcal{H}$ , the series

$$y = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i$$

is convergent by [Bessel's inequality](#). Moreover,

$$\langle x - y, e_j \rangle = \langle x, e_j \rangle - \langle y, e_j \rangle = \langle x, e_j \rangle - \langle x, e_j \rangle = 0.$$

By completeness,  $x - y = 0$ , so  $x = y$ . Hence (3) holds.

(3) *implies* (1). Assume that for every  $x \in \mathcal{H}$ ,

$$x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i.$$

It remains to show that  $\text{span}(S)$  is dense in  $\mathcal{H}$ . For every  $x \in \mathcal{H}$ , we have

$$x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i.$$

Let

$$x_n = \sum_{i=1}^n \langle x, e_i \rangle e_i.$$

The asserted series identity means precisely that its partial sums  $x_n$  converge to  $x$ . Since each  $x_n$  belongs to  $\text{span}(S)$ , this span is dense in  $\mathcal{H}$ , and  $S$  is a basis.  $\square$

## Lecture 14 — Tuesday, February 24, 2015

**Fact 14.1** If  $x_n \rightarrow x$  in a Banach space, then  $\|x_n\| \rightarrow \|x\|$ .

**Proof.** By (reverse) triangle inequality,

$$|\|x_n\| - \|x\|| \leq \|x_n - x\|.$$

If  $x_n \rightarrow x$ , then  $\|x_n - x\| \rightarrow 0$ , so  $\|x_n\| \rightarrow \|x\|$ . □

**Remark 14.2** Let  $S = \{e_n : n \in \mathbb{N}\}$  be an orthonormal basis for  $\mathcal{H}$ . Let  $x \in \mathcal{H}$ . By Theorem 13.9, we have

$$x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i.$$

Then

$$\|x\|^2 = \sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2$$

where the sum is understood as the limit of its partial sums. Thus

$$x_N = \sum_{i=1}^N \langle x, e_i \rangle e_i \rightarrow x$$

where  $\|x_N\| \rightarrow \|x\|$ . But

$$\|x_N\| = \sqrt{\sum_{i=1}^N |\langle x, e_i \rangle|^2}$$

and

$$\|x\| = \lim_{N \rightarrow \infty} \|x_N\| = \sqrt{\sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2}.$$

We now prove that this expansion is unique. Suppose  $(\alpha_i), (\beta_i) \in \ell^2$  and

$$\sum_{i=1}^{\infty} \alpha_i e_i = \sum_{i=1}^{\infty} \beta_i e_i,$$

so

$$\sum_{i=1}^{\infty} (\alpha_i - \beta_i) e_i = 0.$$

Fix  $j \in \mathbb{N}$ . For each  $N \geq j$ , define

$$u_N := \sum_{i=1}^N (\alpha_i - \beta_i) e_i.$$

Then  $u_N \rightarrow 0$  in  $\mathcal{H}$ , and by orthonormality,

$$\langle u_N, e_j \rangle = \alpha_j - \beta_j \text{ for } N \geq j.$$

The map  $T_{e_j}(v) := \langle v, e_j \rangle$  is continuous by [Lemma 13.2](#), so

$$\alpha_j - \beta_j = \lim_{N \rightarrow \infty} \langle u_N, e_j \rangle = \left\langle \lim_{N \rightarrow \infty} u_N, e_j \right\rangle = \langle 0, e_j \rangle = 0.$$

Hence  $\alpha_j = \beta_j$ . Since  $j$  was arbitrary,  $\alpha_i = \beta_i$  for every  $i \in \mathbb{N}$ . Therefore the expansion of  $x$  in the orthonormal basis  $S$  is unique.

## Existence of Orthonormal Bases

### Theorem 14.3

1. Every Hilbert space admits an orthonormal basis.
2. If  $\mathcal{H}$  is a separable Hilbert space, then every orthonormal subset of  $\mathcal{H}$  is countable.

**Proof.** For (1), we apply Zorn's lemma. Let  $\mathcal{O} = \{\text{all orthonormal subsets of } \mathcal{H}\}$ , which is nonempty because it contains the empty set. Partially order  $\mathcal{O}$  by inclusion. If  $\mathcal{C}$  is a chain in  $\mathcal{O}$ , then  $S = \bigcup_{C \in \mathcal{C}} C$  is again an orthonormal set: any two of its elements already belong to a common member of the chain. Hence every chain in  $\mathcal{O}$  has an upper bound in  $\mathcal{O}$ . By Zorn's lemma, there is a maximal element  $S_0 \in \mathcal{O}$ .

It remains to prove that  $S_0$  is an orthonormal basis for  $\mathcal{H}$ . Since  $S_0$  is orthonormal, it suffices to show that it is complete. Let  $x \in \mathcal{H}$  and assume  $\langle x, e \rangle = 0$  for every  $e \in S_0$ . If  $x \neq 0$ , then  $S_0 \cup \{x/\|x\|\}$  is orthonormal, contradicting the maximality of  $S_0$ . Hence  $x = 0$ , so  $S_0$  is complete and therefore is an orthonormal basis.

For (2), let  $S$  be an orthonormal subset of  $\mathcal{H}$ . Since  $\mathcal{H}$  is separable, there is a countable dense subset  $D$  of  $\mathcal{H}$ . For each  $e \in S$ , there exists  $d \in D$  such that  $\|e - d\| < 1/2$ . If  $e, e' \in S$  are distinct, then the reverse triangle inequality gives

$$\|d - d'\| \geq \|e - e'\| - \|e - d\| - \|e' - d'\| > 1 - 1/2 - 1/2 = 0,$$

so  $d \neq d'$ . Thus the map  $e \mapsto d$  is an injection from  $S$  to  $D$ , so  $S$  is countable.  $\square$

**Corollary 14.4** If  $\mathcal{H}$  is an infinite-dimensional separable Hilbert space, then  $\mathcal{H}$  is isometrically isomorphic to  $\ell^2$ .

**Proof.** By Theorem 14.3(1),  $\mathcal{H}$  admits an orthonormal basis  $S$ . By Theorem 14.3(2), every orthonormal subset of a separable Hilbert space is countable, so we may write  $S = \{e_n\}_{n=1}^\infty$ . Define

$$U : \mathcal{H} \rightarrow \ell^2, \quad U(x) := (\langle x, e_n \rangle)_{n=1}^\infty.$$

The map  $U$  is linear by linearity of the inner product in its first variable. Because  $\{e_n\}_{n=1}^\infty$  is an orthonormal basis, Theorem 13.9 gives

$$\|x\|^2 = \sum_{n=1}^\infty |\langle x, e_n \rangle|^2 = \|U(x)\|_2^2,$$

so  $U$  is an isometry.

To prove surjectivity, let  $\beta = (\beta_n)_{n=1}^\infty \in \ell^2$ . Since  $\sum_{n=1}^\infty |\beta_n|^2 < \infty$  and  $\{e_n\}$  is orthonormal, Proposition 12.12 implies that

$$x := \sum_{n=1}^\infty \beta_n e_n$$

converges in  $\mathcal{H}$ . For each fixed  $j$ , continuity of  $v \mapsto \langle v, e_j \rangle$  gives

$$\langle x, e_j \rangle = \left\langle \sum_{n=1}^\infty \beta_n e_n, e_j \right\rangle = \beta_j.$$

Hence  $U(x) = \beta$ , so  $U$  is onto.

Therefore  $U$  is a linear surjective isometry from  $\mathcal{H}$  onto  $\ell^2$ ; that is, an isometric isomorphism.

□

**Example 14.5** Let  $\mathcal{H} = L^2([0, 1])$ , and for  $n \in \mathbb{Z}$  let  $\chi_n(x) = e^{2\pi i n x}$ . Then  $S = \{\chi_n : n \in \mathbb{Z}\}$  is orthonormal in  $L^2([0, 1])$ . To see this, let  $n, m \in \mathbb{Z}$  be arbitrary. Then

$$\langle \chi_n, \chi_m \rangle = \int_0^1 e^{2\pi i n x} e^{-2\pi i m x} dx = \int_0^1 e^{2\pi i (n-m)x} dx.$$

If  $m = n$ , then  $\langle \chi_n, \chi_m \rangle = 1$ . If  $m \neq n$ , then

$$\begin{aligned} \langle \chi_n, \chi_m \rangle &= \int_0^1 (\cos(2\pi(n-m)x) + i \sin(2\pi(n-m)x)) dx \\ &= \frac{\sin(2\pi(n-m)x)}{2\pi(n-m)} \Big|_0^1 + i \frac{-\cos(2\pi(n-m)x)}{2\pi(n-m)} \Big|_0^1 \\ &= 0. \end{aligned}$$

We will show that  $\{\chi_n : n \in \mathbb{Z}\}$  is an orthonormal basis for  $L^2([0, 1])$ . Once this is established, [Theorem 13.9](#) will give, for every  $f \in L^2([0, 1])$ ,

$$f = \sum_{n \in \mathbb{Z}} \langle f, \chi_n \rangle \chi_n,$$

with convergence in  $L^2([0, 1])$ . This means that  $f = \lim_{N \rightarrow \infty} f_N$ , where

$$f_N = \sum_{n=-N}^N \langle f, \chi_n \rangle \chi_n.$$

Then

$$\|f\|_2^2 = \sum_{n=-\infty}^{\infty} |\langle f, \chi_n \rangle|^2 \tag{14.1}$$

(14.1) is called **Parseval's identity**.

**Definition 14.6** For  $f \in L^2([0, 1])$ , the  $n$ th **Fourier coefficient** of  $f$  is

$$\hat{f}(n) := \langle f, \chi_n \rangle = \int_0^1 f(x) e^{-2\pi i n x} dx.$$

The series

$$\sum_{n \in \mathbb{Z}} \hat{f}(n) \chi_n$$

is called the **Fourier series** of  $f$ .

## 4. Fourier Analysis on the Circle

### The Circle Group and Trigonometric Polynomials

Let  $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$  be the circle group. We identify  $\mathbb{T}$  with  $[0, 1)$  (or  $[0, 1]$  with the endpoints 0 and 1 identified), via the map  $\theta \mapsto e^{2\pi i \theta}$ . With addition modulo 1, this identification makes  $[0, 1)$  into a compact group. Let  $dm$  be Lebesgue measure restricted to  $[0, 1)$ ; that is,

$$\int_{\mathbb{T}} f \, dm = \int_0^1 f(x) \, dx$$

for integrable functions on  $\mathbb{T}$ . We use  $\mathbb{T}$  because functions on  $\mathbb{T}$  correspond naturally to 1-periodic functions on  $\mathbb{R}$ .

Figure 8 makes the identification explicit: moving one unit along the parameter interval makes one full turn and returns to the same point of the circle.

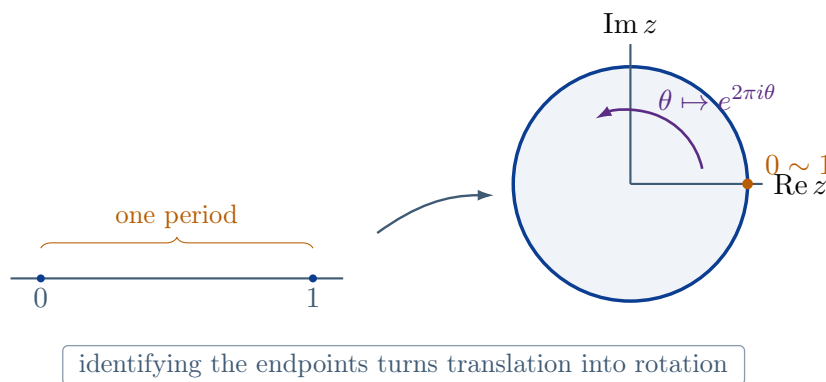


FIGURE 8

**Definition 14.7** For  $1 \leq p < \infty$ , define  $L^p(\mathbb{T})$  to be the space of equivalence classes of

measurable functions on  $[0, 1)$  such that

$$\|f\|_p = \left( \int_0^1 |f|^p dx \right)^{1/p} < \infty.$$

Let  $C(\mathbb{T})$  denote the space of continuous 1-periodic functions on  $[0, 1]$ , and let  $L^\infty(\mathbb{T})$  denote the space of equivalence classes of essentially bounded measurable functions on  $[0, 1]$ .

**Theorem 14.8** Then  $C(\mathbb{T}) \subsetneq L^\infty(\mathbb{T}) \subsetneq L^p(\mathbb{T})$  for  $1 \leq p < \infty$ .

**Exercise 14.9** Prove that the space  $C(\mathbb{T})$  is  $\|\cdot\|_p$ -dense in  $L^p(\mathbb{T})$  for  $1 \leq p < \infty$ .

Recall that  $L^q(\mathbb{T}) \subseteq L^p(\mathbb{T})$  when  $p \leq q$ .

**Definition 14.10** Let  $f \in L^1(\mathbb{T})$ . Then define the  $n$ th **Fourier coefficient** of  $f$  to be

$$\hat{f}(n) := \int_0^1 f(x) e^{-2\pi i n x} dx.$$

The integral

$$\int_0^1 f(x) e^{-2\pi i n x} dx$$

is finite by [Hölder's inequality](#), since  $f \in L^1(\mathbb{T})$  and  $\chi_{-n} = \overline{\chi_n} \in L^\infty(\mathbb{T})$ . The formal series

$$S[f] \sim \sum_{n=-\infty}^{\infty} \hat{f}(n) \chi_n$$

is called the **Fourier series** of  $f$ .

**Question 14.11** This raises natural questions. Does  $S[f]$  converge to  $f$  in any sense? What if  $f$  is nice? What does “nice” even mean? What does anything mean? People living in the 1800s were deeply troubled by these questions.

**Definition 14.12** A function of the form

$$\sum_{n=-N}^N a_n \chi_n$$

is called a **trigonometric polynomial**.  $\text{Trig}(\mathbb{T}) = \{\text{all trigonometric polynomials on } \mathbb{T}\} = \text{span}(S)$ , where  $S = \{\chi_n : n \in \mathbb{Z}\}$ .

**Proposition 14.13** The trigonometric polynomials have the following properties:

1.  $\text{Trig}(\mathbb{T}) \subseteq C(\mathbb{T})$ .
2.  $\text{Trig}(\mathbb{T})$  is an algebra:  $\chi_n \cdot \chi_m = \chi_{n+m} \in \text{span}(S)$ .
3.  $\text{Trig}(\mathbb{T})$  is closed under conjugation:  $\overline{\chi_n} = \chi_{-n}$ .
4.  $\text{Trig}(\mathbb{T})$  separates points.
5.  $\text{Trig}(\mathbb{T})$  is  $\|\cdot\|_\infty$ -dense in  $(C(\mathbb{T}), \|\cdot\|_\infty)$ , and  $\|\cdot\|_p$ -dense in  $L^p(\mathbb{T})$  for every  $1 \leq p < \infty$ .

**Proof.** For (1), let  $N \in \mathbb{N}$  and  $a_{-N}, \dots, a_N \in \mathbb{C}$ . Then

$$\sum_{n=-N}^N a_n \chi_n(x) = \sum_{n=-N}^N a_n e^{2\pi i n x}$$

is continuous as a finite sum of continuous functions.

For (2), let  $N, M \in \mathbb{N}$  and  $a_{-N}, \dots, a_N, b_{-M}, \dots, b_M \in \mathbb{C}$ . Then

$$\left( \sum_{n=-N}^N a_n \chi_n \right) \cdot \left( \sum_{m=-M}^M b_m \chi_m \right) = \sum_{n=-N}^N \sum_{m=-M}^M a_n b_m \chi_{n+m}$$

is a trigonometric polynomial.

For (3), let  $N \in \mathbb{N}$  and  $a_{-N}, \dots, a_N \in \mathbb{C}$ . Then

$$\overline{\sum_{n=-N}^N a_n \chi_n} = \sum_{n=-N}^N \overline{a_n \chi_n} = \sum_{n=-N}^N \overline{a_n} \chi_{-n}$$

is a trigonometric polynomial.

For (4), let  $x, y \in \mathbb{T}$  with  $x \neq y$ . Then  $\chi_1(x) = e^{2\pi i x} \neq e^{2\pi i y} = \chi_1(y)$ , so  $\text{Trig}(\mathbb{T})$  separates points.

For (5), the Stone–Weierstrass theorem implies that  $\text{Trig}(\mathbb{T})$  is  $\|\cdot\|_\infty$ -dense in  $C(\mathbb{T})$ . Since  $\|\cdot\|_p \leq \|\cdot\|_\infty$  on  $\mathbb{T}$ , it follows that  $\text{Trig}(\mathbb{T})$  is also  $\|\cdot\|_p$ -dense in  $L^p(\mathbb{T})$  for  $1 \leq p < \infty$ .

□

**Corollary 14.14**  $S$  is an orthonormal basis for  $L^2(\mathbb{T})$ , where  $S = \{\chi_n : n \in \mathbb{Z}\}$ .

**Proof.** By Proposition 14.13,  $\text{Trig}(\mathbb{T}) = \text{span}(S)$  is dense in  $L^2(\mathbb{T})$ . Since  $S$  is orthonormal, it follows that  $S$  is an orthonormal basis for  $L^2(\mathbb{T})$ . □

**Corollary 14.15 (Parseval’s identity)** For all  $f \in L^2(\mathbb{T})$ ,

$$f = \sum_{n=-\infty}^{\infty} \hat{f}(n) \chi_n$$

with  $\|\cdot\|_2$ -convergence. Consequently,

$$\|f\|_2 = \sqrt{\sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2}$$

**Proof.** By Corollary 14.14,  $S$  is an orthonormal basis for  $L^2(\mathbb{T})$ . By Theorem 13.9, for every  $f \in L^2(\mathbb{T})$ ,

$$f = \sum_{n=-\infty}^{\infty} \langle f, \chi_n \rangle \chi_n = \sum_{n=-\infty}^{\infty} \hat{f}(n) \chi_n$$

with  $\|\cdot\|_2$ -convergence. Moreover,

$$\|f\|_2^2 = \sum_{n=-\infty}^{\infty} |\langle f, \chi_n \rangle|^2 = \sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2.$$

□

## Lecture 15 — Thursday, February 26, 2015

**Definition 15.1** We define

$$\ell^2(\mathbb{Z}) = \{(a_n)_{n \in \mathbb{Z}} : \sum_{n \in \mathbb{Z}} |a_n|^2 < \infty\},$$

with the inner product

$$\langle (a_n), (b_n) \rangle = \sum_{n \in \mathbb{Z}} a_n \overline{b_n}$$

and the norm

$$\|(a_n)\|_2 = \sqrt{\sum_{n \in \mathbb{Z}} |a_n|^2}.$$

The **Fourier transform**  $\mathcal{F} : L^2(\mathbb{T}) \rightarrow \ell^2(\mathbb{Z})$  is given by  $f \mapsto (\hat{f}(n))_{n \in \mathbb{Z}}$ .

**Proposition 15.2** The Fourier transform is an isometric isomorphism of Hilbert spaces:

1.  $\mathcal{F}(f + g) = \mathcal{F}(f) + \mathcal{F}(g)$ .
2.  $\mathcal{F}(cf) = c\mathcal{F}(f)$ .
3.  $\mathcal{F}$  is injective.
4.  $\mathcal{F}$  is surjective.
5.  $\mathcal{F}$  is an isometry.

**Proof.** For (1), let  $n \in \mathbb{Z}$ . Then

$$\begin{aligned} \widehat{f + g}(n) &= \int_0^1 (f(x) + g(x)) e^{-2\pi i n x} dx \\ &= \int_0^1 f(x) e^{-2\pi i n x} dx + \int_0^1 g(x) e^{-2\pi i n x} dx \\ &= \hat{f}(n) + \hat{g}(n). \end{aligned}$$

For (2), the result follows from linearity of the integral.

For (3), suppose that  $f \in L^2(\mathbb{T})$  and  $\hat{f}(n) = 0$  for every  $n$ . Then

$$\|f\|_2^2 = \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 = 0,$$

so  $f = 0$  in  $L^2(\mathbb{T})$ .

For (4), take a sequence  $(a_n)_{n \in \mathbb{Z}} \in \ell^2(\mathbb{Z})$ . The general Hilbert space theory gives

$$f = \sum_{n=-\infty}^{\infty} a_n \chi_n$$

with convergence in  $L^2(\mathbb{T})$ . Continuity of the inner product gives

$$\hat{f}(j) = \langle f, \chi_j \rangle = a_j$$

for every  $j \in \mathbb{Z}$ , so  $\mathcal{F}(f) = (a_n)_{n \in \mathbb{Z}}$ . Finally, (5) follows from Parseval's identity:

$$\|\mathcal{F}(f)\|_2^2 = \sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2 = \|f\|_2^2.$$

□

### Fourier Coefficients on $L^1(\mathbb{T})$

Recall that for  $f \in L^1(\mathbb{T})$  and  $n \in \mathbb{Z}$ , the  $n$ th Fourier coefficient of  $f$  is

$$\hat{f}(n) = \int_0^1 f(x) e^{-2\pi i n x} dx,$$

and the Fourier series of  $f$  is

$$S[f] \sim \sum_{n=-\infty}^{\infty} \hat{f}(n) \chi_n.$$

Note that even when  $f$  is real-valued, the Fourier coefficients  $\hat{f}(n)$  need not be real, since  $\chi_n$  is complex-valued for  $n \neq 0$ .

**Proposition 15.3** The Fourier coefficients of functions in  $L^1(\mathbb{T})$  have the following properties:

1.  $\widehat{f+g}(n) = \hat{f}(n) + \hat{g}(n).$
2.  $\widehat{cf}(n) = c\hat{f}(n).$
3.  $\widehat{\hat{f}}(n) = \overline{\hat{f}(-n)}.$
4. Let  $f \in L^1(\mathbb{T})$ , let  $x \in [0, 1]$ , and define  $f_x(y) := f(y - x) \pmod{1}$ . Then

$$\hat{f}_x(n) = \chi_{-n}(x)\hat{f}(n) = e^{-2\pi i n x}\hat{f}(n).$$

5.  $|\hat{f}(n)| \leq \|f\|_1.$

**Proof.** For (1),

$$\widehat{f+g}(n) = \int_0^1 (f(t) + g(t))e^{-2\pi i n t} dt = \int_0^1 f(t)e^{-2\pi i n t} dt + \int_0^1 g(t)e^{-2\pi i n t} dt = \hat{f}(n) + \hat{g}(n).$$

For (2),

$$\widehat{cf}(n) = \int_0^1 cf(t)e^{-2\pi i n t} dt = c \int_0^1 f(t)e^{-2\pi i n t} dt = c\hat{f}(n).$$

For (3), using conjugation through the integral,

$$\widehat{\hat{f}}(n) = \int_0^1 \overline{f(t)}e^{-2\pi i n t} dt = \overline{\int_0^1 f(t)e^{2\pi i n t} dt} = \overline{\hat{f}(-n)}.$$

For (4),

$$\begin{aligned} \hat{f}_x(n) &= \int_0^1 f(t-x)e^{-2\pi i n t} dt \\ &= \int_0^1 f(t-x)e^{-2\pi i n(t-x+x)} dt \\ &= e^{-2\pi i n x} \int_0^1 f(t-x)e^{-2\pi i n(t-x)} dt. \end{aligned}$$

Now use translation invariance of Lebesgue measure on  $\mathbb{T}$  (equivalently, the change of variable

$u = t - x \bmod 1$ ) to get

$$\int_0^1 f(t - x) e^{-2\pi i n(t-x)} dt = \int_0^1 f(u) e^{-2\pi i n u} du = \hat{f}(n).$$

Hence

$$\hat{f}_x(n) = e^{-2\pi i n x} \hat{f}(n) = \chi_{-n}(x) \hat{f}(n).$$

For (5), since  $|e^{-2\pi i n t}| = 1$ ,

$$|\hat{f}(n)| = \left| \int_0^1 f(t) e^{-2\pi i n t} dt \right| \leq \int_0^1 |f(t)| dt = \|f\|_1.$$

□

**Definition 15.4** The space  $\ell^\infty(\mathbb{Z})$  is

$$\ell^\infty(\mathbb{Z}) = \{(a_n)_{n \in \mathbb{Z}} : a_n \in \mathbb{C}, \|a\|_\infty = \sup |a_n| < \infty\}.$$

Thus, if  $f \in L^1(\mathbb{T})$ , then  $(\hat{f}(n))_{n \in \mathbb{Z}} \in \ell^\infty(\mathbb{Z})$  by Proposition 15.3(5).

**Theorem 15.5** If  $f_k \rightarrow f$  in  $L^1(\mathbb{T})$ , then  $\hat{f}_k(n) \rightarrow \hat{f}(n)$  for all  $n \in \mathbb{Z}$ .

**Proof.**  $|\hat{f}_k(n) - \hat{f}(n)| = |\widehat{f_k - f}(n)| \leq \|f_k - f\|_1 \rightarrow 0$  as  $k \rightarrow \infty$ .

□

**Theorem 15.6 (Riemann–Lebesgue lemma: circle version)** If  $f \in L^1(\mathbb{T})$ , then  $\hat{f}(n) \rightarrow 0$  as  $n \rightarrow \pm\infty$ .

**Proof.** Let  $\varepsilon > 0$ . Let  $p \in \text{Trig}(\mathbb{T})$  such that  $\|f - p\|_1 < \varepsilon$ . Write

$$p = \sum_{n=-N}^N a_n \chi_n.$$

For  $|m| > N$ , orthogonality gives

$$\hat{p}(m) = \langle p, \chi_m \rangle = \sum_{n=-N}^N a_n \langle \chi_n, \chi_m \rangle = 0,$$

and hence  $\hat{p}(n) = 0$  for all  $|n| > N$ . Therefore, for  $|n| > N$ ,

$$|\hat{f}(n)| = |\hat{f}(n) - \hat{p}(n)| \leq \|f - p\|_1 < \varepsilon.$$

Therefore  $\hat{f}(n) \rightarrow 0$  as  $n \rightarrow \pm\infty$ . □

**Remark 15.7** The [Riemann–Lebesgue lemma](#) has the following immediate trigonometric consequences.

1. If  $f \in L^1(\mathbb{T})$ , then

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) \cos(2\pi nx) \, dx = 0,$$

and

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) \sin(2\pi nx) \, dx = 0.$$

2. For  $B \in \mathbb{R}$ ,

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_0^1 f(x) \sin(2\pi nx + B) \, dx &= \lim_{n \rightarrow \infty} \int_0^1 f(x) \sin(2\pi nx) \cos(B) \, dx \\ &\quad + \lim_{n \rightarrow \infty} \int_0^1 f(x) \cos(2\pi nx) \sin(B) \, dx \\ &= 0. \end{aligned}$$

So far, the Fourier transform sends

$$L^1(\mathbb{T}) \rightarrow c_0(\mathbb{Z}), \quad f \mapsto (\hat{f}(n))_{n \in \mathbb{Z}},$$

where

$$c_0(\mathbb{Z}) = \{(a_n)_{n \in \mathbb{Z}} \in \ell^\infty(\mathbb{Z}) : \lim_{n \rightarrow \pm\infty} a_n = 0\}.$$

**Question 15.8** Does the Fourier series of  $f$ ,  $S[f]$ , actually converge to  $f$ ? If so, in what sense? Well? Say something!

**Definition 15.9** For  $n \in \mathbb{N}$ , the **Dirichlet kernel** is

$$D_n := \sum_{k=-n}^n \chi_k.$$

The  $n$ th partial sum of the Fourier series of  $f$  is

$$S_n(f) := \sum_{k=-n}^n \hat{f}(k) \chi_k,$$

and we want to understand whether  $S_n(f)$  converges to  $f$  for sufficiently regular functions.

**Proposition 15.10** For  $f \in L^1(\mathbb{T})$  and  $x \in \mathbb{T}$ ,

$$S_n(f)(x) = f * D_n(x),$$

where  $*$  is **convolution** in  $L^1(\mathbb{T})$ .

**Proof.** We have

$$\begin{aligned} S_n(f)(x) &= \sum_{k=-n}^n \hat{f}(k) \chi_k(x) \\ &= \sum_{k=-n}^n \left( \int_0^1 f(t) e^{-2\pi i k t} dt \right) e^{2\pi i k x} \\ &= \int_0^1 f(t) \left( \sum_{k=-n}^n e^{-2\pi i k t} e^{2\pi i k x} \right) dt \\ &= \int_0^1 f(t) \left( \sum_{k=-n}^n e^{2\pi i k (x-t)} \right) dt \\ &= \int_0^1 f(t) D_n(x-t) dt \\ &= f * D_n(x). \end{aligned}$$

□

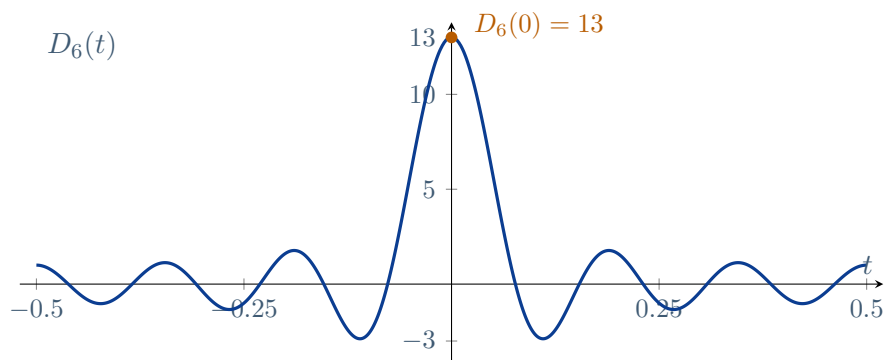
The convolution  $*$  is the operation that turns  $L^1(\mathbb{T})$  into an algebra and makes partial sums accessible through kernels. If  $D_n$  behaved like an identity, then we would expect  $S_n(f)(x) = f * D_n(x)$  to approach  $f(x)$  for continuous  $f$ .

It remains to compute an explicit formula for  $D_n$ .

**Proposition 15.11** For  $n \in \mathbb{N}$  and  $t \in \mathbb{T}$ ,

$$D_n(t) = \begin{cases} \frac{\sin(2\pi(n+1/2)t)}{\sin(\pi t)} & \text{for } t \neq 0, \\ 2n+1, & t = 0. \end{cases}$$

The profile in Figure 9 makes two important features visible:  $D_N$  has a tall central peak, but it also has oscillatory lobes of both signs away from the origin.



the side lobes change sign rather than forming a positive averaging kernel

FIGURE 9

**Proof.** For  $t \neq 0$ , we have

$$\begin{aligned} \sin(\pi t)D_n(t) &= \left( \frac{e^{\pi i t} - e^{-\pi i t}}{2i} \right) \sum_{k=-n}^n e^{2\pi i k t} \\ &= \frac{\sum_{k=-n}^n e^{2\pi i (k+1/2)t} - \sum_{k=-n}^n e^{2\pi i (k-1/2)t}}{2i} \\ &= \frac{e^{2\pi i (n+1/2)t} - e^{-2\pi i (n+1/2)t}}{2i} \\ &= \sin(2\pi(n+1/2)t) \end{aligned}$$

Thus,

$$D_n(t) = \frac{\sin(2\pi(n+1/2)t)}{\sin(\pi t)}, \quad t \neq 0.$$

The value  $D_n(0) = 2n + 1$  is the limit of this quotient as  $t \rightarrow 0$ . □

## Lecture 16 — Tuesday, March 03, 2015

### Dirichlet Kernels and Partial Sums

We now fix  $x \in \mathbb{T}$  and pursue two goals:

1. Give a condition for convergence of  $S_N(f)(x)$  when  $f$  is continuous.
2. Show that Fourier series of continuous functions do not necessarily converge at every  $x \in \mathbb{T}$ .

**Proposition 16.1** Let  $f \in L^1(\mathbb{T})$ , let  $x \in \mathbb{T} \cong [0, 1)$ , and define  $f_x(t) := f(t - x) \pmod{1}$ . Then the following hold:

1.

$$\lim_{n \rightarrow \infty} \int_0^1 f(t) \cos(2\pi nt) \, dt = \lim_{n \rightarrow \infty} \int_0^1 f(t) \sin(2\pi nt) \, dt = 0.$$

2. The Dirichlet kernel is

$$D_N = \sum_{k=-N}^N \chi_k,$$

and

$$D_N(t) = \begin{cases} \frac{\sin(2\pi(N+1/2)t)}{\sin(\pi t)}, & t \neq 0, \\ 2N + 1, & t = 0. \end{cases}$$

3.  $\|f_x\|_1 = \|f\|_1$ .

4.

$$S_N(f)(x) = \sum_{k=-N}^N \hat{f}(k) \chi_k(x) = \int_0^1 f(t) D_N(x - t) \, dt = \int_0^1 f(x + t) D_N(t) \, dt.$$

Since  $D_N$  is even, we may also use  $D_N(t) = D_N(-t)$  in these representations.

**Proof.** For (1), apply the [Riemann–Lebesgue lemma](#) to  $f$  and use

$$\cos(2\pi nt) = \frac{e^{2\pi int} + e^{-2\pi int}}{2} \quad \text{and} \quad \sin(2\pi nt) = \frac{e^{2\pi int} - e^{-2\pi int}}{2i}.$$

Statement (2) was proved in [Proposition 15.11](#).

For (3), translation invariance of Lebesgue measure on  $\mathbb{T}$  gives

$$\|f_x\|_1 = \int_0^1 |f(t-x)| \, dt = \int_0^1 |f(u)| \, du = \|f\|_1.$$

For (4),

$$S_N(f)(x) = \sum_{k=-N}^N \hat{f}(k) \chi_k(x)$$

by definition of the  $N$ th partial Fourier sum. Using

$$D_N(x-t) = \sum_{k=-N}^N \chi_k(x) \chi_{-k}(t),$$

we obtain

$$\begin{aligned} \int_0^1 f(t) D_N(x-t) \, dt &= \sum_{k=-N}^N \chi_k(x) \int_0^1 f(t) \chi_{-k}(t) \, dt \\ &= \sum_{k=-N}^N \hat{f}(k) \chi_k(x) = S_N(f)(x). \end{aligned}$$

Finally, substitute  $u = x + t \pmod{1}$  and use translation invariance:

$$\int_0^1 f(t) D_N(x-t) \, dt = \int_0^1 f(x+t) D_N(t) \, dt.$$

□

Fix  $f \in C(\mathbb{T})$  and  $x \in \mathbb{T}$ . We ask when  $S_N(f)(x)$  has a limit as  $N \rightarrow \infty$ . We calculate

$$\begin{aligned} S_N(f)(x) &= \int_0^1 f(x+t) D_N(t) dt = \int_0^1 f(x+t) \frac{\sin(2\pi(N+1/2)t)}{\sin(\pi t)} dt \\ &= \int_0^1 f(x+t) \frac{\sin(2\pi Nt) \cos(\pi t) + \sin(\pi t) \cos(2\pi Nt)}{\sin(\pi t)} dt \\ &= \int_0^1 f(x+t) \cos(\pi t) \frac{\sin(2\pi Nt)}{\sin(\pi t)} dt + \int_0^1 f(x+t) \cos(2\pi Nt) dt \end{aligned}$$

Since  $f \in L^1(\mathbb{T})$ , the right-hand integral tends to 0 by the [Riemann–Lebesgue lemma](#). Thus the convergence question is reduced to the left-hand integral

$$\int_0^1 f(x+t) \frac{\cos(\pi t)}{\sin(\pi t)} \sin(2\pi Nt) dt = \int_0^1 f_{-x}(t) \frac{\cos(\pi t)}{\sin(\pi t)} \sin(2\pi Nt) dt.$$

**Proposition 16.2 (Riemann localization criterion)** Let  $f \in C(\mathbb{T})$ , let  $x \in \mathbb{T}$ , and fix  $\delta \in (0, 1/2)$ . Then  $S_N(f)(x)$  converges as  $N \rightarrow \infty$  if and only if

$$\lim_{N \rightarrow \infty} 2 \int_0^\delta \frac{f(x+t) + f(x-t)}{2} \frac{\sin(2\pi Nt)}{\pi t} dt$$

exists.

**Proof.** Define

$$J_N := \int_0^1 f_{-x}(t) \frac{\cos(\pi t)}{\sin(\pi t)} \sin(2\pi Nt) dt \quad \text{and} \quad C_N := \int_0^1 f(x+t) \cos(2\pi Nt) dt.$$

By the preceding computation,  $S_N(f)(x) = J_N + C_N$ . Since  $f \in L^1(\mathbb{T})$ , [Riemann–Lebesgue](#) gives  $C_N \rightarrow 0$ , so  $S_N(f)(x)$  converges if and only if  $J_N$  converges.

Split

$$J_N = I_N^{(1)} + I_N^{(2)} + I_N^{(3)},$$

where the terms are the integrals over  $[0, \delta]$ ,  $[\delta, 1 - \delta]$ , and  $[1 - \delta, 1]$ , respectively.

For the middle term, let

$$h_\delta(t) := f_{-x}(t) \frac{\cos(\pi t)}{\sin(\pi t)} \mathbf{1}_{[\delta, 1-\delta]}(t).$$

Then  $h_\delta \in L^1(\mathbb{T})$ , and

$$I_N^{(2)} = \int_0^1 h_\delta(t) \sin(2\pi Nt) dt \rightarrow 0$$

by [Riemann–Lebesgue](#).

For  $I_N^{(3)}$ , use  $t = 1 - s$  and periodicity on  $\mathbb{T}$ :

$$I_N^{(3)} = \int_0^\delta f_{-x}(1-s) \frac{\cos(\pi(1-s))}{\sin(\pi(1-s))} \sin(2\pi N(1-s)) ds = \int_0^\delta f_{-x}(-s) \frac{\cos(\pi s)}{\sin(\pi s)} \sin(2\pi Ns) ds.$$

Hence

$$I_N^{(1)} + I_N^{(3)} = 2 \int_0^\delta A_x(t) \frac{\cos(\pi t)}{\sin(\pi t)} \sin(2\pi Nt) dt, \quad A_x(t) := \frac{f(x+t) + f(x-t)}{2}.$$

Define

$$g_\delta(t) := 2A_x(t) \left( \frac{1}{\pi t} - \frac{\cos(\pi t)}{\sin(\pi t)} \right) \mathbb{1}_{(0,\delta)}(t),$$

extended by 0 outside  $(0, \delta)$ . Since  $f$  is continuous,  $A_x$  is bounded. Also,

$$\frac{1}{\pi t} - \frac{\cos(\pi t)}{\sin(\pi t)} \rightarrow 0 \quad (t \downarrow 0),$$

so  $g_\delta \in L^1(\mathbb{T})$ . Therefore

$$\int_0^1 g_\delta(t) \sin(2\pi Nt) dt \rightarrow 0$$

by [Riemann–Lebesgue](#). Consequently, the convergence of

$$2 \int_0^\delta A_x(t) \frac{\cos(\pi t)}{\sin(\pi t)} \sin(2\pi Nt) dt$$

is equivalent to the convergence of

$$2 \int_0^\delta A_x(t) \frac{\sin(2\pi Nt)}{\pi t} dt.$$

Since  $I_N^{(2)} \rightarrow 0$ , this is also equivalent to the convergence of  $J_N$ , hence equivalent to the convergence of  $S_N(f)(x)$ .  $\square$

The [Riemann localization theorem](#) states that the convergence of  $S_N(f)(x)$  when  $N \rightarrow \infty$  depends

only on the behavior of  $f$  near  $x$ .

Our updated goal is to show that there exists  $f \in C(\mathbb{T})$  such that  $S_N(f)(x)$  does not converge.

## Divergence and the Uniform Boundedness Principle

Let  $(X, \|\cdot\|_X)$  and  $(Y, \|\cdot\|_Y)$  be Banach spaces over a field  $\mathbb{K}$ . Recall that a linear operator  $T : X \rightarrow Y$  is a map satisfying  $T(\alpha x + y) = \alpha T(x) + T(y)$  for all  $\alpha \in \mathbb{K}$  and  $x, y \in X$ .

**Definition 16.3** The **operator norm** of a linear operator  $T : X \rightarrow Y$  is

$$\|T\|_{\text{op}} := \sup_{\|x\|_X=1} \|Tx\|_Y.$$

An operator  $T$  with  $\|T\|_{\text{op}} < \infty$  is called a **bounded linear operator**.

If  $T$  is bounded, then  $\|Tx\|_Y \leq \|T\|_{\text{op}}\|x\|_X$ . Also, every linear operator satisfies  $T(0) = 0$ .

**Theorem 16.4** A linear operator  $T : X \rightarrow Y$  is continuous if and only if it is bounded.

**Proof.** *Forward implication.* Suppose that  $T$  is bounded, so  $\|T\|_{\text{op}} < \infty$ . If  $x_n \rightarrow x$  in  $X$ , then

$$\|T(x_n) - T(x)\|_Y = \|T(x_n - x)\|_Y \leq \|T\|_{\text{op}}\|x_n - x\|_X \rightarrow 0.$$

Thus  $T$  is continuous.

*Reverse implication.* Suppose that  $T$  is continuous, in particular at 0. Then there exists  $\delta > 0$  such that  $\|x\|_X < \delta$  implies  $\|Tx\|_Y \leq 1$ . Hence

$$\|T\|_{\text{op}} = \sup_{\|x\|_X=1} \frac{2}{\delta} \left\| T\left(\frac{\delta}{2}x\right) \right\|_Y \leq \frac{2}{\delta}.$$

□

**Theorem 16.5** Fix  $N \in \mathbb{N}$ . Define  $S_N : L^1(\mathbb{T}) \rightarrow L^1(\mathbb{T})$  by

$$f \mapsto S_N(f) = \sum_{k=-N}^N \hat{f}(k) \chi_k.$$

Then

$$\|S_N\|_{\text{op}} = \|D_N\|_1.$$

**Proof.** The operator  $S_N$  is a continuous (or equivalently, bounded, by [Theorem 16.4](#)), linear operator on  $L^1(\mathbb{T})$ . For  $f \in L^1(\mathbb{T})$ ,

$$\begin{aligned} \|S_N(f)\|_1 &= \int_0^1 |S_N(f)(x)| \, dx \\ &= \int_0^1 \left| \int_0^1 f(t) D_N(x-t) \, dt \right| \, dx \\ &\leq \int_0^1 \int_0^1 |f(t)| \cdot |D_N(x-t)| \, dt \, dx \\ &= \int_0^1 |f(t)| \left( \int_0^1 |D_N(x-t)| \, dx \right) \, dt \\ &= \|f\|_1 \|D_N\|_1, \end{aligned}$$

where the integrals were interchanged via [Tonelli's theorem](#). Thus,

$$\|S_N\|_{\text{op}} \leq \|D_N\|_1.$$

## Lecture 17 — Thursday, March 05, 2015

**Proof continued.** We now prove the reverse inequality  $\|S_N\|_{\text{op}} \geq \|D_N\|_1$ . It is enough to find functions  $f_k \in L^1(\mathbb{T})$  such that  $\|f_k\|_1 = 1$  and  $S_N(f_k) \rightarrow D_N$  in the  $L^1$ -norm. Indeed, for every  $k$ ,

$$\|S_N\|_{\text{op}} \geq \|S_N(f_k)\|_1.$$

Letting  $k \rightarrow \infty$  gives  $\|S_N\|_{\text{op}} \geq \|D_N\|_1$ , which completes the proof of equality.

For the approximation, let  $A_k = [0, 1/k] \cup [1 - 1/k, 1]$  and  $f_k = (k/2)\mathbb{1}_{A_k}$ . Then

$$\|f_k\|_1 = \int_{[0,1/k] \cup [1-1/k,1]} \frac{k}{2} dx = 1.$$

It remains to show convergence.

$$\begin{aligned} \|S_N(f_k) - D_N\|_1 &= \int_0^1 |S_N(f_k)(x) - D_N(x)| dx \\ &= \int_0^1 \left| \int_0^1 f_k(t) D_N(x-t) dt - D_N(x) \right| dx \\ &= \int_0^1 \left| \int_0^1 f_k(t) (D_N(x-t) - D_N(x)) dt \right| dx \\ &\leq \int_0^1 \int_0^1 |f_k(t)| \cdot |D_N(x-t) - D_N(x)| dt dx. \end{aligned}$$

Since  $D_N$  is uniformly continuous and 1-periodic, for sufficiently large  $k$  we have  $|D_N(x-t) - D_N(x)| < \varepsilon$  for all  $x \in [0, 1]$  and all  $t \in A_k$ . For such  $k$ ,

$$\begin{aligned} \|S_N(f_k) - D_N\|_1 &= \int_0^1 \int_{A_k} |f_k(t)| \cdot |D_N(x-t) - D_N(x)| dt dx \\ &\leq \int_0^1 \|f_k\|_1 \cdot \varepsilon \\ &= \varepsilon \cdot \|f_k\|_1 \\ &= \varepsilon. \end{aligned}$$

Since this holds for every  $\varepsilon > 0$ , we have  $\|S_N(f_k) - D_N\|_1 \rightarrow 0$ . □

Next, recall the [uniform boundedness principle](#) from PMATH 351:

**Theorem 17.1 (Uniform boundedness principle)** Let  $X$  and  $Y$  be Banach spaces, and let  $\mathcal{F}$  be a collection of bounded linear operators from  $X$  to  $Y$ . If

$$\sup_{T \in \mathcal{F}} \|Tx\|_Y < \infty$$

for every  $x \in X$ , then

$$\sup_{T \in \mathcal{F}} \|T\|_{\text{op}} < \infty.$$

Equivalently, if  $\sup_{T \in \mathcal{F}} \|T\|_{\text{op}} = \infty$ , then there exists  $x \in X$  such that  $\sup_{T \in \mathcal{F}} \|Tx\|_Y = \infty$ .

**Theorem 17.2** There exists  $f \in L^1(\mathbb{T})$  for which the Fourier series is not  $L^1$ -convergent. Equivalently, the sequence  $S_N(f)$  need not converge in  $L^1(\mathbb{T})$ .

**Proof.** Take  $X = L^1(\mathbb{T})$ ,  $Y = L^1(\mathbb{T})$ , and

$$\mathcal{F} = \{S_N : L^1(\mathbb{T}) \rightarrow L^1(\mathbb{T}) : N \in \mathbb{N}\}.$$

We have  $\|S_N\|_{\text{op}} = \|D_N\|_1$  by [Theorem 16.5](#). We claim that there is a constant  $C > 0$  such that  $\|D_N\|_1 \geq C \log N$ . Assuming this claim, we have

$$\sup_N \|S_N\|_{\text{op}} \geq \sup_N C \log N = \infty.$$

By the [uniform boundedness principle](#), there exists  $f \in L^1(\mathbb{T})$  such that  $\sup_N \|S_N(f)\|_1 = \infty$ . Hence  $\{S_N(f)\}_{N \in \mathbb{N}}$  cannot be convergent in  $L^1(\mathbb{T})$ .

It remains to prove the claim. Fix  $N \in \mathbb{N}$ . Then

$$\begin{aligned} \|D_N\|_1 &= \int_0^1 \left| \frac{\sin(\pi(2N+1)t)}{\sin(\pi t)} \right| dt \\ &\geq \sum_{j=0}^{4N+1} \frac{2(2N+1)}{\pi(j+1)} \int_{\frac{j}{2(2N+1)}}^{\frac{j+1}{2(2N+1)}} |\sin(\pi(2N+1)t)| dt \\ &= \frac{2}{\pi^2} \sum_{j=0}^{4N+1} \frac{1}{j+1}. \end{aligned}$$

The harmonic sum is bounded below by  $\log(4N+2)$ , so  $\|D_N\|_1 \geq C \log N$  for some absolute constant  $C > 0$  and every  $N \geq 2$ .  $\square$

**Corollary 17.3** There exists  $C > 0$  such that

$$\|D_N\|_1 \geq C \log N$$

for every  $N \geq 2$ . In particular,  $\|D_N\|_1 \rightarrow \infty$ .

**Proof.** This estimate was proved at the end of the proof of [Theorem 17.2](#). □

The following lemma will be used to transfer divergence from  $L^1(\mathbb{T})$  to continuous functions.

**Lemma 17.4** There exists  $g_N \in C(\mathbb{T})$  such that  $\|g_N\|_\infty = 1$  and  $|S_N(g_N)(0)| \geq \|D_N\|_1/2$ .

**Proof.** The function

$$D_N(t) = \frac{\sin(2\pi(N + 1/2)t)}{\sin(\pi t)}$$

is real-valued, and  $\text{sgn}(D_N)$  takes the values  $\pm 1$  except at finitely many points. Define a continuous  $g_N$  so that  $|g_N| \leq 1$  everywhere and  $\|g_N\|_\infty = 1$ , and so that  $g_N$  agrees with  $\text{sgn}(D_N)$  outside small intervals around the discontinuities of  $\text{sgn}(D_N)$ . Since  $D_N \in L^1(\mathbb{T})$ , absolute continuity of the integral lets us choose the union  $U$  of these intervals so that

$$\int_U |D_N(t)| dt < \varepsilon.$$

Then

$$\begin{aligned} |S_N(g_N)(0)| &= \left| \int_0^1 g_N(t) D_N(t) dt \right| \\ &= \left| \int_U g_N(t) D_N(t) dt + \int_{[0,1] \setminus U} |D_N(t)| dt \right|. \end{aligned}$$

Since  $|g_N(t)D_N(t)| \leq |D_N(t)|$  for all  $t$ , we have

$$\begin{aligned} |S_N(g_N)(0)| &\geq \|D_N\|_1 - \int_U |D_N(t)| dt - \left| \int_U g_N(t) D_N(t) dt \right| \\ &\geq \|D_N\|_1 - 2\varepsilon. \end{aligned}$$

Taking  $\varepsilon < \|D_N\|_1/4$  gives  $|S_N(g_N)(0)| \geq \|D_N\|_1/2$ . □

## Lecture 18 — Tuesday, March 10, 2015

Consider the contrapositive form of the [uniform boundedness principle](#): if  $X$  and  $Y$  are Banach spaces and  $\mathcal{F}$  is a family of bounded linear operators from  $X$  to  $Y$  with  $\sup_{T \in \mathcal{F}} \|T\|_{\text{op}} = \infty$ , then there exists  $x \in X$  such that  $\sup_{T \in \mathcal{F}} \|Tx\|_Y = \infty$ . Recall that we constructed functions  $g_n \in C(\mathbb{T})$  with  $\|g_n\|_{\infty} = 1$  and  $|S_n(g_n)(0)| \geq c \log n$  for some  $c > 0$ .

**Theorem 18.1** There exists  $f \in C(\mathbb{T})$  such that  $S_n(f)(0)$  does not converge.

**Proof.** Take  $X = C(\mathbb{T})$  and  $Y = \mathbb{C}$ . Let  $\mathcal{F} = \{T_n : C(\mathbb{T}) \rightarrow \mathbb{C}\}$  where

$$T_n(f) = S_n(f)(0) = \sum_{k=-n}^n \hat{f}(k) \chi_k(0).$$

For every  $n \in \mathbb{N}$ ,  $T_n$  is bounded, because

$$\|T_n\|_{\text{op}} = \sup_{\|f\|_{\infty}=1} |T_n(f)| = \sup_{\|f\|_{\infty}=1} \left| \int_0^1 f(t) D_n(-t) dt \right| \leq \sup_{\|f\|_{\infty}=1} \|f\|_{\infty} \|D_n\|_1 = \|D_n\|_1 < \infty$$

by [Hölder's inequality](#). Using the functions  $g_n \in C(\mathbb{T})$  constructed in [Lemma 17.4](#), we have

$$\|T_n\|_{\text{op}} \geq |T_n(g_n)| = |S_n(g_n)(0)| \geq c \log(n).$$

Thus  $\|T_n\|_{\text{op}} \rightarrow \infty$  as  $n \rightarrow \infty$ . By the [uniform boundedness principle](#), there exists  $f \in C(\mathbb{T})$  such that

$$\sup_n |T_n(f)| = \infty.$$

Since  $T_n(f) = S_n(f)(0)$ , the sequence  $S_n(f)(0)$  is unbounded and therefore divergent.  $\square$

**Remark 18.2** We can, of course, replace 0 by any point of  $\mathbb{T}$  in [Theorem 18.1](#), since the Fourier partial sums are translation invariant.

## Convolutions

We will use the following results on iterated integrals. **Tonelli's theorem** says that if  $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$  is nonnegative and measurable, then

$$\int_0^1 \int_0^1 f(x, y) \, dy \, dx = \int_0^1 \int_0^1 f(x, y) \, dx \, dy.$$

**Fubini's theorem** says that if  $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$  is integrable, then

$$\int_0^1 \int_0^1 f(x, y) \, dx \, dy = \int_0^1 \int_0^1 f(x, y) \, dy \, dx.$$

**Definition 18.3** For  $f, g \in L^1(\mathbb{T})$ , we define

$$(f * g)(x) := \int_0^1 f(t)g(x - t) \, dt,$$

and call the function  $f * g$  the **convolution** of  $f$  and  $g$ .

When  $g$  is a localized, smooth averaging kernel, the operation can be read as a moving weighted average of  $f$ , as in [Figure 10](#). This is the intuition behind the summability kernels introduced later.

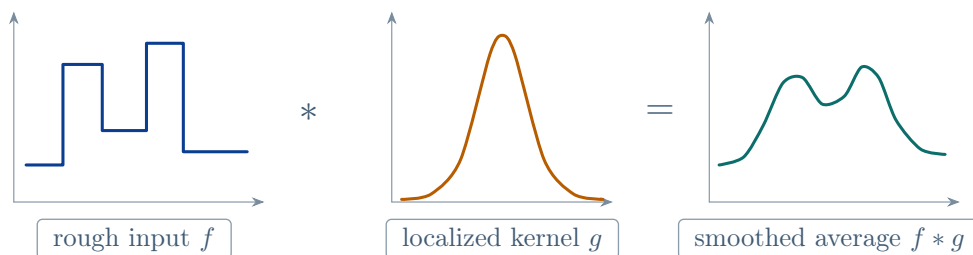


FIGURE 10

**Proposition 18.4** Let  $f, g \in L^1(\mathbb{T})$ . The convolution has the following properties:

1.  $f * g \in L^1(\mathbb{T})$  and

$$\|f * g\|_1 \leq \|f\|_1 \|g\|_1.$$

2.  $f * g = g * f$  almost everywhere.

3. If  $g \in L^\infty(\mathbb{T})$ , then  $f * g \in L^\infty(\mathbb{T})$  and

$$\|f * g\|_\infty \leq \|f\|_1 \|g\|_\infty.$$

4. The partial sums satisfy

$$S_n(f) = f * D_n.$$

5. If  $g = \chi_k$  for some  $k \in \mathbb{Z}$ , then

$$f * g(x) = \hat{f}(k) \chi_k(x).$$

6. For every  $n \in \mathbb{Z}$ ,

$$\widehat{f * g}(n) = \hat{f}(n) \hat{g}(n).$$

**Proof.** For (1), by [Tonelli's theorem](#),

$$\begin{aligned} \|f * g\|_1 &= \int_0^1 \left| \int_0^1 f(t) g(x-t) dt \right| dx \\ &\leq \int_0^1 \int_0^1 |f(t)| |g(x-t)| dt dx \\ &= \int_0^1 |f(t)| \left( \int_0^1 |g(x-t)| dx \right) dt \\ &= \int_0^1 |f(t)| \|g\|_1 dt = \|f\|_1 \|g\|_1 < \infty. \end{aligned}$$

Hence  $f * g \in L^1(\mathbb{T})$ .

For (2), for almost every  $x$ ,

$$(g * f)(x) = \int_0^1 g(t) f(x-t) dt.$$

Set  $u = x - t \pmod{1}$ . By translation invariance of Lebesgue measure on  $\mathbb{T}$ ,

$$(g * f)(x) = \int_0^1 g(x-u) f(u) du = \int_0^1 f(u) g(x-u) du = (f * g)(x).$$

For (3), for almost every  $x$ ,

$$|f * g(x)| = \left| \int_0^1 f(t)g(x-t) dt \right| \leq \int_0^1 |f(t)| |g(x-t)| dt \leq \|g\|_\infty \|f\|_1.$$

Taking the essential supremum in  $x$  gives the claimed norm bound.

For (4), from the kernel representation of partial sums,

$$S_n(f)(x) = \int_0^1 f(t)D_n(x-t) dt = (f * D_n)(x).$$

For (5), if  $g = \chi_k$ , then

$$\begin{aligned} (f * \chi_k)(x) &= \int_0^1 f(t)\chi_k(x-t) dt \\ &= \int_0^1 f(t)e^{2\pi i k(x-t)} dt \\ &= e^{2\pi i kx} \int_0^1 f(t)e^{-2\pi i kt} dt \\ &= \chi_k(x)\hat{f}(k). \end{aligned}$$

For (6), by Fubini's theorem,

$$\begin{aligned} \widehat{f * g}(n) &= \int_0^1 (f * g)(x)\chi_{-n}(x) dx \\ &= \int_0^1 \int_0^1 f(t)g(x-t)\chi_{-n}(x) dt dx \\ &= \int_0^1 f(t) \left( \int_0^1 g(x-t)\chi_{-n}(x) dx \right) dt. \end{aligned}$$

With  $u = x - t \pmod{1}$ ,

$$\int_0^1 g(x-t)\chi_{-n}(x) dx = \int_0^1 g(u)\chi_{-n}(u+t) du = \chi_{-n}(t) \int_0^1 g(u)\chi_{-n}(u) du = \chi_{-n}(t)\hat{g}(n).$$

Therefore,

$$\widehat{f * g}(n) = \hat{g}(n) \int_0^1 f(t)\chi_{-n}(t) dt = \hat{g}(n)\hat{f}(n).$$

□

**Corollary 18.5**  $L^1(\mathbb{T})$  does not have an identity for convolution; that is, there is no  $g \in L^1(\mathbb{T})$  satisfying  $f * g = f$  for all  $f \in L^1(\mathbb{T})$ .

**Proof.** Suppose there exists  $g \in L^1(\mathbb{T})$  with  $f * g = f$  for every  $f \in L^1(\mathbb{T})$ . Then

$$\widehat{f * g}(n) = \hat{f}(n)$$

for every  $n$ , and hence  $\hat{f}(n)\hat{g}(n) = \hat{f}(n)$  for every  $n$ . Taking  $f = \chi_n$  gives  $\hat{f}(n) = 1$ , so  $\hat{g}(n) = 1$  for all  $n \in \mathbb{Z}$ . This contradicts the [Riemann–Lebesgue lemma](#), since  $\hat{g}(n) \not\rightarrow 0$ .  $\square$

**Definition 18.6** The **Fourier transform** on the circle is the map  $\mathcal{F} : L^1(\mathbb{T}) \rightarrow c_0(\mathbb{Z})$  given by

$$\mathcal{F}(f) = (\hat{f}(n))_{n \in \mathbb{Z}}.$$

On  $L^1(\mathbb{T})$ , addition, multiplication, and scalar multiplication are defined by

$$(f + g)(x) = f(x) + g(x), \quad (f * g)(x) = \int_0^1 f(t)g(x - t) dt, \quad \text{and} \quad (cf)(x) = cf(x),$$

and the norm is

$$\|f\|_1 = \int_0^1 |f(x)| dx.$$

On  $c_0(\mathbb{Z})$ , addition, multiplication, and scalar multiplication are defined pointwise: for  $a = (a_n)$  and  $b = (b_n)$  in  $c_0(\mathbb{Z})$ ,

$$a + b = (a_n + b_n), \quad ab = (a_n b_n), \quad \text{and} \quad ca = (ca_n),$$

and the norm is

$$\|a\|_\infty = \sup_n |a_n|.$$

With these operations, the Fourier transform satisfies

$$\mathcal{F}(f + g) = \mathcal{F}(f) + \mathcal{F}(g) \quad \text{and} \quad \mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g).$$

Moreover,  $\|\mathcal{F}(f)\|_\infty \leq \|f\|_1$ , so  $\mathcal{F}$  is continuous. Thus  $\mathcal{F}$  is a linear homomorphism of algebras.

## Summability Kernels and the Fejér Kernel

**Definition 18.7** A sequence  $\{g_n\} \subset C(\mathbb{T})$  is called a **summability kernel** if it satisfies the following three properties:

1.  $\int_0^1 g_n(t) dt = 1$  for every  $n$ .
2. There exists  $M > 0$  such that  $\|g_n\|_1 \leq M$  for every  $n$ .
3. For every  $\delta$  such that  $0 < \delta < 1/2$ ,

$$\int_{\delta}^{1-\delta} |g_n(t)| dt \longrightarrow 0 \text{ as } n \rightarrow \infty.$$

If  $g_n(t) \geq 0$  for every  $n \in \mathbb{N}$  and  $t \in \mathbb{T}$ , then  $\{g_n\}$  is a **positive summability kernel**.

Figure 11 compares two summability kernels,  $g_{n_1}$  in blue and  $g_{n_2}$  in orange, on  $\mathbb{T}$ , where  $n_1 < n_2$ .

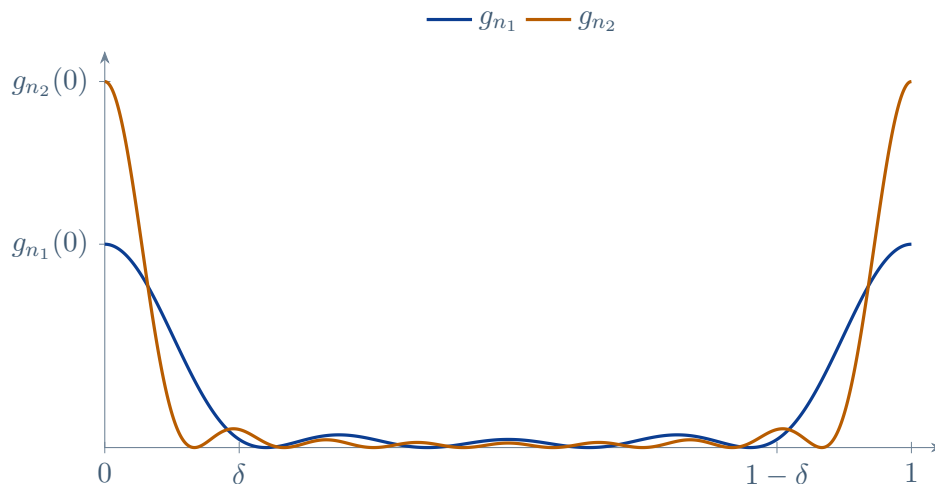


FIGURE 11

**Example 18.8** The functions

$$K_n = \frac{n}{2} \mathbf{1}_{[0,1/n] \cup [1-1/n,1]}$$

satisfy the analytic approximate-identity conditions in the definition, except for continuity. In the proof of [Theorem 16.5](#), they served as an approximate identity to show that  $S_n(K_n) \rightarrow D_n$  in the  $L^1$  norm.

## Lecture 19 — Thursday, March 12, 2015

Recall that  $\mathcal{F} : L^1(\mathbb{T}) \rightarrow c_0(\mathbb{Z})$  defined by  $f \mapsto \hat{f} = (\hat{f}(n))_{n \in \mathbb{Z}}$  is the Fourier transform of  $f$ . Multiplication on  $L^1(\mathbb{T})$  is handled through convolution:

$$f * g(x) = \int_0^1 f(t)g(x-t) dt,$$

with  $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$ . The map  $\mathcal{F}$  is injective but not surjective.

**Question 19.1** Can  $f$  be reconstructed from  $\{\hat{f}(n)\}_{n \in \mathbb{Z}}$ ?

One strategy is to consider the partial sums

$$S_N(f) = \sum_{n=-N}^N \hat{f}(n) \chi_n$$

and ask whether  $S_N(f)$  converges to  $f$ , and in what sense.

We can summarize what we already know. By [Theorem 17.2](#), there exists  $f \in L^1(\mathbb{T})$  such that  $S_N(f)$  does not converge to  $f$  in  $\|\cdot\|_1$ . By [Corollary 14.15](#), if  $f \in L^2(\mathbb{T})$ , then  $S_N(f)$  does converge to  $f$  in  $\|\cdot\|_2$ . By [Theorem 18.1](#), there exists  $f \in C(\mathbb{T})$  such that  $S_N(f)(0)$  does not converge to  $f(0)$ , and hence  $S_N(f)$  does not converge to  $f$  in  $\|\cdot\|_\infty$ .

**Theorem 19.2** Let  $\{K_n\}$  be a summability kernel, and let  $f \in C(\mathbb{T})$ . Then  $K_n * f \rightarrow f$  uniformly (that is,  $\|K_n * f - f\|_\infty \rightarrow 0$ ).

**Proof.** Let  $x \in [0, 1]$ . Then

$$|K_n * f(x) - f(x)| = \left| \int_0^1 K_n(t) f(x-t) dt - f(x) \int_0^1 K_n(t) dt \right|$$

$$\begin{aligned}
&= \left| \int_0^1 K_n(t)(f(x-t) - f(x)) dt \right| \\
&\leq \int_0^1 |K_n(t)| \cdot |f(x-t) - f(x)| dt.
\end{aligned}$$

Since  $f \in C(\mathbb{T})$ , it is uniformly continuous in the circle metric. Let  $M = \sup_n \|K_n\|_1$ , and choose  $0 < \delta < 1/2$  such that

$$d_{\mathbb{T}}(x, y) < \delta \implies |f(x) - f(y)| < \frac{\varepsilon}{2M}.$$

Then choose  $N$  such that

$$\int_{\delta}^{1-\delta} |K_n(t)| dt < \frac{\varepsilon}{4(\|f\|_{\infty} + 1)}$$

whenever  $n \geq N$ . If  $t \in [0, \delta] \cup [1 - \delta, 1]$ , then  $d_{\mathbb{T}}(x - t, x) < \delta$ ; for  $t$  near 1, this follows from periodicity because  $f(x - t) = f(x + (1 - t))$ . Hence, for  $n \geq N$ ,

$$\begin{aligned}
|K_n * f(x) - f(x)| &\leq \frac{\varepsilon}{2M} \int_{[0, \delta] \cup [1-\delta, 1]} |K_n(t)| dt \\
&\quad + 2\|f\|_{\infty} \int_{\delta}^{1-\delta} |K_n(t)| dt \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.
\end{aligned}$$

The bound is independent of  $x$ , so  $K_n * f \rightarrow f$  uniformly.  $\square$

**Theorem 19.3** Let  $\{K_n\}$  be a summability kernel, and let  $f \in L^1(\mathbb{T})$ . Then  $K_n * f \rightarrow f$  in  $L^1$  (that is,  $\|K_n * f - f\|_1 \rightarrow 0$ ).

**Proof.** Let  $\varepsilon > 0$ . Since  $C(\mathbb{T})$  is  $\|\cdot\|_1$ -dense in  $L^1(\mathbb{T})$ , there exists  $g \in C(\mathbb{T})$  such that  $\|f - g\|_1 < \varepsilon$ . Then

$$\begin{aligned}
\|K_n * f - f\|_1 &= \|K_n * f - K_n * g + K_n * g - g + g - f\|_1 \\
&\leq \|K_n * (f - g)\|_1 + \|K_n * g - g\|_1 + \|g - f\|_1 \\
&\leq \|K_n\|_1 \|f - g\|_1 + \|K_n * g - g\|_{\infty} + \|g - f\|_1 \\
&\leq M\varepsilon + \|K_n * g - g\|_{\infty} + \varepsilon,
\end{aligned}$$

where  $M = \sup_{n \in \mathbb{N}} \|K_n\|_1$ . By [Theorem 19.2](#),  $\|K_n * g - g\|_{\infty} \rightarrow 0$ . Therefore

$$\limsup_{n \rightarrow \infty} \|K_n * f - f\|_1 \leq (M + 1)\varepsilon.$$

Since  $\varepsilon > 0$  was arbitrary,  $K_n * f \rightarrow f$  in  $L^1$ . □

**Remark 19.4** The Dirichlet kernel

$$D_N = \sum_{n=-N}^N \chi_n.$$

satisfies

$$\int_0^1 D_N(t) dt = \hat{D}_N(0) = 1,$$

but by [Corollary 17.3](#),  $\|D_N\|_1 \rightarrow \infty$ , so  $D_N$  is not a summability kernel. This explains why  $S_N(f)$  need not converge to  $f$  in  $L^1$ . We therefore look for a better kernel.

**Definition 19.5** The **Fejér kernel** is defined by

$$F_N = \sum_{n=-N}^N \left(1 - \frac{|n|}{N+1}\right) \chi_n.$$

**Lemma 19.6** Let  $n \in \mathbb{N}$ . Then

$$F_n(t) = \begin{cases} \frac{1}{n+1} \frac{\sin^2((n+1)\pi t)}{\sin^2(\pi t)}, & t \neq 0 \text{ in } \mathbb{T}, \\ n+1, & t = 0 \text{ in } \mathbb{T}. \end{cases}$$

**Proof.** We have

$$\sin^2(\pi t) = \left( \frac{e^{\pi i t} - e^{-\pi i t}}{2i} \right)^2 = \frac{e^{2\pi i t} + e^{-2\pi i t} - 2}{-4} = \frac{1}{2} - \frac{1}{4}(\chi_1 + \chi_{-1}).$$

Also,

$$\begin{aligned} \sin^2(\pi t) \cdot F_n(t) &= \left( \frac{1}{2} - \frac{1}{4}(\chi_1 + \chi_{-1}) \right) \sum_{j=-n}^n \left( 1 - \frac{|j|}{n+1} \right) \chi_j \\ &= \frac{1}{2} \sum_{j=-n}^n \left( 1 - \frac{|j|}{n+1} \right) \chi_j \end{aligned}$$

$$\begin{aligned}
& -\frac{1}{4} \sum_{j=-n}^n \left(1 - \frac{|j|}{n+1}\right) \chi_{j+1} \\
& -\frac{1}{4} \sum_{j=-n}^n \left(1 - \frac{|j|}{n+1}\right) \chi_{j-1}.
\end{aligned}$$

Expanding and collecting terms, the coefficient of  $\chi_k$  is

$$\frac{1}{2} \left(1 - \frac{|k|}{n+1}\right) - \frac{1}{4} \left(1 - \frac{|k-1|}{n+1}\right) - \frac{1}{4} \left(1 - \frac{|k+1|}{n+1}\right).$$

For  $k \neq 0, n+1, -(n+1)$ , this simplifies to 0. Specifically, for  $k = 0$ , the coefficient is

$$\frac{1}{2} - \frac{1}{4} \left(1 - \frac{1}{n+1}\right) - \frac{1}{4} \left(1 - \frac{1}{n+1}\right) = \frac{1}{2} - \frac{1}{2} + \frac{1}{2(n+1)} = \frac{1}{2(n+1)}.$$

Thus,

$$\begin{aligned}
\sin^2(\pi t) F_n(t) &= \frac{1}{2(n+1)} \chi_0 - \frac{1}{4(n+1)} \chi_{n+1} - \frac{1}{4(n+1)} \chi_{-(n+1)} \\
&= \frac{1}{4(n+1)} (2 - \chi_{n+1} - \chi_{-(n+1)}) \\
&= \frac{1}{4(n+1)} |1 - e^{2\pi i(n+1)t}|^2 \\
&= \frac{\sin^2((n+1)\pi t)}{n+1}.
\end{aligned}$$

This proves the quotient formula away from 0. At 0, continuity gives

$$F_n(0) = \lim_{t \rightarrow 0} \frac{1}{n+1} \frac{\sin^2((n+1)\pi t)}{\sin^2(\pi t)} = n+1.$$

□

**Remark 19.7** The Fejér kernel has the following properties:

1.  $\hat{F}_N(0) = 1$ .
2.  $F_N = \frac{1}{N+1} (D_0 + D_1 + \cdots + D_N)$ .
3.  $F_N \geq 0$ .

$$4. \|F_N\|_1 = \int_0^1 F_N(t) dt = \hat{F}_N(0) = 1.$$

$$5. \text{ If } 0 < \delta < 1/2, \text{ then } \int_\delta^{1-\delta} F_N(t) dt \longrightarrow 0 \text{ as } N \rightarrow \infty.$$

Thus,  $\{F_N\}$  is a summability kernel.

**Proof.** For (1), by the definition

$$F_N = \sum_{n=-N}^N \left(1 - \frac{|n|}{N+1}\right) \chi_n,$$

the coefficient of  $\chi_0$  is 1. Hence  $\hat{F}_N(0) = 1$ .

For (2), for each  $j \in \{-N, \dots, N\}$ , the term  $\chi_j$  appears in  $D_k$  exactly when  $|j| \leq k \leq N$ , which occurs for  $N - |j| + 1$  values of  $k$ . Therefore

$$\frac{1}{N+1} \sum_{k=0}^N D_k = \sum_{j=-N}^N \frac{N - |j| + 1}{N+1} \chi_j = \sum_{j=-N}^N \left(1 - \frac{|j|}{N+1}\right) \chi_j = F_N.$$

For (3), by [Lemma 19.6](#),

$$F_N(t) = \frac{1}{N+1} \frac{\sin^2((N+1)\pi t)}{\sin^2(\pi t)}.$$

The right-hand side is nonnegative for every  $t$  (interpreting the quotient at zeros of  $\sin(\pi t)$  by continuity), so  $F_N \geq 0$ .

For (4), we combine (3) with (1). Since  $F_N \geq 0$ ,

$$\|F_N\|_1 = \int_0^1 |F_N(t)| dt = \int_0^1 F_N(t) dt = \hat{F}_N(0) = 1.$$

For (5), fix  $\delta \in (0, 1/2)$ . By [Lemma 19.6](#),

$$F_N(t) = \frac{1}{N+1} \frac{\sin^2((N+1)\pi t)}{\sin^2(\pi t)} \leq \frac{1}{N+1} \frac{1}{\sin^2(\pi t)} \leq \frac{1}{N+1} \frac{1}{\sin^2(\pi \delta)}$$

for all  $t \in [\delta, 1 - \delta]$ , because  $\sin(\pi t) \geq \sin(\pi\delta) > 0$  on that interval. Integrating gives

$$\int_{\delta}^{1-\delta} F_N(t) dt \leq \frac{1 - 2\delta}{(N + 1) \sin^2(\pi\delta)} \xrightarrow{N \rightarrow \infty} 0.$$

Thus all three summability kernel conditions hold for  $\{F_N\}$ , so  $\{F_N\}$  is a summability kernel.  $\square$

## Lecture 20 — Tuesday, March 17, 2015

**Notation 20.1** We define the **Fejér means** of  $f$  by

$$\sigma_n(f) := F_n * f = \left( \sum_{j=-n}^n \left( 1 - \frac{|j|}{n+1} \right) \chi_j \right) * f = \sum_{j=-n}^n \left( 1 - \frac{|j|}{n+1} \right) \hat{f}(j) \chi_j.$$

**Corollary 20.2**  $\{F_n\}_{n \in \mathbb{N}}$  is a summability kernel. Consequently, we have the following:

1. If  $f \in L^1(\mathbb{T})$ , then  $\sigma_n(f) \xrightarrow{\|\cdot\|_1} f$ .
2. If  $f \in C(\mathbb{T})$ , then  $\sigma_n(f) \xrightarrow{\|\cdot\|_\infty} f$ .

**Proof.** (1) follows from [Theorem 19.3](#), and (2) follows from [Theorem 19.2](#).  $\square$

**Remark 20.3** [Corollary 20.2\(2\)](#) proves that  $\text{Trig}(\mathbb{T})$  is dense in  $C(\mathbb{T})$ . We will show later that if  $f \in C(\mathbb{T})$  and  $S_n f(x_0)$  converges, then it converges to the same limit as  $\sigma_n(f)(x_0)$  does, which is  $f(x_0)$ .

**Theorem 20.4** The Fourier transform  $\mathcal{F} : L^1(\mathbb{T}) \rightarrow c_0(\mathbb{Z})$  is injective.

**Proof.** Suppose there exists  $f \in L^1(\mathbb{T})$  with  $\hat{f}(n) = 0$  for all  $n \in \mathbb{Z}$ . Then

$$\sigma_n(f) = F_n * f = \sum_{j=-n}^n \left(1 - \frac{|j|}{n+1}\right) \hat{f}(j) \chi_j = 0.$$

By [Corollary 20.2\(1\)](#),  $\sigma_n(f) \rightarrow f$  in  $\|\cdot\|_1$ , which implies that  $f = 0$  almost everywhere. This is the **uniqueness theorem**: if  $f, g \in L^1(\mathbb{T})$  and  $\hat{f}(n) = \hat{g}(n)$  for every  $n$ , then  $\widehat{f-g}(n) = 0$  for every  $n$ , and hence  $f = g$  almost everywhere.  $\square$

Recall that the [Riemann–Lebesgue lemma](#) says that if  $f \in L^1(\mathbb{T})$ , then  $\hat{f} \in c_0(\mathbb{Z})$ . Also,  $f \in L^2(\mathbb{T})$  if and only if  $\hat{f} \in \ell^2(\mathbb{Z})$ .

Recall the [open mapping theorem](#) from PMATH 352:

**Theorem 20.5 (Open mapping theorem)** If  $X$  and  $Y$  are Banach spaces and  $T : X \rightarrow Y$  is bounded, linear, and surjective, then  $T$  is open.

**Corollary 20.6** If  $T$  is continuous, linear, and bijective, then  $T^{-1}$  is continuous as well.

**Proof.** The inverse  $T^{-1}$  is linear because

$$T^{-1}(\alpha Tx + Ty) = T^{-1}(T(\alpha x + y)) = \alpha x + y = \alpha T^{-1}(Tx) + T^{-1}(Ty).$$

Moreover,  $T^{-1}$  is continuous because inverse images under  $T^{-1}$  are images under  $T$ , and  $T$  is open by the [open mapping theorem](#). Since  $T^{-1}$  is linear and continuous, it is bounded.  $\square$

**Theorem 20.7** The Fourier transform  $\mathcal{F} : L^1(\mathbb{T}) \rightarrow c_0(\mathbb{Z})$  is not surjective.

**Proof.** Suppose, toward a contradiction, that  $\mathcal{F}$  is surjective. By [Theorem 20.4](#),  $\mathcal{F}$  is injective, so  $\mathcal{F}$  is bijective.

Since  $L^1(\mathbb{T})$  and  $c_0(\mathbb{Z})$  are Banach spaces and  $\mathcal{F}$  is bounded and bijective, the inverse map  $\mathcal{F}^{-1} : c_0(\mathbb{Z}) \rightarrow L^1(\mathbb{T})$  is continuous by [Corollary 20.6](#). Since  $\mathcal{F}^{-1}$  is linear and continuous, it is bounded. In particular, there exists  $M > 0$  such that

$$\|\mathcal{F}^{-1}(\mathcal{F}(f))\|_1 \leq M \|\mathcal{F}(f)\|_\infty$$

for all  $f \in L^1(\mathbb{T})$ . But taking  $f = D_N$  gives

$$\|D_N\|_1 \leq M\|\mathcal{F}(D_N)\|_\infty = M$$

for every  $N$ , because

$$\mathcal{F}(D_N)(n) = \hat{D}_N(n) = \begin{cases} 1 & \text{if } |n| \leq N, \\ 0 & \text{if } |n| > N. \end{cases}$$

This is a contradiction, since  $\|D_N\|_1 \rightarrow \infty$  by [Corollary 17.3](#). Hence  $\mathcal{F}$  is not surjective.  $\square$

## Pointwise Convergence of Fourier Series

What have we learned? If  $f \in L^1(\mathbb{T})$ , then  $\sigma_n(f) \rightarrow f$  in  $\|\cdot\|_1$ . If  $f \in C(\mathbb{T})$ , then  $\sigma_n(f) \rightarrow f$  in  $\|\cdot\|_\infty$ . However, there exists  $f \in C(\mathbb{T})$  such that  $S_n(f)(0)$  does not converge to  $f(0)$ , and hence  $S_n(f)$  does not converge to  $f$  in  $\|\cdot\|_\infty$ .

**Question 20.8** A natural question remains: what can we say about pointwise convergence when  $f$  is not continuous?

**Theorem 20.9 (Fejér's theorem)** Let  $f \in L^1(\mathbb{T})$  and assume that

$$w_f(x_0) := \lim_{h \rightarrow 0} \frac{f(x_0 + h) + f(x_0 - h)}{2}$$

exists. Then  $\sigma_n(f)(x_0) \rightarrow w_f(x_0)$ .

**Proof.** Since  $\int_0^1 F_n(t) dt = 1$ , we have

$$|\sigma_n(f)(x_0) - w_f(x_0)| = |F_n * f(x_0) - w_f(x_0)| = \left| \int_0^1 F_n(t)(f(x_0 - t) - w_f(x_0)) dt \right|.$$

Split the integral into the two neighborhoods  $[0, \delta]$  and  $[1 - \delta, 1]$  of 0 in  $\mathbb{T}$ , together with the middle interval  $[\delta, 1 - \delta]$ . By the symmetry of  $F_n$ ,

$$\begin{aligned} & \left| \int_0^\delta F_n(t)(f(x_0 - t) - w_f(x_0)) dt + \int_{1-\delta}^1 F_n(t)(f(x_0 - t) - w_f(x_0)) dt \right| \\ & \leq 2 \int_0^\delta F_n(t) \left| \frac{f(x_0 + t) + f(x_0 - t)}{2} - w_f(x_0) \right| dt. \end{aligned}$$

Let  $\delta > 0$  be such that  $|t| < \delta$  implies

$$\left| \frac{f(x_0 + t) + f(x_0 - t)}{2} - w_f(x_0) \right| < \varepsilon.$$

For this choice of  $\delta$ , the contribution from the two neighborhoods of 0 is bounded by

$$2\varepsilon \int_0^\delta F_n(t) dt \leq 2\varepsilon \|F_n\|_1 = 2\varepsilon.$$

For the middle interval, [Lemma 19.6](#) gives

$$F_n(t) = \frac{1}{n+1} \frac{\sin^2((n+1)\pi t)}{\sin^2(\pi t)} \leq \frac{1}{(n+1)\sin^2(\pi\delta)}$$

whenever  $\delta \leq t \leq 1 - \delta$ . Hence

$$\begin{aligned} \int_\delta^{1-\delta} F_n(t) |f(x_0 - t) - w_f(x_0)| dt &\leq \frac{1}{(n+1)\sin^2(\pi\delta)} \int_\delta^{1-\delta} (|f(x_0 - t)| + |w_f(x_0)|) dt \\ &\leq \frac{\|f\|_1 + |w_f(x_0)|}{(n+1)\sin^2(\pi\delta)}. \end{aligned}$$

Choose  $N_0$  such that

$$\frac{\|f\|_1 + |w_f(x_0)|}{(N_0 + 1)\sin^2(\pi\delta)} < \varepsilon.$$

Then the contribution from the middle interval is less than  $\varepsilon$  for every  $n \geq N_0$ . Combining the contributions from the two neighborhoods of 0 and the middle interval, we have

$$|\sigma_n(f)(x_0) - w_f(x_0)| \leq 2\varepsilon + \varepsilon = 3\varepsilon$$

for all  $n \geq N_0$ , and the result follows.  $\square$

### Corollary 20.10

1. If  $f \in L^1(\mathbb{T})$  is continuous at  $x_0$ , then  $\sigma_n(f)(x_0) \rightarrow f(x_0)$ .
2. If  $\lim_{h \rightarrow 0^+} f(x_0 + h) = f(x_0^+)$  and  $\lim_{h \rightarrow 0^+} f(x_0 - h) = f(x_0^-)$  exist, then

$$\sigma_n(f)(x_0) \rightarrow (f(x_0^+) + f(x_0^-))/2.$$

**Proof.** If  $f$  is continuous at  $x_0$ , then  $w_f(x_0) = f(x_0)$  by [Theorem 20.9](#). If  $f(x_0^+)$  and  $f(x_0^-)$  exist,

then  $w_f(x_0) = (f(x_0^+) + f(x_0^-))/2$ . □

## Lecture 21 — Thursday, March 19, 2015

**Theorem 21.1** If  $S_n f(x)$  converges, then it converges to the same limit as  $\sigma_n(f)(x)$ .

**Proof.** Suppose  $S_n(f)(t_0) \rightarrow g(t_0)$ . Since

$$\sigma_n(f)(t_0) = \frac{1}{n+1} \sum_{j=0}^n S_j(f)(t_0),$$

we estimate

$$\begin{aligned} |\sigma_n(f)(t_0) - g(t_0)| &= \left| \frac{S_0(f)(t_0) + \cdots + S_n(f)(t_0)}{n+1} - g(t_0) \right| \\ &= \frac{1}{n+1} \left| \sum_{j=0}^n (S_j(f)(t_0) - g(t_0)) \right| \\ &\leq \frac{1}{n+1} \left[ \sum_{j=0}^{N-1} |S_j(f)(t_0) - g(t_0)| + \sum_{j=N}^n |S_j(f)(t_0) - g(t_0)| \right] \end{aligned}$$

Let  $N$  be such that  $|S_j(f)(t_0) - g(t_0)| < \varepsilon$  for all  $j \geq N$ . Then

$$|\sigma_n(f)(t_0) - g(t_0)| \leq \frac{1}{n+1} \sum_{j=0}^{N-1} |S_j(f)(t_0) - g(t_0)| + \frac{n-N+1}{n+1} \varepsilon$$

and the first term tends to 0 as  $n \rightarrow \infty$  while the second is at most  $\varepsilon$ . So for  $n$  large enough,

$$|\sigma_n(f)(t_0) - g(t_0)| \leq 2\varepsilon.$$

Hence  $\sigma_n(f)(t_0) \rightarrow g(t_0)$ . □

**Corollary 21.2** If  $f$  is continuous at  $t_0$ , or if  $f$  has one-sided limits at  $t_0$ , and if  $\{S_n(f)(t_0)\}$  converges, then

$$S_n(f)(t_0) \rightarrow \frac{f(t_0^+) + f(t_0^-)}{2}.$$

**Proof.** This follows from [Theorems 20.9](#) and [21.1](#). □

**Theorem 21.3** If  $f \in L^1(\mathbb{T})$  and is differentiable at  $t_0$ , then  $S_n(f)(t_0) \rightarrow f(t_0)$  as  $n \rightarrow \infty$ .

**Proof.** Since  $\int_0^1 D_n(t) dt = 1$ , we have

$$\begin{aligned} |S_n(f)(t_0) - f(t_0)| &= \left| \int (f(t_0 - t) - f(t_0)) D_n(t) dt \right| \\ &= \left| \int \frac{f(t_0 - t) - f(t_0)}{\sin(\pi t)} \sin(\pi(2n+1)t) dt \right| \end{aligned}$$

It remains to prove that

$$g(t) = \frac{f(t_0 - t) - f(t_0)}{\sin(\pi t)}$$

belongs to  $L^1(\mathbb{T})$ . Since

$$\lim_{t \rightarrow 0} \frac{f(t_0 - t) - f(t_0)}{t} = -f'(t_0) \quad \text{and} \quad \lim_{t \rightarrow 0} \frac{t}{\sin(\pi t)} = \frac{1}{\pi},$$

the function  $g$  is bounded near 0. By periodicity, the same argument shows that  $g$  is bounded near 1. On every compact subinterval of  $(0, 1)$ , the denominator is bounded away from 0, so  $g$  is integrable there. Hence  $g \in L^1(\mathbb{T})$ .

$$\int_0^1 |g(t)| dt = \int_0^\delta |g(t)| dt + \int_\delta^{1-\delta} |g(t)| dt + \int_{1-\delta}^1 |g(t)| dt$$

Choose  $\delta > 0$  such that  $|g(t)| < |f'(t_0)|/\pi + 1$  whenever  $0 < t < \delta$  or  $1 - \delta < t < 1$ . Then

$$\int_0^1 |g(t)| dt < \infty.$$

Therefore,

$$\int g(t) \sin(\pi(2n+1)t) dt \rightarrow 0$$

by the [Riemann–Lebesgue lemma](#), and hence  $S_n(f)(t_0) \rightarrow f(t_0)$ .  $\square$

**Remark 21.4** In the proof of [Theorem 21.3](#), we only used differentiability to prove  $g \in L^1(\mathbb{T})$ .

**Corollary 21.5** If  $f$  is constant on an interval, then  $S_n(f)(t_0)$  converges to  $f(t_0)$  for all  $t_0$  in the interior of the interval.

**Proof.** If  $f$  is constant on an interval, then  $f$  is differentiable at every point in the interior of the interval, with derivative 0. Hence  $S_n(f)(t_0) \rightarrow f(t_0)$  for every  $t_0$  in the interior of the interval by [Theorem 21.3](#).  $\square$

**Definition 21.6** A function  $f$  satisfies the **Lipschitz condition of order  $\alpha$**  at a point  $t_0$ , where  $\alpha \geq 0$ , if there exist constants  $c > 0$  and  $\delta > 0$  such that

$$|f(t_0 - t) - f(t_0)| < c|t|^\alpha$$

whenever  $|t| < \delta$ .

**Remark 21.7**

1. It is harder to be Lipschitz for larger  $\alpha$ : if  $f$  is Lipschitz of order  $\alpha$  at  $t_0$ , then  $f$  is Lipschitz of order  $\beta$  at  $t_0$  for every  $0 \leq \beta \leq \alpha$ .
2.  $f$  being Lipschitz of order  $\alpha$  at  $t_0$  with  $\alpha = 0$  means that  $f$  is bounded close to  $t_0$ .
3. If  $f$  is Lipschitz of order  $\alpha$  at  $t_0$  for some  $\alpha > 0$ , then  $f$  is continuous at  $t_0$ . Indeed, if  $f$  is Lipschitz of order  $\alpha$  at  $t_0$ , then there are constants  $c > 0$  and  $\delta > 0$  such that

$$|f(t_0 - t) - f(t_0)| < c|t|^\alpha$$

whenever  $|t| < \delta$ . Letting  $t \rightarrow 0$  gives  $f(t_0 - t) \rightarrow f(t_0)$ , so  $f$  is continuous at  $t_0$ .

4. If  $f$  is differentiable at  $t_0$ , then  $f$  is Lipschitz of order 1. Indeed, differentiability gives a  $\delta > 0$  such that

$$|f(t_0 - t) - f(t_0)| \leq (|f'(t_0)| + 1)|t|$$

whenever  $|t| < \delta$ . Thus differentiability at  $t_0$  implies the Lipschitz condition of order

1 at  $t_0$ , which implies the Lipschitz condition of order  $\alpha$  for every  $0 \leq \alpha \leq 1$ , and in particular implies continuity at  $t_0$ .

We can generalize [Theorem 21.3](#) as follows.

**Theorem 21.8** If  $f \in L^1(\mathbb{T})$  and is Lipschitz of order  $\alpha$  for some  $\alpha > 0$  at  $t_0$ , then  $S_n(f)(t_0) \rightarrow f(t_0)$ .

**Proof.** As in the proof of [Theorem 21.3](#), it is enough to show that

$$g(t) = \frac{f(t_0 - t) - f(t_0)}{\sin(\pi t)}$$

belongs to  $L^1(\mathbb{T})$ . Since  $f$  is Lipschitz of order  $\alpha$  at  $t_0$ , there are constants  $c > 0$  and  $\delta > 0$  such that

$$|f(t_0 - t) - f(t_0)| \leq c|t|^\alpha$$

whenever  $|t| < \delta$ . Decreasing  $\delta$  if necessary, we may assume that  $0 < \delta < 1/2$  and that

$$|\sin(\pi t)| \geq 2|t|$$

whenever  $|t| < \delta$ . Hence, for  $0 < t < \delta$ ,

$$|g(t)| \leq \frac{c}{2} t^{\alpha-1}.$$

The same estimate holds near 1, with  $t$  replaced by  $1 - t$ , because  $t = 1$  represents the same point as  $t = 0$  on  $\mathbb{T}$ . On the interval  $[\delta, 1 - \delta]$ , the denominator is bounded away from 0, and  $f(t_0 - t) - f(t_0)$  is integrable. Therefore

$$\int_0^1 |g(t)| dt < \infty,$$

since  $\int_0^\delta t^{\alpha-1} dt < \infty$  when  $\alpha > 0$ . □

**Definition 21.9** We say that  $f$  has a **derivative at  $t_0$  from the right** if

$$\lim_{h \rightarrow 0^+} \frac{f(t_0 + h) - f(t_0^+)}{h}$$

exists, where  $f(t_0^+) = \lim_{h \rightarrow 0^+} f(t_0 + h)$ . Similarly, we say that  $f$  has a **derivative at  $t_0$  from the left** if

$$\lim_{h \rightarrow 0^+} \frac{f(t_0 - h) - f(t_0^-)}{-h}$$

exists, where  $f(t_0^-) = \lim_{h \rightarrow 0^+} f(t_0 - h)$ .

**Corollary 21.10** If  $f \in L^1(\mathbb{T})$  is continuous at  $t_0$  and has left and right derivatives at  $t_0$ , then  $S_n(f)(t_0) \rightarrow f(t_0)$  as  $n \rightarrow \infty$ .

**Proof.** Since  $f$  is continuous at  $t_0$ , we have  $f(t_0^+) = f(t_0^-) = f(t_0)$ . Also,  $f$  is differentiable from the left and right at  $t_0$ .

The limits

$$\lim_{h \rightarrow 0^+} \frac{f(t_0 + h) - f(t_0)}{h} \quad \text{and} \quad \lim_{h \rightarrow 0^+} \frac{f(t_0 - h) - f(t_0)}{-h}$$

exist. Hence  $f$  is Lipschitz of order 1, and so [Theorem 21.8](#) gives  $S_n(f)(t_0) \rightarrow f(t_0)$ .  $\square$

**Example 21.11** Let  $f(x) = x^2 - x$  on  $[0, 1]$ . Since  $f(0) = 0 = f(1)$ , this defines a continuous periodic function on  $\mathbb{T}$ . At  $t_0 = 0$ , the right derivative is

$$(2x - 1)|_{x=0} = -1,$$

while the left derivative, computed using the value at  $x = 1$ , is

$$(2x - 1)|_{x=1} = 1.$$

Thus  $f$  is not differentiable at 0, but it has left and right derivatives there. By [Theorem 21.8](#),  $S_n(f)(0) \rightarrow f(0) = 0$ . Therefore

$$\sum_{n \in \mathbb{Z}} \hat{f}(n) = 0 \tag{21.1}$$

by [Corollary 21.2](#). For  $n \neq 0$ , we compute  $\hat{f}(n)$  by integration by parts:

$$\begin{aligned}\hat{f}(n) &= \int_0^1 (x^2 - x)e^{-2\pi inx} dx \\ &= \frac{(x^2 - x)e^{-2\pi inx}}{-2\pi in} \Big|_0^1 - \int_0^1 \frac{(2x - 1)e^{-2\pi inx}}{-2\pi in} dx \\ &= \frac{1}{2\pi in} \int_0^1 (2x - 1)e^{-2\pi inx} dx \\ &= \frac{(2x - 1)e^{-2\pi inx}}{-4\pi^2 n^2} \Big|_0^1 - \int_0^1 \frac{2e^{-2\pi inx}}{-4\pi^2 n^2} dx \\ &= \frac{1}{2\pi^2 n^2}.\end{aligned}$$

On the other hand,

$$\hat{f}(0) = \int_0^1 (x^2 - x) dx = \left[ \frac{x^3}{3} - \frac{x^2}{2} \right]_0^1 = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}.$$

Inserting these into [\(21.1\)](#) gives

$$\sum_{n \in \mathbb{Z} \setminus \{0\}} \frac{1}{n^2} = \frac{2\pi^2}{6},$$

and therefore

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Surprise!

**Remark 21.12** There is a sneakier way to arrive at the solution to the Basel problem. Take  $f(x) = x$  on  $[0, 1]$ . Then

$$\|f\|_2^2 = \int_0^1 x^2 dx = \frac{1}{3}.$$

For all  $n \neq 0$ ,

$$\hat{f}(n) = \int_0^1 xe^{-2\pi inx} dx = \frac{xe^{-2\pi inx}}{-2\pi in} \Big|_0^1 - \int_0^1 \frac{e^{-2\pi inx}}{-2\pi in} dx = \frac{-1}{2\pi in}.$$

We also have

$$\hat{f}(0) = \int_0^1 x \, dx = \frac{1}{2}.$$

By Parseval's identity,

$$\sum_{n \in \mathbb{Z} \setminus \{0\}} \left| \frac{-1}{2\pi i n} \right|^2 + \left| \frac{1}{2} \right|^2 = \frac{1}{3}.$$

Therefore

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Lecture 22 — Tuesday, March 24, 2015

## 5. Fourier Analysis on $\mathbb{R}$ and Applications

### Fourier Transform on $\mathbb{R}$

The major difference between the circle and the real line is that  $L^p(\mathbb{T}) \subseteq L^1(\mathbb{T})$ , whereas  $L^p(\mathbb{R}) \not\subseteq L^1(\mathbb{R})$  for  $p \neq 1$ . Recall that  $L^1(\mathbb{R})$  is the space of Lebesgue measurable functions on  $\mathbb{R}$  with

$$\|f\|_1 = \int_{\mathbb{R}} |f| \, dx.$$

For  $f, g \in L^1(\mathbb{R})$ , define their convolution by

$$(f * g)(x) = \int_{\mathbb{R}} f(y)g(x - y) \, dy.$$

As on  $\mathbb{T}$ , we can prove that the function  $y \mapsto f(y)g(x - y)$  is integrable for almost every  $x$ . Moreover,  $f * g \in L^1(\mathbb{R})$  and  $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$ .

**Definition 22.1** For  $\xi \in \mathbb{R}$ , define  $\chi_\xi : \mathbb{R} \rightarrow \mathbb{T}$  by  $\chi_\xi(x) = e^{2\pi i \xi x}$ .

We have

$$\chi_\xi(x + y) = \chi_\xi(x)\chi_\xi(y) \quad \text{and} \quad \chi_\xi(0) = 1.$$

Moreover,

$$|\chi_\xi(x) - \chi_\xi(y)| \leq 2\pi|\xi| |x - y|,$$

so  $\chi_\xi$  is uniformly continuous.

**Definition 22.2** A continuous function  $\phi : \mathbb{R} \rightarrow \mathbb{T}$  is called a **character** if  $\phi$  is a homomorphism:  $\phi(x + y) = \phi(x)\phi(y)$ .

**Proposition 22.3** If  $\phi$  is a character, then there exists  $\xi \in \mathbb{R}$  such that  $\phi = \chi_\xi$ .

**Proof.** If  $\phi$  is a character, then

$$\phi(0) = \phi(0 + 0) = \phi(0) \cdot \phi(0).$$

Thus  $\phi(0) = 1$ . By continuity, there exists  $\varepsilon > 0$  such that

$$\phi((- \varepsilon, \varepsilon)) \subseteq \{w \in \mathbb{C} : \operatorname{Re}(w) > 0\}.$$

Write  $\phi(\varepsilon) = e^{2\pi i \varepsilon y}$ , where  $y$  is the unique element of  $[-1/(4\varepsilon), 1/(4\varepsilon)]$  with this property. Since  $\phi(\varepsilon/2)^2 = \phi(\varepsilon)$  and  $\phi(\varepsilon/2)$  lies in the right half-plane, we have  $\phi(\varepsilon/2) = e^{2\pi i \varepsilon y/2}$ . Iterating gives

$$\phi(\varepsilon/2^n) = e^{2\pi i \varepsilon y/2^n}$$

for every  $n \in \mathbb{N}$ . Therefore, for  $m \in \mathbb{Z}$ ,

$$\phi(m\varepsilon/2^n) = \phi(\varepsilon/2^n)^m = e^{2\pi i m \varepsilon y/2^n}.$$

The dyadic multiples  $m\varepsilon/2^n$  are dense in  $\mathbb{R}$ , and continuity of  $\phi$  extends this formula to every  $x \in \mathbb{R}$ . Hence  $\phi(x) = e^{2\pi i y x}$  for every  $x$ , so  $\phi = \chi_y$ .  $\square$

**Remark 22.4**  $\chi_\xi \notin L^p(\mathbb{R})$  for  $1 \leq p < \infty$ .

**Definition 22.5** Let  $f \in L^1(\mathbb{R})$  and  $\xi \in \mathbb{R}$ . The **Fourier transform** of  $f$  at  $\xi$  is

$$\hat{f}(\xi) = \int_{\mathbb{R}} f(x) \overline{\chi_\xi(x)} dx = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} dx.$$

Observe that

$$|\hat{f}(\xi)| \leq \int_{\mathbb{R}} |f(x)| \cdot |e^{-2\pi i \xi x}| dx = \int_{\mathbb{R}} |f(x)| dx < \infty,$$

so  $\hat{f}(\xi)$  is well-defined.

**Proposition 22.6** If  $f \in L^1(\mathbb{R})$ , then  $\hat{f}$  is uniformly continuous.

**Proof.** For every  $\xi, h \in \mathbb{R}$ ,

$$|\hat{f}(\xi + h) - \hat{f}(\xi)| \leq \int_{\mathbb{R}} |f(x)| |e^{-2\pi i h x} - 1| dx.$$

The right-hand side is independent of  $\xi$  and tends to zero as  $h \rightarrow 0$  by the [dominated convergence theorem](#), since it is bounded by  $2|f(x)|$ . Hence  $\hat{f}$  is uniformly continuous.  $\square$

## Basic Properties and the Riemann–Lebesgue Lemma

**Proposition 22.7** Let  $f, g \in L^1(\mathbb{R})$  and let  $\alpha \in \mathbb{C}$ . The Fourier transform has the following properties:

1.  $\widehat{f + g} = \hat{f} + \hat{g}$ .
2.  $\widehat{\alpha f} = \alpha \hat{f}$ .
3.  $\widehat{f * g} = \hat{f} \cdot \hat{g}$ .
4. If  $f_y(x) := f(x - y)$ , then  $\hat{f}_y(\xi) = \chi_\xi(-y) \hat{f}(\xi)$ .
5. If  $\tilde{f}(x) := f(-x)$ , then  $\widehat{\tilde{f}}(\xi) = \hat{f}(-\xi)$ .

**Proof.** For (1), linearity of the integral gives

$$\widehat{f + g}(\xi) = \int_{\mathbb{R}} (f(x) + g(x)) e^{-2\pi i \xi x} dx = \hat{f}(\xi) + \hat{g}(\xi).$$

For (2), again by linearity,

$$\widehat{\alpha f}(\xi) = \int_{\mathbb{R}} \alpha f(x) e^{-2\pi i \xi x} dx = \alpha \hat{f}(\xi).$$

For (3), since  $f, g \in L^1(\mathbb{R})$ , Fubini's theorem applies:

$$\widehat{f * g}(\xi) = \int_{\mathbb{R}} \left( \int_{\mathbb{R}} f(t) g(x-t) dt \right) e^{-2\pi i \xi x} dx = \int_{\mathbb{R}} f(t) \left( \int_{\mathbb{R}} g(x-t) e^{-2\pi i \xi x} dx \right) dt.$$

Set  $u = x - t$ . Then

$$\int_{\mathbb{R}} g(x-t) e^{-2\pi i \xi x} dx = e^{-2\pi i \xi t} \int_{\mathbb{R}} g(u) e^{-2\pi i \xi u} du = e^{-2\pi i \xi t} \hat{g}(\xi).$$

Therefore,

$$\widehat{f * g}(\xi) = \hat{g}(\xi) \int_{\mathbb{R}} f(t) e^{-2\pi i \xi t} dt = \hat{f}(\xi) \hat{g}(\xi).$$

For (4), with  $u = x - y$ ,

$$\begin{aligned} \hat{f}_y(\xi) &= \int_{\mathbb{R}} f(x-y) e^{-2\pi i \xi x} dx \\ &= \int_{\mathbb{R}} f(u) e^{-2\pi i \xi(u+y)} du \\ &= e^{-2\pi i \xi y} \int_{\mathbb{R}} f(u) e^{-2\pi i \xi u} du \\ &= \chi_{\xi}(-y) \hat{f}(\xi). \end{aligned}$$

For (5), with  $u = -x$ ,

$$\widehat{\tilde{f}}(\xi) = \int_{\mathbb{R}} f(-x) e^{-2\pi i \xi x} dx = \int_{\mathbb{R}} f(u) e^{2\pi i \xi u} du = \hat{f}(-\xi).$$

□

**Lemma 22.8** Let  $f \in L^1(\mathbb{R})$  be continuous and let

$$F(x) = \int_{-\infty}^x f(t) dt.$$

If  $F \in L^1(\mathbb{R})$ , then  $\widehat{F}(\xi) = \widehat{f}(\xi)/(2\pi i\xi)$  for every  $\xi \neq 0$ .

**Proof.** Integration by parts gives

$$(\widehat{F})(\xi) = F(x) \frac{e^{-2\pi i\xi x}}{-2\pi i\xi} \Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{e^{-2\pi i\xi x}}{2\pi i\xi} f(x) \, dx.$$

The boundary term vanishes, so

$$\widehat{F}(\xi) = \frac{\widehat{f}(\xi)}{2\pi i\xi}.$$

□

**Corollary 22.9** If  $f$  is continuously differentiable and  $f, f' \in L^1(\mathbb{R})$ , then

$$\widehat{f}'(\xi) = 2\pi i\xi \widehat{f}(\xi).$$

**Proof.** Apply [Lemma 22.8](#) to  $f'$  and  $f$ .

□

**Theorem 22.10 (Riemann–Lebesgue lemma: real version)** Let  $f \in L^1(\mathbb{R})$ . Then  $\lim_{|\xi| \rightarrow \infty} \widehat{f}(\xi) = 0$ .

**Proof.** First assume that  $f$  is compactly supported and continuously differentiable. By [Corollary 22.9](#),

$$\widehat{f}'(\xi) = 2\pi i\xi \widehat{f}(\xi).$$

Thus

$$|2\pi i\xi \widehat{f}(\xi)| = |\widehat{f}'(\xi)| = \left| \int_{\mathbb{R}} f'(x) e^{-2\pi i\xi x} \, dx \right| \leq \int_{\mathbb{R}} |f'(x)| \, dx = \|f'\|_1.$$

It follows that  $\widehat{f}(\xi) \rightarrow 0$  as  $|\xi| \rightarrow \infty$  in this case.

Now let  $f \in L^1(\mathbb{R})$  be arbitrary, and let  $\varepsilon > 0$ . Choose a compactly supported continuously differentiable function  $g$  such that  $\|f - g\|_1 < \varepsilon$ . Then

$$|\widehat{f}(\xi) - \widehat{g}(\xi)| = |\widehat{f - g}(\xi)| \leq \|f - g\|_1 < \varepsilon,$$

and the first part gives  $\hat{g}(\xi) \rightarrow 0$  as  $|\xi| \rightarrow \infty$ . Hence

$$\limsup_{|\xi| \rightarrow \infty} |\hat{f}(\xi)| \leq \varepsilon.$$

Since  $\varepsilon > 0$  was arbitrary,  $\hat{f}(\xi) \rightarrow 0$  as  $|\xi| \rightarrow \infty$ . □

**Definition 22.11** Let  $C_0(\mathbb{R})$  be the space of complex-valued continuous functions on  $\mathbb{R}$  which vanish at infinity. The **Fourier transform** is the map

$$\mathcal{F} : L^1(\mathbb{R}) \rightarrow C_0(\mathbb{R}), \quad f \mapsto \hat{f}.$$

|                | In $L^1(\mathbb{R})$       | In $C_0(\mathbb{R})$        |
|----------------|----------------------------|-----------------------------|
| Addition       | $(f + g)(x) = f(x) + g(x)$ | $(f + g)(x) = f(x) + g(x)$  |
| Multiplication | $f * g$ (convolution)      | $(fg)(x) = f(x) \cdot g(x)$ |
| Norm           | $\ f\ _1$                  | $\ f\ _\infty$              |
| Involution     | $f^*$                      | $\overline{f}$              |

## Summability Kernels and Fourier Inversion

**Definition 22.12** A family of continuous functions  $\{k_\lambda\}_{\lambda \in I} \subseteq L^1(\mathbb{R})$  is called a **summability kernel** if it satisfies the following properties:

1.  $\int_{\mathbb{R}} k_\lambda(x) dx = 1$  for every  $\lambda$ .
2.  $\sup_{\lambda \in I} \|k_\lambda\|_1 < \infty$ .
3.  $\lim_{\lambda \rightarrow \infty} \int_{|x| > \delta} |k_\lambda(x)| dx = 0$  for all  $\delta > 0$ .

**Example 22.13** For a continuous function  $k \in L^1(\mathbb{R})$  and  $\lambda > 0$ , define  $k_\lambda(x) = \lambda k(\lambda x)$ . If

$$\int_{\mathbb{R}} k(x) dx = 1,$$

then  $\{k_\lambda\}_{\lambda > 0}$  is a summability kernel.

**Proof.** For  $\lambda > 0$ ,

$$\int_{\mathbb{R}} k_{\lambda}(x) \, dx = \int_{\mathbb{R}} \lambda k(\lambda x) \, dx = \int_{\mathbb{R}} k(u) \, du = 1$$

by the change of variables  $u = \lambda x$ . Also

$$\|k_{\lambda}\|_1 = \int_{\mathbb{R}} \lambda |k(\lambda x)| \, dx = \int_{\mathbb{R}} |k(u)| \, du = \|k\|_1,$$

so the  $L^1$  norms are uniformly bounded. Finally, for any  $\delta > 0$ ,

$$\int_{|x|>\delta} |k_{\lambda}(x)| \, dx = \int_{|u|>\lambda\delta} |k(u)| \, du \xrightarrow{\lambda \rightarrow \infty} 0$$

by absolute integrability of  $k$ . These are exactly the summability kernel conditions.  $\square$

## Lecture 23 — Thursday, March 26, 2015

### Plancherel and the Inverse Transform

**Example 23.1** Two useful summability kernels are the following.

1. The **Fejér kernel** is

$$F_{\omega}(x) = \int_{-\omega}^{\omega} \left(1 - \frac{|\xi|}{\omega}\right) e^{2\pi i \xi x} \, d\xi.$$

The family  $\{F_{\omega}\}_{\omega>0}$  is a summability kernel, and

$$\begin{aligned} F_{\omega} * f(x) &= \int_{\mathbb{R}} f(y) F_{\omega}(x - y) \, dy \\ &= \int_{\mathbb{R}} f(y) \left( \int_{-\omega}^{\omega} \left(1 - \frac{|\xi|}{\omega}\right) e^{2\pi i \xi(x-y)} \, d\xi \right) dy \\ &= \int_{-\omega}^{\omega} \left(1 - \frac{|\xi|}{\omega}\right) \hat{f}(\xi) e^{2\pi i \xi x} \, d\xi. \end{aligned}$$

2. The **Gauss kernel** is

$$G_{\omega}(x) = \int_{\mathbb{R}} e^{-\pi \xi^2 / \omega^2} e^{2\pi i \xi x} \, d\xi.$$

In this case,  $G_\omega(x) = \omega e^{-\pi\omega^2 x^2}$ . Then  $\{G_\omega\}$  is a summability kernel, and

$$\begin{aligned} G_\omega * f(x) &= \int_{\mathbb{R}} f(y) G_\omega(x-y) dy \\ &= \int_{\mathbb{R}} f(y) \left( \int_{\mathbb{R}} e^{-\pi\xi^2/\omega^2} e^{2\pi i\xi(x-y)} d\xi \right) dy \\ &= \int_{\mathbb{R}} \left( \int_{\mathbb{R}} f(y) e^{-2\pi i\xi y} dy \right) e^{-\pi\xi^2/\omega^2} e^{2\pi i\xi x} d\xi \\ &= \int_{\mathbb{R}} e^{-\pi\xi^2/\omega^2} \hat{f}(\xi) e^{2\pi i\xi x} d\xi. \end{aligned}$$

**Theorem 23.2** Let  $\{k_\alpha\}_{\alpha \in I}$  be a summability kernel. Then the following hold:

1. If  $f \in L^1(\mathbb{R})$ , then  $k_\alpha * f \xrightarrow{\alpha \rightarrow \infty} f$  in  $L^1(\mathbb{R})$ .
2. If  $f \in C_0(\mathbb{R})$ , then  $k_\alpha * f \rightarrow f$  in  $C_0(\mathbb{R})$ .
3. If  $f \in L^p(\mathbb{R})$ ,  $1 \leq p < \infty$ , then  $k_\alpha * f \xrightarrow{\alpha \rightarrow \infty} f$  in  $L^p(\mathbb{R})$ .

**Proof.** The proofs of (1) and (2) are identical to the corresponding arguments on  $\mathbb{T}$ . We will just accept (3). □

As a consequence, if  $f \in L^1(\mathbb{R})$ , then

$$F_\omega * f(x) = \int_{-\omega}^{\omega} \left(1 - \frac{|\xi|}{\omega}\right) \hat{f}(\xi) e^{2\pi i\xi x} d\xi \longrightarrow f$$

in  $L^1(\mathbb{R})$ .

**Corollary 23.3** The map  $\mathcal{F} : L^1(\mathbb{R}) \rightarrow C_0(\mathbb{R})$  is injective.

**Proof.** Suppose  $f \in L^1(\mathbb{R})$  with  $\hat{f}(\xi) = 0$  for all  $\xi \in \mathbb{R}$ . Then  $F_\omega * f = 0$  for all  $\omega > 0$ . Then  $f = \lim_{\omega \rightarrow \infty} F_\omega * f = 0$  in  $L^1(\mathbb{R})$ , so  $f = 0$ . □

We next restrict the Fourier transform to  $L^2(\mathbb{R})$ . Initially this gives

$$\mathcal{F} : L^2(\mathbb{R}) \cap L^1(\mathbb{R}) \rightarrow C_0(\mathbb{R}).$$

The space  $L^2(\mathbb{R}) \cap L^1(\mathbb{R})$  contains all continuous compactly supported functions on  $\mathbb{R}$ , and is therefore  $\|\cdot\|_2$ -dense in  $L^2(\mathbb{R})$ .

Recall that  $L^2(\mathbb{R})$  is a Hilbert space with inner product

$$\langle f, g \rangle = \int_{\mathbb{R}} f(x) \overline{g(x)} \, dx$$

and norm  $\|f\|_2 = \sqrt{\langle f, f \rangle}$ .

**Theorem 23.4** The map  $\mathcal{F} : L^2(\mathbb{R}) \cap L^1(\mathbb{R}) \rightarrow C_0(\mathbb{R})$  extends to an isometric isomorphism  $\mathcal{F} : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ . In particular, we have

$$\|\mathcal{F}f\|_2 = \|f\|_2 \quad (\text{Plancherel's identity})$$

and

$$\langle f, g \rangle = \langle \mathcal{F}f, \mathcal{F}g \rangle \quad (\text{Parseval's identity}).$$

**Proof.** Let  $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ . Recall that  $G_\omega * f \rightarrow f$  in  $L^2(\mathbb{R})$  since  $\{G_\omega\}$  is a summability kernel. Then

$$\begin{aligned} \|f\|_2^2 &= \langle f, f \rangle \\ &= \lim_{\omega \rightarrow \infty} \langle f, G_\omega * f \rangle \\ &= \lim_{\omega \rightarrow \infty} \int f(x) \overline{G_\omega * f(x)} \, dx \\ &= \lim_{\omega \rightarrow \infty} \int \widehat{f}(\xi) \overline{\widehat{f}(\xi)} e^{-\pi \xi^2 / \omega^2} \, d\xi \\ &= \int |\widehat{f}(\xi)|^2 \, d\xi \\ &= \|\mathcal{F}f\|_2^2, \end{aligned}$$

which establishes [Plancherel's identity](#) for  $f$ . Thus  $\mathcal{F}$  is an isometry on  $L^1 \cap L^2(\mathbb{R})$ .

The subspace  $L^1 \cap L^2(\mathbb{R})$  is  $\|\cdot\|_2$ -dense in  $L^2(\mathbb{R})$ . Because  $\mathcal{F} : L^1 \cap L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$  is an isometry, it extends uniquely to an isometry  $\mathcal{F} : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ . In particular, it is injective. [Parseval's](#)

[identity](#) now follows from the **polarization identity** applied to the isometry:

$$\langle f, g \rangle = \frac{1}{4} (\|f + g\|_2^2 - \|f - g\|_2^2 + i\|f + ig\|_2^2 - i\|f - ig\|_2^2),$$

and the same formula with  $f, g$  replaced by  $\mathcal{F}f, \mathcal{F}g$  gives

$$\langle f, g \rangle = \langle \mathcal{F}f, \mathcal{F}g \rangle.$$

Surjectivity will follow from the [Fourier inversion formula](#), which we will prove next.  $\square$

**Definition 23.5** For  $g \in L^1(\mathbb{R})$ , define the **inverse Fourier transform**

$$(\mathcal{J}g)(x) = \int_{\mathbb{R}} g(\xi) e^{2\pi i \xi x} d\xi.$$

**Theorem 23.6 (Fourier inversion formula)**

1. If  $f, \hat{f} \in L^1(\mathbb{R})$ , then  $f$  agrees almost everywhere with the continuous function  $\mathcal{J}\hat{f}$ .
2. If  $g, \mathcal{J}g \in L^1(\mathbb{R})$ , then  $g$  agrees almost everywhere with the continuous function  $\mathcal{F}(\mathcal{J}g)$ .

Thus, the Fourier transform  $\mathcal{F} : L^1(\mathbb{R}) \rightarrow C_0(\mathbb{R})$  is injective, and its inverse is given by  $\mathcal{J}$  on the range of  $\mathcal{F}$ , so the following diagram commutes:

$$\begin{array}{ccc} L^1(\mathbb{R}) & \xrightarrow{\mathcal{F}} & C_0(\mathbb{R}) \\ \mathcal{F}^{-1} \uparrow & & \uparrow \mathcal{F} \\ \mathcal{F}(L^1(\mathbb{R})) & \xrightarrow{\mathcal{J}} & \mathcal{J}(\mathcal{F}(L^1(\mathbb{R}))) \end{array}$$

**Proof.** For (1), let  $f, \hat{f} \in L^1(\mathbb{R})$ . Recall that  $F_{\omega} * f \rightarrow f$  in  $L^1(\mathbb{R})$ . Choose a sequence  $\omega_k \rightarrow \infty$  along which  $F_{\omega_k} * f \rightarrow f$  pointwise almost everywhere. For each  $x \in \mathbb{R}$ ,

$$\begin{aligned} \lim_{k \rightarrow \infty} (F_{\omega_k} * f)(x) &= \lim_{k \rightarrow \infty} \int_{-\omega_k}^{\omega_k} \left(1 - \frac{|\xi|}{\omega_k}\right) \hat{f}(\xi) e^{2\pi i \xi x} d\xi \\ &= \int_{\mathbb{R}} \hat{f}(\xi) e^{2\pi i \xi x} d\xi \end{aligned}$$

$$= (\mathcal{J}\hat{f})(x)$$

by the [dominated convergence theorem](#). Along the chosen sequence, the left-hand side equals  $f(x)$  almost everywhere. Thus  $f = \mathcal{J}\hat{f}$  almost everywhere. Since the inverse transform of an  $L^1$  function belongs to  $C_0(\mathbb{R})$ , this gives the claimed continuous representative.

The proof of (2) is the same argument with the signs in the exponent reversed.

We can now finish the surjectivity part of [Theorem 23.4](#). If  $g \in C_c^\infty(\mathbb{R})$ , then  $\mathcal{J}g$  is a Schwartz function, so  $\mathcal{J}g \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ , and (2) gives

$$\mathcal{F}(\mathcal{J}g) = g.$$

Thus the range of the  $L^2$  Fourier transform contains  $C_c^\infty(\mathbb{R})$ , which is dense in  $L^2(\mathbb{R})$ . The range of an isometry from a complete space is closed, so the range is all of  $L^2(\mathbb{R})$ .  $\square$

**Corollary 23.7**  $\mathcal{F} : L^1(\mathbb{R}) \rightarrow C_0(\mathbb{R})$  is a norm-decreasing (that is,  $\|\hat{f}\|_\infty \leq \|f\|_1$ ) injective algebra homomorphism (that is,  $\widehat{f+g} = \hat{f} + \hat{g}$  and  $\widehat{f*g} = \hat{f} \cdot \hat{g}$ ) with dense range.

**Proof.** The norm estimate is immediate:

$$\|\hat{f}\|_\infty = \sup_{\xi \in \mathbb{R}} \left| \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} dx \right| \leq \int_{\mathbb{R}} |f(x)| dx = \|f\|_1.$$

Hence  $\mathcal{F}$  is norm-decreasing. Linearity follows from linearity of the integral, and the convolution identity proved earlier gives

$$\widehat{f * g} = \hat{f} \hat{g},$$

so  $\mathcal{F}$  is an algebra homomorphism from convolution on  $L^1(\mathbb{R})$  to pointwise multiplication on  $C_0(\mathbb{R})$ . Injectivity is [Corollary 23.3](#).

For density of the range, let  $g \in C_c^\infty(\mathbb{R})$ . Then  $\mathcal{J}g$  is a Schwartz function and therefore belongs to  $L^1(\mathbb{R})$ . By [Theorem 23.6\(2\)](#),

$$\mathcal{F}(\mathcal{J}g) = g.$$

Hence the range contains  $C_c^\infty(\mathbb{R})$ , which is dense in  $C_0(\mathbb{R})$  in the uniform norm.  $\square$

## Lecture 24 — Thursday, April 02, 2015

## The Heat Equation

Let us solve the heat equation on  $\mathbb{R}$  using the Fourier transform. This is a partial differential equation which models the flow of heat in a medium.

For the following formal calculation, we assume that  $u_t$ ,  $u_x$ , and  $u_{xx}$  belong to  $L^1(\mathbb{R})$ . We write  $\hat{u} = \mathcal{F}u$ .

**Definition 24.1** Let  $u(x, t)$  be the amount of heat at point  $x$  at time  $t$ . The **heat equation** is

$$u_t = \alpha u_{xx}$$

for  $\alpha > 0$  with initial condition  $u(x, 0) = \phi(x)$ .

We start with two observations:

1. For  $k \in \mathbb{R}$ , let

$$\mathcal{F}u(k, t) = \int_{\mathbb{R}} u(x, t) e^{-2\pi i x k} dx,$$

Then the [dominated convergence theorem](#) gives

$$\begin{aligned} \mathcal{F}u_t(k, t) &= \int_{\mathbb{R}} u_t(x, t) e^{-2\pi i x k} dx \\ &= \int_{\mathbb{R}} \lim_{h \rightarrow 0} \frac{u(x, t+h) - u(x, t)}{h} e^{-2\pi i x k} dx \\ &= \lim_{h \rightarrow 0} \int_{\mathbb{R}} \frac{u(x, t+h) - u(x, t)}{h} e^{-2\pi i x k} dx \\ &= \lim_{h \rightarrow 0} \frac{\hat{u}(k, t+h) - \hat{u}(k, t)}{h} = \partial_t \hat{u}(k, t), \end{aligned}$$

2. We have

$$\begin{aligned} \mathcal{F}u_{xx}(k, t) &= \int_{\mathbb{R}} u_{xx}(x, t) e^{-2\pi i x k} dx \\ &= \left[ u_x(x, t) e^{-2\pi i x k} \right]_{-\infty}^{\infty} - \int_{\mathbb{R}} (-2\pi i k) u_x(x, t) e^{-2\pi i x k} dx \end{aligned}$$

$$\begin{aligned}
&= (2\pi ik)^2 \int_{\mathbb{R}} u(x, t) e^{-2\pi i x k} dx \\
&= -(2\pi)^2 k^2 \hat{u}(k, t).
\end{aligned}$$

We now take the Fourier transform of the heat equation:

$$\partial_t \hat{u}(k, t) = -\alpha(2\pi)^2 k^2 \hat{u}(k, t),$$

or equivalently,

$$\partial_t \hat{u}(k, t) + \alpha(2\pi)^2 k^2 \hat{u}(k, t) = 0.$$

This is a linear differential equation, and those things are easy to solve! Using an integrating factor,

$$\partial_t \left( \hat{u}(k, t) e^{\alpha(2\pi)^2 k^2 t} \right) = 0.$$

Hence  $\hat{u}(k, t) e^{\alpha(2\pi)^2 k^2 t}$  is constant in  $t$ . Using the initial condition at  $t = 0$  gives

$$\hat{u}(k, 0) = \hat{\phi}(k).$$

Therefore,

$$\hat{u}(k, t) = e^{-\alpha(2\pi)^2 k^2 t} \hat{\phi}(k).$$

We look for  $S(x, t)$  such that  $\hat{S}(k, t) = e^{-\alpha(2\pi)^2 k^2 t}$ . By the [Fourier inversion theorem](#),

$$S(x, t) = \int_{\mathbb{R}} \hat{S}(k, t) e^{2\pi i k x} dk = \int_{\mathbb{R}} e^{-\alpha(2\pi)^2 k^2 t} e^{2\pi i k x} dk = \frac{1}{\sqrt{4\pi\alpha t}} e^{-\frac{x^2}{4\alpha t}}$$

by a tedious but straightforward transformation of the standard Gaussian integral. Then

$$\hat{u}(k, t) = \hat{S}(k, t) \hat{\phi}(k),$$

so

$$u(x, t) = (S(\cdot, t) * \phi)(x) = \int_{-\infty}^{\infty} \phi(y) \frac{1}{\sqrt{4\pi\alpha t}} e^{-\frac{(x-y)^2}{4\alpha t}} dy.$$

The kernels in [Figure 12](#) show what the convolution does: as time increases, the same unit mass spreads over a wider region and its peak falls.

**Remark 24.2** The solution above can be interpreted as  $\mathbb{E}[\phi(X)]$  where  $X \sim \mathcal{N}(x, 2\alpha t)$ . This probabilistic interpretation connects the heat equation to Brownian motion and stochastic processes, where the diffusion of heat can be seen as the random movement of particles.

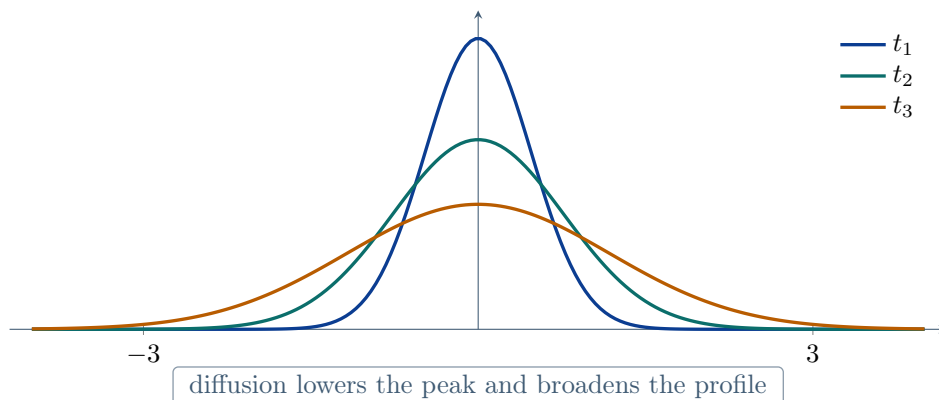


FIGURE 12

### Translation-Invariant Subspaces of $L^2(\mathbb{R})$

For  $f \in L^2(\mathbb{R})$  and  $x \in \mathbb{R}$ , we write  $f_x(y) = f(y - x)$  in what follows.

**Definition 24.3** We say that a subspace  $\mathcal{U}$  of  $L^2(\mathbb{R})$  is **translation-invariant** if it satisfies the following:

1.  $\mathcal{U}$  is a closed subspace of  $L^2(\mathbb{R})$  (that is, it is closed in  $\|\cdot\|_2$  and for all  $f, g \in \mathcal{U}$  and for all  $c \in \mathbb{C}$ ,  $f + cg \in \mathcal{U}$ ).
2. If  $f \in \mathcal{U}$ , then  $f_x \in \mathcal{U}$  for all  $x \in \mathbb{R}$ .

The trivial examples of translation-invariant subspaces are  $\{0\}$  and  $L^2(\mathbb{R})$ . To find less trivial examples, first observe that

$$\begin{aligned}
 \hat{f}_x(k) &= \int_{\mathbb{R}} f(y - x) e^{-2\pi i y k} dy \\
 &= \int_{\mathbb{R}} f(y - x) e^{-2\pi i (y - x + x) k} dy \\
 &= \int_{\mathbb{R}} f(y - x) e^{-2\pi i (y - x) k} e^{-2\pi i x k} dy \\
 &= e^{-2\pi i x k} \hat{f}(k).
 \end{aligned}$$

Let  $\chi_x(k) = e^{-2\pi i x k}$ . Then we have the familiar identity  $\hat{f}_x = \chi_x \hat{f}$ .

Let  $\mathcal{U}$  be a translation-invariant subspace of  $L^2(\mathbb{R})$ , and define  $\hat{\mathcal{U}} = \{\hat{f} : f \in \mathcal{U}\}$ . The translation-

invariance of  $\mathcal{U}$  implies that for all  $g \in \hat{\mathcal{U}}$  and for all  $x \in \mathbb{R}$ , we have  $\chi_x g \in \hat{\mathcal{U}}$ . Recall that the Fourier transform  $\mathcal{F} : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$  is linear, surjective, and isometric by [Theorem 23.4](#). Therefore  $\hat{\mathcal{U}}$  is a closed subspace of  $L^2(\mathbb{R})$  with the following invariance property:

$$g \in \hat{\mathcal{U}} \quad \text{and} \quad x \in \mathbb{R} \implies \chi_x g \in \hat{\mathcal{U}}.$$

Now take a closed subset  $F \subseteq \mathbb{R}$  and consider

$$V_F = \{f \in L^2(\mathbb{R}) : f = 0 \text{ almost everywhere on } \mathbb{R} \setminus F\}.$$

Then  $V_F$  is a closed subspace of  $L^2(\mathbb{R})$  with the invariance property above. Hence

$$\mathcal{U} = \{f \in L^2(\mathbb{R}) : \hat{f} = 0 \text{ almost everywhere on } \mathbb{R} \setminus F\}$$

is a translation-invariant subspace of  $L^2(\mathbb{R})$ .

This matters for representation theory. A **representation** of a group  $G$  is a homomorphism  $\pi : G \rightarrow GL_n$ , so  $\pi(xy) = \pi(x)\pi(y)$  and  $\pi(e) = I_{n \times n}$ . We often want to classify the **irreducible representations** of a group (i.e., representations that have no nontrivial invariant subspaces). If  $G$  is compact, then any translation-invariant subspace of  $L^2(G)$  gives rise to representations that are closely related to irreducible representations of  $G$  (and vice versa). If  $G$  is noncompact, then the translation-invariant subspaces of  $L^2(G)$  are more complicated, and this is one of the reasons why the representation theory of noncompact groups is more difficult than that of compact groups.

## Appendix: Lecture and Tutorial Index

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